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STUDIES OF SURFACE ROUGHNESS AND BLOWING EFFECTS ON HYPERSONIC TURBULENT BOUNDARY LAYERS OVER SLENDER CONES

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Abstract

Studies are presented of the effects of surface roughness and blowing on the aerothermal characteristics of turbulent hypersonic boundary layers over rough transpiration-cooled slender cones. These detailed surface measurements provide the basis for the direct evaluation of the basic modeling of these phenomena in advanced codes. Experimental studies were conducted at Mach numbers of 11, 13 and 15 for turbulent flows over rough blowing models in which the blowing parameter B' was varied from .01 to 5. Three new and unique transducers were developed and successfully used to measure skin friction and local and average heating on the rough transpiration cooled surfaces. Two types of "roughness and blowing" heat transfer gages were developed, a calorimeter gage and a multi-element thin film gage. The calorimeter gage was constructed with a silver calorimeter element containing two roughness elements and insulated injection passages molded into it. A nickel thin film resistance thermometer is to sense the temperature of the calorimeter element. This gage was used successfully to obtain average local heating measurements to determine the effects of mass addition on cone heating. The distribution of heating over and between the roughness elements was determined for a range of blowing rates and freestream conditions with the multi-element thin film gage. The thin film elements were distributed over the roughness elements and on the base between the roughness elements and the injection passages. A new skin friction gage, in which there was blowing through the floating sensing element, was used successfully to measure the effects of blowing and roughness on surface shear. Two roughness elements were molded into the surface of the transducer and injection passages are incorporated into the gage. The results of the measurements with these gages are presented together with comparisons with earlier measurements at lower Mach numbers and semi-empirical prediction techniques.

I. Introduction

In the development of advanced prediction techniques required to design and predict the aerothermal loads, on sophisticated maneuverable airbreathing hypersonic vehicles, there are a number of key flow/field and flow/surface interaction phenomena that must be modelled which require the insight and measurement of detailed experimental study. The modelling of the turbulent flow structure, of the flow over transpiration-cooled and rough ablating surfaces, is an area where a detailed understanding of the mixing process between the injected fluid, the roughness elements, and the fluids at the base of the turbulent boundary layer is required. Large surface ablation resulting from heat transfer rates generated on the windward ray of the ablative heat shield close to the nosetip and on the control surfaces of vehicles flying at high angles of attack will be of critical concern to the designer of missiles that maneuver during re-entry. To develop an accurate predictive capability to describe the ablation rates of the nosetip, heat shield, and control surfaces, it is necessary to understand and model the separate and combined effects of surface blowing and surface roughness. Our recent studies of roughness shape and spacing effects on ablating and non-ablating slender cones (Ref. 1) suggest not only that the subsonic studies are inapplicable to the heating of heat shields in hypersonic flow, but also that the basic modeling of the roughness drag and mechanisms of heating in the theoretical models is highly questionable.

The measurements made in earlier studies of transpiration cooling conducted with transpiration cooled nosetips[1] were designed principally to determine whether blockage effects of mass injection are as large as predicted by the current codes. The measurements on the model with zero blowing, presented in Figure 1, clearly show that the intrinsic roughness of the surface causes heating enhancement factors of over 1.7. In fact, it can be seen by comparing Figures 1 and 2 that the heat transfer measurements on the conically rough hemisphere are in good agreement with those obtained on the non-blowing transpiration-cooled nosetip. However, when a small amount of blowing ($\dot{m}/\rho_e \mu c_H$ = 0.032) was introduced, the heating rates over a major part of the transpiration-cooled model dropped to levels close to those recorded on the smooth model, as shown in Figures 2 and 3. It could be postulated on the basis of these measurements that the initial effect of mass addition from a rough ablating nosetip is to modify the flow around the roughness elements by eliminating the cavity flows between them in such a way that the momentum defect is small. If the effect of mass addition is to remove surface roughness as an important characteristic parameter, a series of questions are posed for the correlation of flight measurements in terms of an effective surface roughness and the computational procedures in which the ablation rate is determined from heating levels enhanced by surface-roughness effects.

Mass addition at the base of the boundary layer results in a reduction in momentum in the wall layer, which, like roughness, will make the boundary layer more susceptible to separation. However, injecting mass into the boundary layer will also severely distort the temperature and velocity distribution at the base of the boundary layer, which has the potential to cause a reduction in heat transfer and skin friction to the wall. Because momentum reduction on rough walls at the base of the boundary layer results principally from the form drag of the roughness elements, the introduction of

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additional mass at the rough wall could significantly change the basic mechanisms that control momentum and energy exchange. Our experience with transpiration-cooled nosetips[1] as well as those of Voisinet[2] on wind tunnel walls suggests that calculating the combined effects of mass addition and surface roughness (as is done in most prediction techniques) by using the product of the 'influence factors', based on measurements of the separate effects of roughness and blowing, is invalid.

To develop and evaluate prediction schemes that will accurately describe the effects of combined roughness, blowing and entropy layer on the aerothermal performance of slender rough ablating cones, we must conduct experimental studies under hypersonic, high-Reynolds-number, and highly cooled-wall conditions. The models should be constructed with well-defined roughness and blowing characteristics. The correlations of the measurements from these studies can then be used to directly evaluate the accuracy of aerothermal models currently in use in "shape change" codes, while individual sets of measurements should be compared with more detailed calculations of combined roughness and blowing on sharp slender cones.

II. Review of Related Studies

Studies of Surface Roughness Effects in Hypersonic Flows

Because most predictive techniques employ an effective sand-grain roughness as the single length scale characterizing roughness size, there continues to remain a key problem in relating the topography of a given surface to a sand-grain height. The experimental studies of Nikuradase[6] and Schlichting[7], both hydraulic pipe flow studies, were principally responsible for the selection of sand-grain roughness as the standard against which to measure relative effects of other types of roughness. Although this standard has been frequently employed, the topographical characteristics of a sand-grain surface have yet to be defined. In fact, because of the experimental difficulties involved in the preparation and inspection of rough surfaces inside a small-diameter pipe, it is surprising that Nikuradase's results are as consistent as reported. For fully developed pipe flows, Nikuradase established that the parameter controlling the similitude of the flows is the roughness Reynolds number $(u_\tau K/\nu_w)$. This parameter $(\tilde{R}_{eK})$ was selected in many subsequent studies to characterize boundary layer flows where other non-dimensional groupings (such as R_s/δ, R_s/θ, etc.) might have been considered more valid.

The Schlichting studies, conducted with roughness of well-defined geometric shapes, provided the first set of measurements which could be reproduced in both experimental and theoretical studies. The results from these studies, together with those from a number of subsequent investigations in subsonic adiabatic flows, were correlated by researchers to yield relationship between an "effective sand-grain height" and parameters which describe the geometric features of the surface. This further perpetuated the use of sand-grain roughness as a standard. The Dirling[8] correlation, which is shown in Figure 4, is one such plot, from which an effective sand-grain roughness height can be determined from knowledge of peak-to-valley roughness height together with the shape and spacing of the roughness elements. While there is little direct supporting experimental evidence, the "lambda" form of the correlating curve is assumed to reflect a sudden change in flow structure from an "open" to a "closed" cavity flow around (between) the roughness elements as the spacing between the roughness elements is varied. Do open and closed cavity flows really exist on rough surfaces constructed from three-dimensional roughness elements? Is the structure of the flow around the elements also dependent upon the local Reynolds number, a parameter not taken into account in any such correlations? If an effective roughness height can be accurately determined from a 'bump curve', this dimension must be combined with key fluid dynamic properties to yield a non-dimensional parameter or groups of parameters with which to characterize the flow. The roughness Reynolds number $\tilde{R}_{eK}\left(=\frac{U_\tau K}{\nu_w}\right)$ originally used by Nikuradase, and the non-dimensional roughness heights K/δ^*, K/θ_m and K/θ_t (where δ^*, θ_m, and θ_t are the displacement, momentum, and thermal energy thickness respectively), have all been used to correlate the aerothermal effects associated with boundary layers over rough re-entry vehicles. To date, however, no single parameter or combinations of parameters (e.g., $\tilde{R}_{eK}$, $\frac{K}{\theta_m}$) has been used with any great success to describe the general similitude of turbulent boundary layers in supersonic and hypersonic flows over rough, highly cooled walls.

The studies of Dirling[8] ettermann[9], Dvorak[10], Simpson[11] and more recently Lin[12] and Finson[13] have provided further insight into the basic effects of roughness shape and spacing on the characteristics of the rough wall boundary layer and skin friction and heating to a rough surface. Dvorak combined the effects of roughness shape and spacing into a single parameter λ (the roughness density), defined as shown in Figure 5. He linked the downward shift in the velocity profile $\Delta U/U_\tau$ to a combination of roughness Reynolds number $R_{eK}(=U_\tau K/\nu_w)$ and λ through the relationship

$$\frac{U}{U_\tau} = \frac{1}{K}\ln\left(\frac{y U_\tau}{\nu}\right) + A - \frac{\Delta U}{U_\tau}\left(\frac{K U_\tau}{\nu_w}, \lambda\right) \quad (1)$$

In incompressible flows the smooth regime, where the surface shear is entirely due to viscous shear, is defined by $\tilde{R}_{eK} < 5$. For $\tilde{R}_{eK} > 30$, the surface shear is composed of form drag on the roughness elements combined with viscous shear. When $\tilde{R}_{eK} > 70$, the surface shear results principally from drag, and viscosity is no longer a factor in controlling the velocity profile. For fully rough flows, Equation 1 can be rewritten

$$\frac{U}{U_\tau} = \frac{1}{K}\ln\left(\frac{y}{K}\right) + A - \left[\frac{\Delta U}{U_\tau} - \frac{1}{K}\ln\left(\frac{U_\tau K}{\nu_w}\right)\right] \quad (2)$$

where

$$\frac{\Delta U}{U_\tau} - \frac{1}{K}\ln\left(\frac{K U_\tau}{\nu_w}\right) = f(\lambda) \quad (3)$$

which is a function of the roughness density. Here it should be noted that $f(\lambda) = A - U_R^*$ where U_R^* is the velocity close to the top of the roughness elements and A (= 5) is the smooth wall constant. Now the definition of Nikuradse's sand-grain roughness is basically

$$\frac{U}{U_\tau} = \frac{1}{K}\ln\left(\frac{y}{K_s}\right) + 8.5$$

Hence, combining equations 2 and 3 we obtain

$$f(\lambda) = -\left[3 + \frac{1}{K}\ln\left(\frac{K}{K_s}\right)\right] \quad (4)$$

Betterman and Dvorak suggest that the subsonic measurements should be correlated by two relationships:

For

$$1 < \lambda < 4.68, \; f(\lambda) = 17.35\,(1.625 \log_{10} \lambda - 1) \quad (5)$$

And for

$$\lambda > 4.68, \; f(\lambda) = -5.95\,(1.103 \log_{10} \lambda - 1) \quad (6)$$

The existence of two regions has been rationalized on physical grounds, supported by experimental studies in two-dimensional flows. There is serious debate about whether a significant change in flow structure occurs as three-dimensional roughness elements are drawn together. Certainly the measurements with stone roughness do not exhibit such a trend. This is particularly unfortunate, since Nikuradse's data falls on the line constructed principally from data obtained on roughness constructed by two-dimensional machined grooves.

Finson's engineering model based on the early concepts of Liepmann and Goddard[14] and his (Finson's) detailed numerical calculations provides a good basis for interpreting the physical phenomena of key importance in rough wall heating, as well as a relatively simple prediction technique. The shear on a rough wall can be expressed as the sum of the viscous and form drag of the rough surface:

$$c_f = c_{f_{base}} + \int_B^T \frac{\rho u^2}{\rho_e u_e^2} \cdot c_D \cdot B(y) \cdot \frac{d(y)}{D^2} \, dy \quad (7)$$

where $B(y)$ is the blockage factor, and $d(y)$ and D are the diameter of the roughness element and the spacing between elements, respectively. From his detailed numerical solutions, Finson showed that ρ and u were relatively constant between the base and top of the roughness element at values ρ_R, u_R close to the top of the roughness. Equation (7) becomes

$$c_f = c_{f_{base}} + \frac{\rho_R U_R^2}{\rho_e u_e^2} \cdot c_D \cdot B\left(\frac{K}{2}\right) \frac{A_p}{A_s} \quad (8)$$

where A_p/A_s is the ratio of projected area of the roughness element in the direction of the flow to total area of the flow on which they stand, and $B(\frac{K}{2})$ is the average vlaue of $B(y)$.

For compressible flows Finson found that

$$U_R/U_e = 0.247 + .234 \log\left(\frac{\rho_R}{\rho_e} \frac{A_s}{A_p}\right)$$

where

$$\frac{\rho_R}{\rho_e} = \left[\frac{T_w}{T_e} + \left(1 + \frac{\gamma - 1}{2} M_e^2 - \frac{T_w}{T_e}\right) \cdot \left(\frac{U_R}{U_e}\right) - \left(\frac{\gamma - 1}{2}\right) M_e^2 \frac{U_R^2}{U_e^2}\right]$$

Therefore, assuming c_{f_B} is the smooth wall heating level, it is possible to relate the rough wall skin friction to the smooth wall value in the generalized form

$$\frac{c_{f_{ROUGH}}}{c_{f_{SMOOTH}}} = 1 + f_1\left(M_e, \frac{T_w}{T_e}, \frac{K}{\theta}\right) \cdot f_2\left(\lambda, B, \left(\frac{K}{D}\right), C_D\right)$$

The relationship for heat transfer is assumed of the same form

$$\frac{C_{H\,ROUGH}}{C_{H\,SMOOTH}} = f_1'\left(M_e, \frac{T_w}{T_e}, \frac{K}{\theta}\right) \cdot f_2'\left(\lambda, B, \frac{K}{D}, C_D\right)$$

If we assume that the product of the blockage factor and are invariant with rough shape and space, then for constant local free stream condition, we get the Dvorak-Simpson Parameter:

$$\frac{C_{H\,ROUGH}}{C_{H\,SMOOTH}} = f\left(\frac{A_p}{A_s}\right)$$

A slightly different form can be obtained by the subsonic blunt body approximation

$$C_D = C_{D\,REF}\,(A_p / A_{WS})$$

and using D/K rather than A_s/A_p to obtain the correlation in terms of the Dirling[15] parameter,

$$\frac{C_{H\,ROUGH}}{C_{H\,SMOOTH}} = f\left(\frac{D}{K} \cdot \left(\frac{A_{WS}}{A_p}\right)^{4/3}\right)$$

where the various areas are illustrated in Figure 36.

In recent studies of the effects of roughness shape and spacing on the heat transfer and skin friction to the roughness, nosetips, frusta and flaps of a typical MRV configuration, Holden used both the Dvorak/Simpson parameter A_s/A_p and the Dirling parameter $\frac{D}{K}\left(\frac{A_p}{A_{WS}}\right)^{4/3}$, and achieved reasonable success in correlations. In these studies, the effects of roughness shape and spacing on the heat transfer and skin friction both for surfaces with sand-grain roughness and those constructed with geometrically well-defined hemispherical and conical roughness elements were examined. Also, heat transfer, skin-friction, and pressure-distribution measurements were obtained on spherical and ablated noseshapes, conical frusta, and control surfaces in hypersonic flow. The results of this work have demonstrated that the low-speed measurements of Nikuradse[16], Schlichting[17], and others, and the correlations of Dirling[18]/Simpson[19], cannot be used directly to predict rough-wall heating and skin friction in supersonic and hypersonic flows over non-adiabatic surfaces. Even the more solidly founded prediction scheme developed by Finson[20], which is based upon his detailed numerical solutions, consistently overpredicts the roughness-enhanced heating levels in high-speed flows. While direct measurements of the skin friction and heat transfer to geometrically well-defined rough surfaces provide the opportunity to more closely evaluate the accuracy of the current shape-change codes and the more fundamental treatments like that of Finson in high Mach number, high Reynolds number flows, there remains a basic need to define more closely the fundamentals of the fluid dynamics which control momentum and energy exchange in high-speed flows over adiabatic and non-adiabatic rough surfaces.

Combined Blowing and Surface Roughness Effects in High-Speed Flows

The models used in the current codes to describe the effects of surface roughness in the presence of surface blowing unfortunately have little foundation in physical modeling and even less support from experimental measurements. While there have been a series of experimental studies from Nikuradase to Holden to determine the effects of surface roughness, skin friction, and heat transfer, little work has been done to examine the separate and combined effects of surface blowing and surface roughness on the heat transfer and skin friction to rough surfaces. In fact, only two studies are available to substantiate theoretical modelings. One study conducted in low speed flow (v = 50 ft/sec) by

Healzer et al,[21] and a second by Voisinet[22] in a wind tunnel wall with Mach 3 airflow in the absence of heat transfer, where the key parameters were orders of magnitude different from those encountered in hypersonic re-entry. In the Healzer studies, the surface was constructed with hemispherical roughness elements while Voisinet employed a porous surface constructed from wire mesh. In both cases, an effective sand-grain roughness height was selected from Nikuradase[23] or Dirling[24] correlations of low speed roughness measurements. In the Healzer studies, the skin friction measurements were deduced indirectly from velocity profiles and momentum-integral techniques. While this reduction technique can be relatively accurate for the non-blowing case, when applied to experiments with surface blowing, the accuracy of the technique is poor. Apparently, this was the case because Healzer did not present correlations of his skin friction measurements. One very interesting feature of the results of the Stanford[25] study is the form of this expression to correlate the effects of heating on rough walls:

$$\frac{St_{BLOWING}}{St_{UNBLOWN}} = \left[\ln\left(1 + \frac{\rho_w V_w}{\rho_e u_e C_H}\right) \Big/ \frac{\rho_w V_w}{\rho_e u_e C_H}\right]^{1.25} \left[1 + \frac{\rho_w V_w}{\rho_e u_e C_H}\right]^{1/4}$$

This is very similar to the form of the correlation developed at Stanford for smooth walls. Here it should be noted that the non-blowing value of heating coefficient was taken at the same enthalpy thickness as in the blowing case, rather than at a common Reynolds number based on scale length. These measurements were compared with a prediction scheme based on a modified Van Driest method for rough walls. However, because they were tailored to this specific experiment, they have not been used in shape change codes.

Voisinet's[26] studies of the combined effects of roughness and blowing were conducted at Mach 6 under adiabatic wall conditions ($q_w = 0$) on a wind tunnel wall. Only surface skin friction measurements were made during Voisinet's studies. This is unfortunate because while modified Reynolds analogy relationships (however poor) have been developed for rough walls, linking skin friction to heat transfer in the presence of both roughness and blowing is a very difficult theoretical task. However, Voisinet's measurements have clearly demonstrated that the combined effects of flowing and roughness on skin friction cannot be described in a simple manner. First, Voisinet found that on the tunnel wall, where the experiments were conducted, the effects of surface roughness alone on skin friction could be correlated in terms of the roughness Reynolds number, Re_K. This is a result consistent with earlier measurements on adiabatic walls by Goddard[27] and Reda[28]. In contrast, measurements on models placed in the flow involving significant levels of heating ($T_w/T_o > 0.5$) have in general correlated better with parameters like K/δ, K/θ or K/δ_T; which, as shown by the theoretical studies of Dvorak and Finson, should have greater relevance to roughness effects on re-entry vehicles. Voisinet's studies demonstrated that the effects of surface roughness and blowing on skin friction cannot be deduced from simple expressions derived from the measurements made of each of the separate effects. Correlations were presented to enable skin friction to be deduced in the presence of combined roughness and blowing. Extrapolating these measurements to flows at higher Mach numbers over highly cooled surfaces, as well as inferring changes in heat transfer from the skin friction measurements, is obviously highly speculative.

A correlation of the measurements on transpiration-cooled "smooth" flat plates and cones at lower Mach numbers are shown in Figure 6. Measurements by Holden[1] on transpiration cooled nosetips and are compared with the earlier measurements in Figure 7. These measurements are compared with the results of current studies in Section IV subheading three.

III. Experimental Program

Objective and Design of the Experimental Study

The objective of this experimental study of blowing and roughness effects in high Reynolds number hypersonic flow was to obtain detailed surface heat transfer and skin friction as well as flow field measurements, to evaluate the models of the surface/flow field interaction. The development of advanced computer codes which, in principal, can be used to describe the macroscopic features at the base of the boundary layer, has basically outmoded the use of surfaces with poorly defined roughness characteristics such as "sand grain" roughness. A review of surface roughness studies suggests that the definition of sand grain roughness is related more to a theoretical criteria based on a measured velocity profile rather than the geometric characteristics of the surface. In fact, Nicaradase[31], who first introduced sand grained roughness based bonding sand grains "end to end" to a surface, noted that sand of the same size from a different geographic location exhibited different roughness characteristics. From the viewpoint of generating measurements with which to provide a definitive evaluation of predictive techniques, it is important that the roughness geometry is well defined. Schlicting was the first researcher to obtain measurements on surfaces whose geometric roughness could be reproduced in both theoretical and experimental studies. From the results of these and subsequent studies in subsonic, adiabatic flows, an empirical relationship was devised between roughness shape and spacing and an "effective sand-grain roughness height." The correlation of the measurements further perpetuated the use of "sand grain roughness" as a standard.

The single parameter characterization used in many "shape-change" codes designed to predict change in the noseshape of an ablating nosetip as it re-enters the atmosphere is based principally on measurements in low speed, low temperature flows at Mach numbers and Reynolds numbers differing significantly from re-entry conditions. In these studies, the surfaces were characterized by parameters such as K/δ, K/θ and Tw/To. The roughness shape parameter (λ), which has been defined in various ways by different investigators, can be generally expressed as the product of the spacing ratio (D/K) and a parameter related to the shape of the roughness element or its drag coefficient. The relationship between K/Ks and the shape parameter λ (the lambda curve shown in Figure 5) exhibits large changes in Ks for small changes in λ. A key question is whether the correlation is sensitive to flow structure between the roughness elements resulting from changes in the local Mach number and Reynolds number. In the present study we elected to use a surface with a spherical roughness characteristic of a sand grain surface with some roughness height.

Experimental Facilities and Test Conditions

The experimental program was conducted in Calspan's 96-inch Shock Tunnel at freestream Mach numbers from 11 to 16, for Reynolds numbers up to 100

x 10^6 and wall-to-freestream stagnation temperature ratios of 0.1 and 0.2. At Mach numbers up to 13, boundary layer transition is complete on the completely smooth models within six inches of the nosetip. For the blowing model at Mach numbers up to 16, transition is within six inches from the nosetip. These studies were conducted at roughness Reynolds numbers (Re_K) from 20 to 300 on the conical forebody and 200 to 2000 on the flap, thus covering transitional to fully rough regime. Non-dimensional blowing rates, $B'_h = \frac{\rho_w V_w}{\rho_e u_e c_H}$, from 0 to 5 were used in the studies.

Models The slender cone model which was used in the experimental studies is shown in Figure 8. The model has a transpiration cooled surface fed from eight high-pressure reservoirs through eight Valcor fast acting valves. Each section of the model is constructed with six zones that run from the front to the rear of the model. This enabled us to vary the blowing circumferentially to simulate the effects of differential blowing resulting from model incidence. To distribute the gas from the reservoirs uniformly to the surface, each model employs a matrix of distribution passages.

As in our earlier studies of the effects of patterned roughness on the aerothermal characteristics of nonporous slender cones, we used hemispherical roughness elements. A molding technique that was developed at Calspan to produce a number of different roughness patterns was used to obtain the smooth and rough surfaces. The low velocity mass addition between each roughness element occured from passages molded in the skin between the flow, which was controlled by sonic orifices at their base. This latter technique was employed very successfully in our earlier transpiration-cooled studies where we were able to obtain precisely controlled blowing conditions. The flow from each hole in the model was controlled by eight sonic orifices in the model skin that were fed from plenum chambers in the model. The flow from each orifice was released into a cylindrical passage molded in the rubber skin between each roughness element. The area ratio between the orifice and circular passage allowed injectant flows from the surface at velocities of approximately 100 ft/sec. While this type of model construction is a painstaking task, the end result, which is an experiment where surface roughness and blowing are completely defined, is worth the effort. Because we used choked orifices over the entire model, mass flow from each model zone was precisely controlled by plenum pressures. Mass flows over the model were unaffected by the distribution of surface pressure on transients associated with tunnel starting. Although spherical roughness elements were used in the current studies, this technique can be used to construct rough surfaces that replicate any ablated surface.

Surface Instrumentation The thin film heat transfer instrumentation developed to measure the surface distribution of heat transfer to the roughness is shown in Figures 9 and 10. The two instrumented hemispherical roughness elements can be rotated in the holder to provide measurements on the windward and leeside surfaces of the element. Also, the gage can be rotated in the holder to provide greater details of the heat transfer at the base of the roughness elements. As in the rest of the model, the blowing velocities were in the range of 80 to 300 ft/sec, generating dynamic pressures of less than 1% of the freestream values.

A blowing calorimeter gage developed specifically for this program is shown in Figure 11. The calorimeter element is minted from a high-purity silver billet and drilled to accept the passages for the blowing holes. The calorimeter element is mounted in a Macor holder and the passages for the coolant are installed with a highly insulating adhesive.

One of the skin friction gages that was designed, developed and constructed is shown in Figure 12. After a number of different approaches were tried, a design was developed with non-metric coolant passages passing through the metric diaphragm with very little clearance. Such tight clearances are allowed because the crystal and rubber support and measuring system on which the diaphragm is mounted is very stiff, so that deflection under load is insignificant. The gage has a linear response up to 0.7 psia and a nominal sensitivity of 20,000 mV/psi.

Pressure measurements were made using flush-mounted Kulite transducers. Pressure gages have been mounted flush with the silicon skin to record the pressure fluctuations, as well as the mean pressure.

Holographic Interferometry

Holographic interferometry was used to make flowfield measurements. Interferograms of complex flowfields provide good qualitative basis for evaluating some of the important phenomena that control the characteristics of these flows. CUBRC's holographic recording system, as shown in Figure 13, was used in this study.[38] Both single plate and dual plate techniques are required to record holograms, which are subsequently used in the playback step to obtain shadowgrams, Schlieren photographs, and interferograms of the tests.

IV. Results and Discussion

The experimental studies were conducted at Mach Numbers of 11 and 13 for a range of blowing rates (B' from 0.01 to 5). Both sharp and blunt nosetip configurations were employed in this study, however the principal emphasis was on flows which were not influenced by entropy swallowing effects. The molecular weight of the injectant was varied by diluting the principal injectant nitrogen with helium to provide a gas with a molecular weight similar to that of a typical ablator. A major objective of the current study was to investigate the effects of blowing on boundary layer separation. Measurements were made for a series of flap angles, which in the absence of blowing, the flow was attached. We also studied the effects of blowing and roughness on flows which were well-separated. A matrix of the runs made in the current experimental program is shown in Figure 14, and the test conditions reported in Figure 15.

Performance of New Heat Transfer and Skin Friction Instrumentation

A significant accomplishment in this study was the development three new types of instrumentation to obtain detailed measurements in the presence of roughness and blowing. To provide information which is of direct value to those attempting to model the macroscopic flow mechanics around the roughness elements, we must provide measurements of the heat transfer distribution around and at the base of the roughness elements. These measurements would be of little use if the gages themselves were not an integral part of the blowing surface with the injected gas passing through and flowing over the sensing element. Clearly, such requirements significantly complicate the design and construction of the gage. However, these very difficult problems were overcome, and a series of very unique measurements were obtained.

Design and Measurements with the New Skin Friction Gages

The skin friction gage, shown schematically in Figure 12, is an acceleration compensated single component force balance in which transpiration cooling passages are vented through the surface of the gage. A floating diaphragm, which in this particular design embodies two roughness elements, is supported flush with the surface on a piezoelectric sensing beam through a single fixture. The diaphragm is stabilized around its perimeter silicon posts. The silicon posts are molded into a gasket that contains a rubber boot which is cemented between the diaphragm and the body of the gage to prevent the hot gases from reaching the crystal beam. A second beam and diaphragm combination is incorporated into the body of the gage to provide a signal used for transducer acceleration compensation. The electrical signals from the sensing and compensation beam are added electrically in such a manner that when the gage is "shaken" in the absence of an air load, the net output is zero. The injection flow through the gage is metered through a series of sonic orifices set in the base of the gage. The geometric and mechanical design features of the gages were refined in a series of bench tests and tunnel studies. The developed gage has a frequency response of 20 kHz and a sensitivity of 20,000 mV/psi. A typical output of this skin friction instrumentation during a run with injection through the gage is shown in Figure 16.

New "Roughness and Blowing" Heat Transfer Instrumentation

During the course of these studies we developed two new gages to measure heat transfer in the presence of roughness and blowing. The first was a calorimeter gage which measured the total heat transfer rate to a segment of the surface. The second was a gage which provided the detailed distribution of heat transfer over the roughness elements and on the base of the model between the roughness elements. In each case, the size and geometry of the gage was selected so that it was a representative segment of the roughness and blowing surface. Thus, we selected a rectangular segment which contained two roughness elements and two transpiration cooling ports.

Calorimeter Instrumentation A photograph of the calorimeter gage is shown in Figure 11. The sensing element is a silver calorimeter slug the temperature of which is measured with a nickel resistance thermometer attached to the rear surface. Two roughness elements are molded into the surface of the calorimeter slug as the element is minted from a pure silver billet. The rear surface of the gage is highly polished, and then coated with several different, thin, electrically insulating, but thermally conducting layers upon which the nickel film is deposited. While this gage is significantly more difficult to construct than one employing a thermocouple sensor peened into the base of the silver slug, attaining a sensitivity of almost an order of magnitude larger that the thermocouple gage is well worth the added effort. The calorimeter elements are cemented into a Macor holder with an ultra-low conductivity, polyurethane adesive which has a thermal diffusion depth of a fraction of a thousand during the 7 millisecond test time. This same adhesive was used to cement the transpiration tubes as they passed through the calorimeter element. Thus, during the test time, there was negligible loss in heating from this source or the evacuated rear surface of the gage. The injectant which passed through the gage was metered through a series of sonic orifices at the base of the passageway. The design and operation of these gages were refined in a number of bench tests and tunnel studies. While dynamic calibrations have been performed with a high powered laser heat source, we have found that computing the sensitivity on the basis of measuring the mass of the pure silver slug and the temperature coefficient of the nickel resistence thermometer consistantly gives the most accurate results. A typical output from the calorimeter gages is shown in Figure 17. The gage has a response time of less than a millisecond. A sliding least-squares technique is used to fit the temperature record, and this curve-fit is differentiated to obtain the temperature rate of change required to calculate the heat transfer rate. Fifteen of the calorimeter gages were constructed and used successfully in this program.

Thin Film Heat Transfer Instrumentation The miniature thin film instrumentation employed in this study enabled, for the first time, detailed distributions of heat transfer to be obtained on the actual roughness elements and on the cone surface adjacent to the injection ports and the roughness elements. Again, the size of the gage was selected to give a representative segment of the surface. Each gage contained 11 individual heat transfer gages. The positioning of the 7 thin films around the base of the roughness elements and around the injection ports was selected to provide distribution of heating in sufficient detail to infer the key aerothermal phenomena (see Figures 9 and 10). Likewise the five measurements over the roughness elements, which could be oriented relative to direction of the free stream, were selected to obtain the heating loads as well as provide evidence to define the micro-structure of the flow. Two injection passages run through the body of the gage, and as in the calorimeter and skin friction gages, the flow through these passages are regulated by sonic orifices at the base of these tubes. The temperature coefficients of the thin film resistance elements are measured in an oil bath. We employ the thermal properties of the pyrex substrate. The heat transfer rate, shown in Figure 18, is derived from the surface temperature/time history using the Rae/Tauble algorithm (Ref. 38).

Interferometer Studies with Blowing and Roughness Model

During the blowing and roughness studies a significant effort was devoted to flow visualization and measurements with holographic interferometry. Both infinite and semi-infinite fringe techniques were used, the first principally for flow visualization, and the second for quantitative flow field measurements. During the course of these studies, measurements were made for a range of compression angles and blowing rates for a rough surface configuration. Both nitrogen and nitrogen/helium mixtures were used as injectant, but clearly, only the interferograms made with the nittrogen can be the subject of quantitative interpretation.

A typical example of an infinite fringe for a small blowing rate and a compression surface angle of 25 degrees is shown in Figure 19. In this photograph, the shock layer and the boundary layer over the cone ahead of the expansion corner are well-defined as is the expansion of the boundary layer around this expansion corner. Boundary layer separation takes place halfway along the flat surfaces ahead of the compression ramp resulting in a well-defined separated region approximately three boundary layer thicknesses in length. The recirculation region is well-defined with holographic interferometry, and in this case there appears to be

multiple bubbles. The separation shock, which is clearly visible when it emerges from the boundary layer, indicates by its changes in curvature that the separation region is unsteady, a feature also observed on smooth non-blowing surfaces. Shown for comparison in Figure 20 is the equivalent non-blowing configuration. Here there is little or no separation and the corner shock is relatively steady. Decreasing the flap angle, again with no blowing, results in a completely attached flow over the model as shown in Figure 21. Increasing the blowing to a level above that used for run 34, we see in Figure 22 that the separation is further extended until the separation for blowing rates of B' = 3 has moved forward to the cone/flat junction as shown in Figure 23. Finally for blowing rates boundary layer blow off occurs on the cone and the flow over the entire model is separated as shown in Figure 24. The finite fringe operating mode is the one from which the most accurate measurements of the distribution of density through the boundary layer and shock layer can be obtained. To obtain a quantitative determination of the density, the assumption must of course be made that the flow is either two dimensional or completely axisymmetric and the latter assumption is potentially true only on the conical section of the model. In Figure 25 we show a finite fringe hologram for a flow with very small blowing in which the cone boundary layer shows no visible evidence of injection at its base. The boundary layer appears to remain attached in the compression corner. Tripling the blowing rate results in a definitive change in the structure at the base of the boundary layer on the cone and the formation of a definitive separated region at the flat/compression surface junction, as shown in Figure 26. For large blowing rates, the structure of the shock layer is completely distorted by blowing with the density increasing from the bow shock to the top of the shear layer before dropping rapidly to a relatively constant density region close to the wall, as shown in Figure 27. Here flow separation has moved forward to almost the cone/flat junction and a large portion of the flap is covered with a separated region. Increasing the blowing further begins to move the bow shock away from the body and the stucture of the complete shock layer is controlled by surface blowing as shown in Figures 28 and 29. Also shown in Figures 30 and 31 are the holograms with a helium/nitrogen mixture for the blowing gas at low and high injection rates respectively. While only qualitative results can be deduced from these photographs, it is clear that blowing with a gas of low molecular weight has a significantly greater effect on the structure of the flow over the cone and compression surface.

Effects of Transpiration Cooling on Surface Skin Friction and Heating and Pressure

The effects of mass addition on the skin friction and heat transfer to the rough cone/flap configuration were investigated at Mach numbers of 11, 13 and 16 for a range of blowing rates (B') from 0 to 10. A test matrix of these studies is given in Figure 14 and a listing of the run conditions are tabulated in Figure 15.

Figure 32 shows the distribution of heat transfer rate to the surface of the rough cone/flap model for a range of blowing rates at Mach 11. The distribution of heating rate on the cone surface and over the roughness elements are shown for each gage station along the model. Here we observe that in the absence of blowing, the peak heating on the roughness elements is typically 10 times the levels at the base of the roughness elements. At the higher blowing the peak-to-base heating ratios have dropped to values close to 3.

Measurements similar to those described above are shown for the Mach 13 condition in Figure 33 a-f, and those for the Mach 15 condition are shown in Figure 34 a-c. We again see that the peak/base heating levels are between 3 and 10 depending upon the blowing rates.

The distribution of heating rate over the roughness elements for a range of low blowing rates plotted as a fraction of maximum heating rate is shown in Figure 35. Here we see that for the lower blowing rates the shape of the distribution becomes more rounded with increasing blowing. At the larger blowing rates shown in Figure 35, the peak heating is now only three times the base value compared with factors of up to ten times for the weakly blowing case.

The variation of peak heat transfer measured on the roughness elements along the cone for a range of blowing rates for the Mach 11 and 13 conditions are shown in Figures 36a and 36b. Similar distributions are shown for heating levels on the floor between the roughness elements and injectors in Figure 37a and 37b. Measurements from the calorimeter gages plotted in this format are shown in Figure 38a and 38b. Clearly the peak heating levels show a larger variation with lambda than those at the base of the roughness elements.

The measurements of heating to the model, normalized by the value for negligiable blowing rate, are plotted versus blowing parameter in Figures 39a, 39b, 40a, 40b, 41a and 41b. We observe a rapid decrease in the heat transfer rates with blowing rates for low values of the blowing parameter, and for blowing parameters above 3 there is little decrease in heat transfer with increased blowing.

One of the unique features of this study is that both skin friction and heat transfer measurements were obtained simultaneously for each of the blowing rates and free stream conditions. Thus, we can provide an analogous set of skin friction plots to those presented above for heat transfer. The variations in skin friction along the cone for a range of blowing rates are shown in Figures 42a and 42b for Mach 11, 13, respectively. These measurements are non-dimensionalized by the zero blowing value and re-plotted to show the variation of skin friction with blowing rate parameters in Figure 43. Again, following a rapid decrease in skin friction with a mass addition rate above B' of 1, there is little decrease in skin friction with increased blowing. We speculate that for blowing rates of B' over 2.0, the boundary layer begins to lift from the surface. This contention is supported by the pressure measurements shown in Figure 44 which suggest boundary "lift off" begins at values of blowing close to 1.

Comparisons with Earlier Measurements and Correlation

The experimental study of Voisinet[2] is the only published study which has provided information in supersonic flow on the effects of combined roughness and blowing on the skin friction to a rough porous surface. Since these studies were conducted for adiabatic wall conditions only skin friction measurements were made in this work. Almost all the theoretical approaches have investigated blowing effects on smooth surfaces, the work of Christoph[32] and Laganelli, et al[33] being the exceptions. The blowing parameter B_f' has been employed by the major segment of the community to correlate the skin friction data, where

$$B_f' = \frac{\dot{m}}{\rho_e u_e c_{f_o}}$$

where the equivalent parameter for heat transfer is

$$B_h' = \frac{\dot{m}}{\rho_e u_e c_{H_o}}$$

In the cases where the molecular weight of the injected gas differs from that of the freestream, parameters $(MW_{inj}/MW_{f_s})^n$ and $(c_{p_{inj}}/c_{p_{f_s}})^m$ are used to modify the skin friction and heat transfer parameters B_f' and B_h' respectively. For turbulent flows $n = 2/3$ and $m = 1$ have been selected. For flows over non-porous surfaces the relationship

$$\frac{c_f}{c_{f_o}} = \frac{B_f'}{(e^{B_f'} - 1)}$$

has been shown to be in good agreement with measurements in incompressible flows[34]. Further studies[35,36] which have included the effects of compressibility suggest that

$$\frac{c_{H_o} - c_H}{c_{H_o}} = \frac{1}{3} B_h'^{\,1/3}$$

is a better representation of the measurements.

In our studies there was a significant surface roughness effect at all the blowing conditions at Mach number significantly larger than the earlier work, therefore we cannot expect the earlier approaches to be totally valid. A comparison between our skin friction measurements and those of Voisinet[2] are shown in Figure 42 together with the contemporary correlations for compressible flows. Here we see that our measurements fall above those of Voisinet[2] and more in line with the correlations of Walker[37] and Langanelli[33]. The heat transfer measurements plotted in an analogous format are shown in Figure 43.

Clearly we will have to await our studies on smooth bodies with and in the absence of blowing before we can make definitive statements on the effects of Mach numbers in these flows.

V. Conclusion

In the studies of blowing and roughness effects on the characteristics of turbulent hypersonic boundary layers a unique set of measurements were obtained which provide the basis for the direct evaluation of the basic modeling of these phenomena effects of roughness and blowing on the aerothermal characteristics of flows over cones friction and local and average heating on rough transpiration-cooled surfaces. Experimental studies conducted at Mach numbers of 11, 13 and 16 for turbulent correlations are presented to show the effects of blowing on the distribution of skin friction and heat transfer over the cone/flap model and on the distribution of heating over the individual roughness elements.

References

1. Holden, M.S., "Studies of Boundary Layer Transition and Surface Roughness Effects in Hypersonic Flow," Final Report, AFOSR Contract No. F49620-82-C-0026, October 1983.

2. Voisinet, R.L.P., "Combined Influence of Roughness and Mass Transfer on Turbulent Skin Friction at Mach 2.9", AIAA Paper 79-0003, 17th Aerospace Sciences Meeting, January 1979.

3. Holden, M.S., "Boundary Layer Displacement and Loading Edge Bluntness Effects on Attached and Separate Laminar Boundary Layer on a Compression Corner, Part II: Experimental Study," AIAA Journal, Vol. 9, No. 1, pp. 84-93, January 1971.

4. Bin, L.Z. & Harvey, J.K., "The Investigation of the Structure of Hypersonic Turbulent Boundary Layers on a 5° Sharp Cone Using the Electron Beam Fluorescence Technique," I.C. Aero Report, 82-02, January 1986.

5. Holden, M.S., Shock Wave Turbulent-Boundary Layer Interaction in Hypersonic Flow, AIAA 77-45.

6. Nikuradse, J., "Stromungsgeretze en rauken Rohren," VDI Forshuaq-shift No. 361, 1933, Translated as NACA TM 1292, 1950.

7. Schlichting, H., "Boundary Layer Theory," 4th Edition, McGraw Hill Book Co., New York, 1960.

8. Dvorak, F.A., "Calculation of Turbulent Boundary Layers with Roughness and Heat Transfer," AIAA Journal, 10, pp. 1447-1451, 1969.

9. Betterman, D., "Contribution of L'etude de la Couche Limite Turbulent li Long Plagnes Rugueness," Rapport 65-5, Centre National de la Recherche Scientifique, Paris, France, 1965.

10. Lewis, M.J., "An Elementary Analysis for Predicting the Momentum- and Heat-Transfer Characteristics of a Hydraulically Rough Surface," Journal of Heat Transfer, 97, pp. 249-254, 1975.

11. Simpson, R.L., "A Generalized Correlation of Roughness Density Effects on the Turbulent Boundary Layer," AIAA Journal, Vol. II, No. 2, pp. 242-244, 1973.

12. Lin, T.C. and Bywater, R.J., "The Evaluation of Selected Turbulence Models for High-Speed Rough-Wall Boundary Layer Calculations," AIAA Paper-80-0132, Pasadena, California, 1980.

13. Finson, J.L., and Wu, P.K.S., "Analysis of Rough Wall Turbulent Heating with Application to Blunted Flight Vehicles," AIAA Paper No. 79-0008, 17th Aerospace Science Meeting, 1979.

14. Goddard, F.E., Jr., "Effect of Uniformly Distributed Roughness on Turbulent Skin-Friction Drag at Supersonic Speeds," Journal of Aerospace Sciences, 26, pp. 1-15, 1959.

15. Dirling, R.B., Jr., "A Method for Computing Rough-Wall Heat Transfer Rates on Reentry Nosetips," MDAC Paper WD 1778, AIAA 8th Thermophysics Conference, Palm Springs, California, 1973.

16. Nikuradse, J., "Stromungsgeretze en rauken Rohren," VDI Forshuaq-shift No. 361, 1933, Translated as NACA TM 1292, 1950.

17. Schlichting, H., "Boundary Layer Theory," 4th Edition, McGraw Hill Book Co., New York, 1960.

18. Dirling, R.B., Jr., "A Method for Computing Rough-Wall Heat Transfer Rates on Reentry Nosetips," MDAC Paper WD 1778, AIAA 8th Thermophysics Conference, Palm Springs, California, 1973.

19. Simpson, R.L., "A Generalized Correlation of Roughness Density Effects on the Turbulent Boundary Layer," AIAA Journal, Vol. II, No. 2, pp. 242-244, 1973.

20. Finson, J.L., and Wu, P.K.S., "Analysis of Rough Wall Turbulent Heating with Application to Blunted Flight Vehicles," AIAA Paper No. 79-0008, 17th Aerospace Science Meeting, 1979.

21. Healzer, J.M., Moffat, R.J. and Kays, W.M., "The Turbulent Boundary Layer on a Rough Porous Plate: Experimental Heat Transfer with Uniform Blowing," Report No. HMT-18, Thermosciences Division Dept. of Mech. Eng., Stanford University, 1974.

22. Voisinet, R.L.P., "Influence of Roughness and Blowing on Compressible Turbulent Boundary Layer Flow," Naval Surface Weapons Center, NSWC Report No. TR-79-153, June 1979.

23. Nikuradse, J., "Stomungsgeretze en rauken Rohren," VDI Forshuaq-shift No. 361, 1933, Translated as NACA TM 1292, 1950.

24. Dirling, R.B., Jr., "A Method for Computing Rough-Wall Heat Transfer Rates on Reentry Nosetips," MDAC Paper WD 1778, AIAA 8th Thermophysics Conference, Palm Springs, California, 1973.

25. Healzer, J.M., Moffat, R.J. and Kays, W.M., "The Turbulent Boundary Layer on a Rough Porous Plate: Experimental Heat Transfer with Uniform Blowing," Report No. HMT-18, Thermosciences Division Dept. of Mech. Eng., Stanford University, 1974.

26. Voisinet, R.L.P., "Influence of Roughness and Blowing on Compressible Turbulent Boundary Layer Flow," Naval Surface Weapons Center, NSWC Report No. TR-79-153, June 1979.

27. Goddard, F.E., Jr., "Effect of Uniformly Distributed Roughness on Turbulent Skin-Friction Drag at Supersonic Speeds," Journal of Aerospace Sciences, 26, pp. 1-15.

28. Reda, D.C., "Compressible Turbulent Skin Friction on Rough and Rough/Wavy Walls in Adiabatic Flow," Naval Ordnance Laboratory, Report No. NOLTR 74-34, Silver Springs, MD., 1974.

29. Finson, J.L., and Wu, P.K.S., "Analysis of Rough Wall Turbulent Heating with Application to Blunted Flight Vehicles," AIAA Paper No. 79-0008, 17th Aerospace Science Meeting, 1979.

30. Lin, T.C. and Bywater, R.J., "The Evaluation of Selected Turbulence Models for High-Speed Rough-Wall Boundary Layer Calculations," AIAA Paper-80-0132, Pasadena, California, 1980.

31. Nikuradse, J., "Stromungsgeretze en rauken Rohren," VDI Forshuaq-shift No. 361, 1933, Translated as NACA TM 1292, 1950.

32. Christoph, G.H., "Low-Of-The-Wall Analysis Revisited for Reentry Vehicle Design," AIAA Paper 85-0905, Williamsburg, Virginia, 1985.

33. Laganelli, A.L., Foganoli, R.P., and Martellucci, A., "The Effects of Mass Transfer and Angle of Attack on Hypersonic Turbulent Boundary Layer Characteristics," AFFDL-TR 75-35, April 1975.

34. Kays, W.M. and Moffat, R.J., "The Behavior of Transpired Turbulent Boundary Layers," Stanford University, Report No. H MT-20, April 1975.

35. Rubesin, M.W., "An Analytical Estimation of the Effect of Transpiration Cooling on the Heat-Transfer and Skin-Friction Characteristics of a Compressible Turbulent Boundary Layer," NACA TN 3341, December 1954.

36. Dorrance, W.H., and Dore, F.J., "The Effect of Mass Transfer on the Compressible Turbulent Boundary-Layer Skin Friction and Heat Transfer," J. Aeronautical Sciences, June 1954.

37. Walker, G.K., "Turbulent Boundary Layers with Mass Addition," G.E. Document No. TFM-8151-021, November 1963.

38. Holden, M.S., "Studies of Aerothermal Loads Generated in Regions of Shock/Shock Interaction in Hypersonic Flow," CUBRC Report, July 1988.

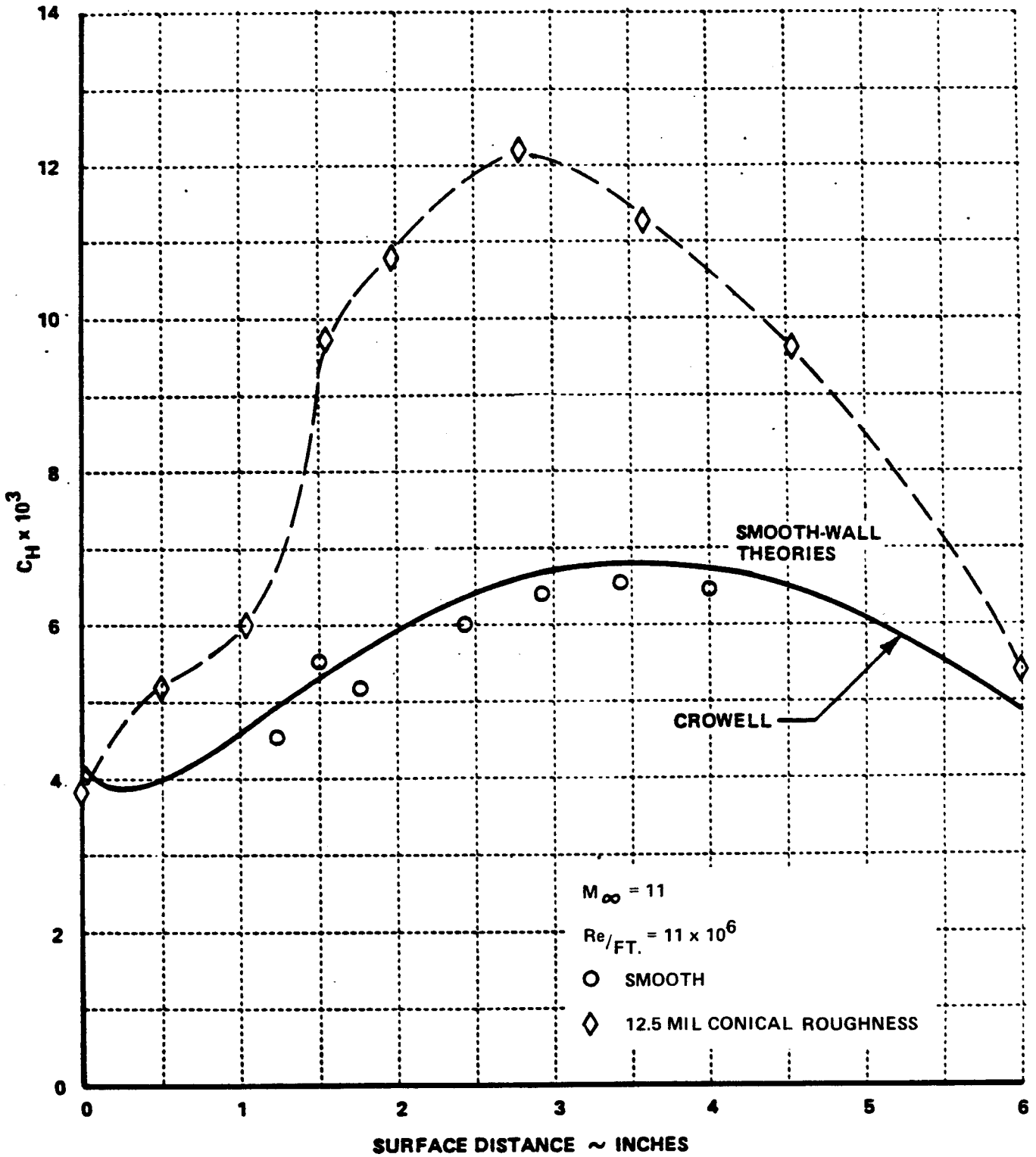


Figure 1 COMPARISON BETWEEN THE TURBULENT THEORIES OF CROWELL, AND SMOOTH-WALL, AND ROUGH-WALL MEASUREMENTS OF HOLDEN ON 12" DIAMETER HEMISPHERE ($M = 11.2$, $Re_D = 11 \times 10^6$, $K = 12.5$)

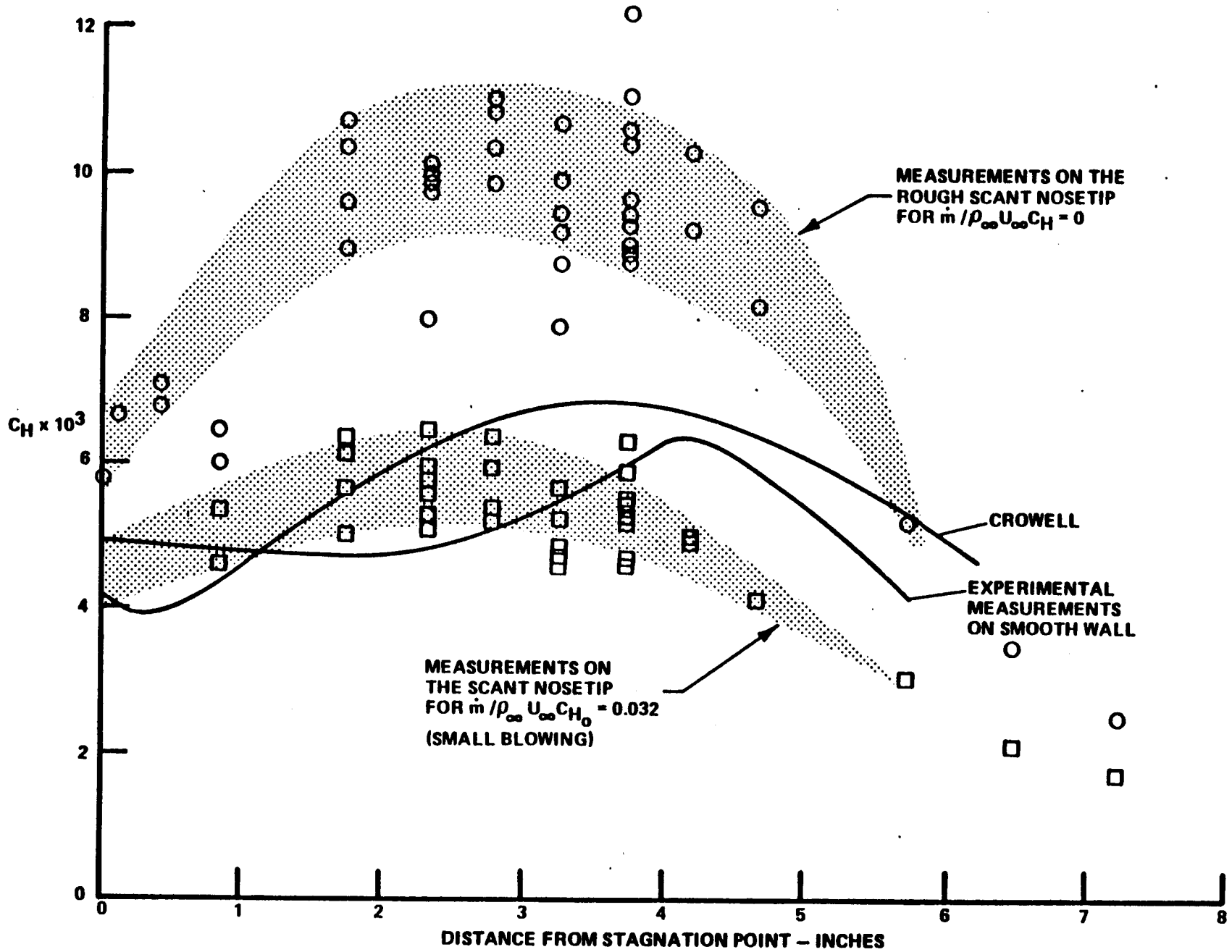


Figure 2 HEAT TRANSFER MEASUREMENTS ON SCANT AND SMOOTH HEMISPHERICAL NOSETIP SHOWING HOW SMALL BLOWING BRINGS DOWN HEATING LEVELS TO SMOOTH-WALL VALUES

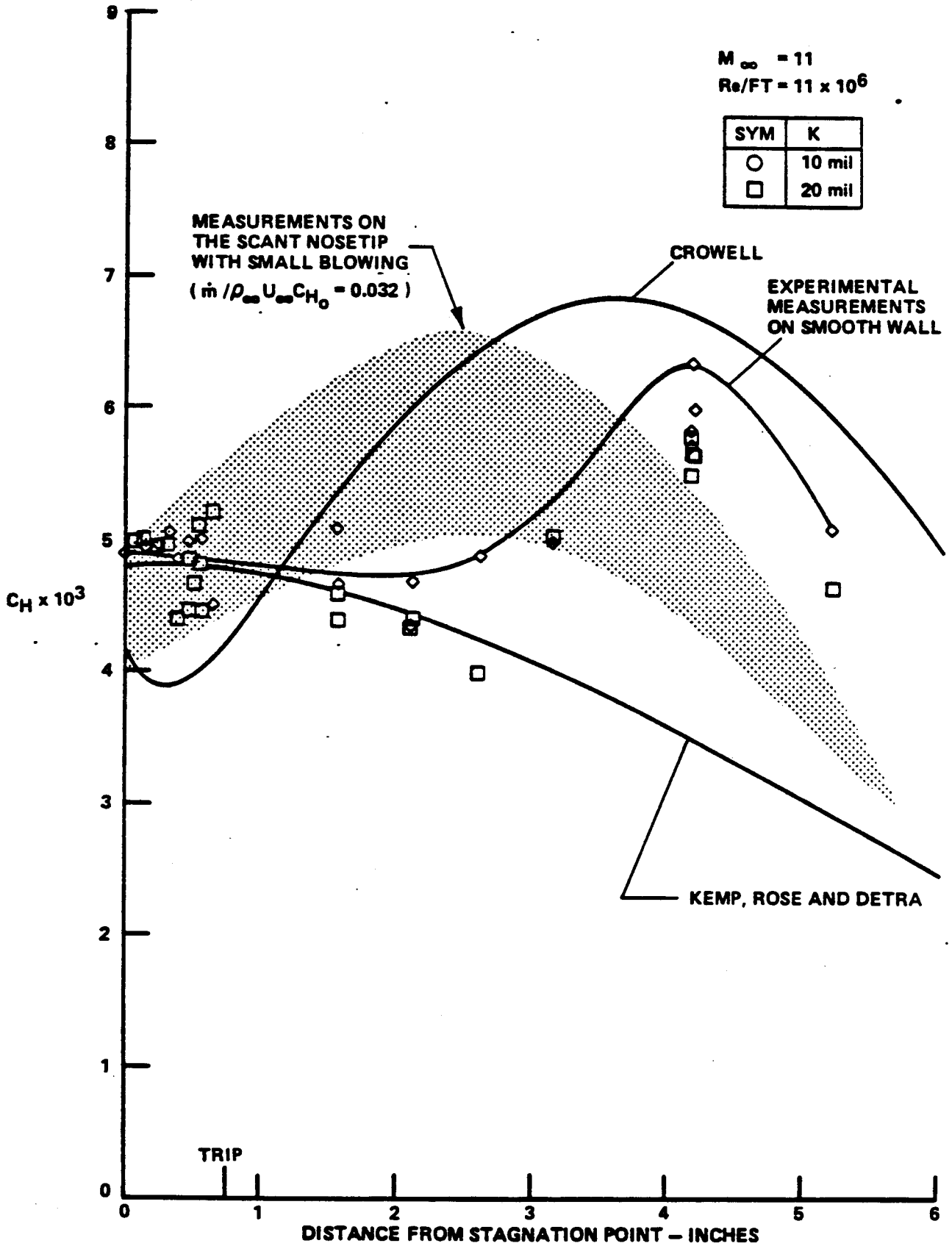


Figure 3 HEAT TRANSFER MEASUREMENTS INDICATING THAT SMALL BLOWING ON ROUGH NOSETIP INITIALLY ACTS TO BRING DOWN HEATING LEVELS TO SMOOTH-WALL VALUES

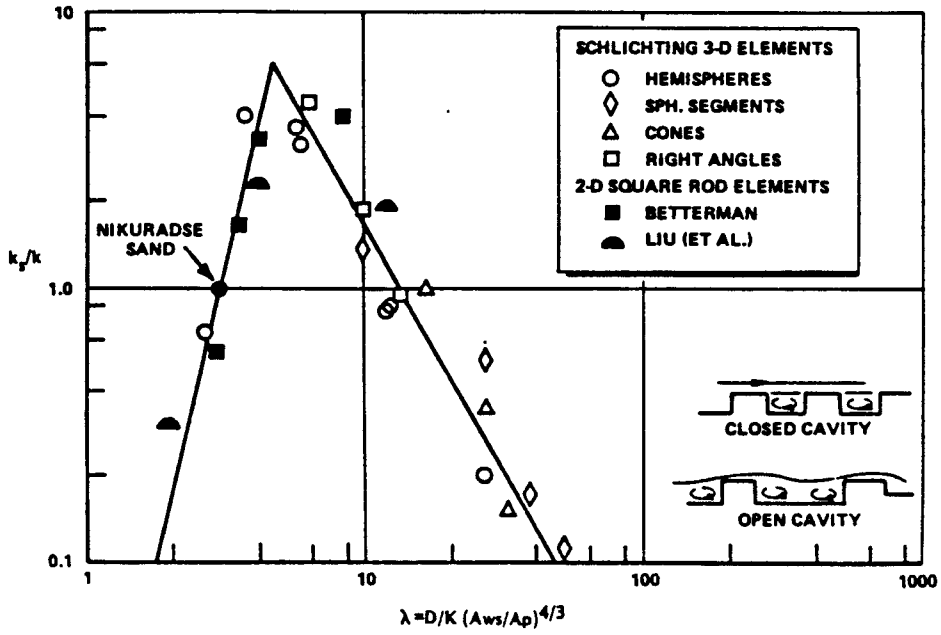
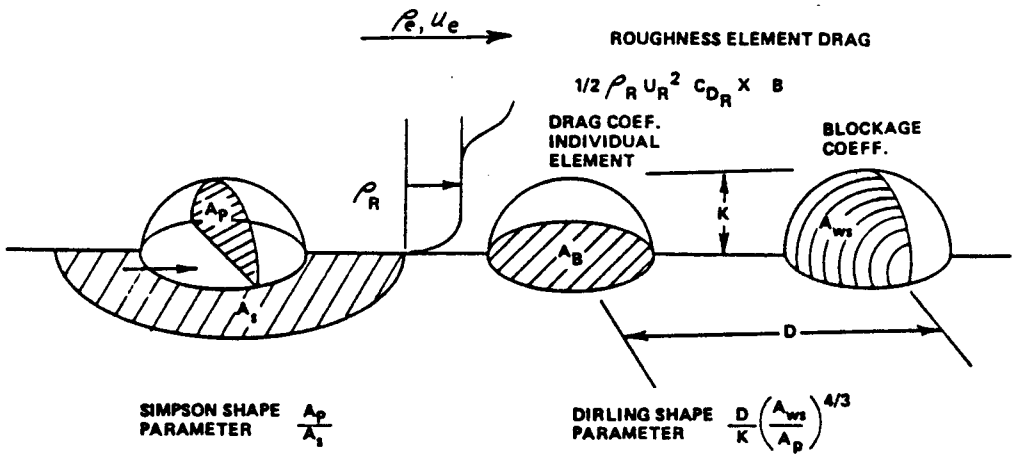


Figure 4 "EFFECTIVE" ROUGHNESS CORRELATIONS FOR DIFFERING ROUGHNESS GEOMETRIES AND SPACINGS



SMOOTH WALL DRAG $\Sigma \Delta D_s$

$$\Delta D_s = \tfrac{1}{2} \rho_e u_e^2 C_{F_s} A_s$$

FORM DRAG OF ROUGHNESS ELEMENT ΔD_R

$$\Delta D_R = \tfrac{1}{2} \rho_R u_R^2 \cdot C_{D_R} A_p \cdot B$$

$$\Delta D_T = \Delta D_s + \Delta D_R$$

$$\Sigma \Delta D_T = \tfrac{1}{2} \rho_e u_e^2 C_{F_R} \Sigma A_s$$

$$\boxed{\frac{C_{F_R}}{C_{F_s}} = 1 + \frac{1}{C_{F_s}} \cdot \frac{\rho_R u_R^2}{\rho_e u_e^2} \cdot B \cdot C_{D_R} \cdot \left(\frac{A_s}{A_p}\right)^{-1}}$$

Figure 5 SIMPLIFIED DRAG MODEL FOR ROUGH-WALL SKIN FRICTION

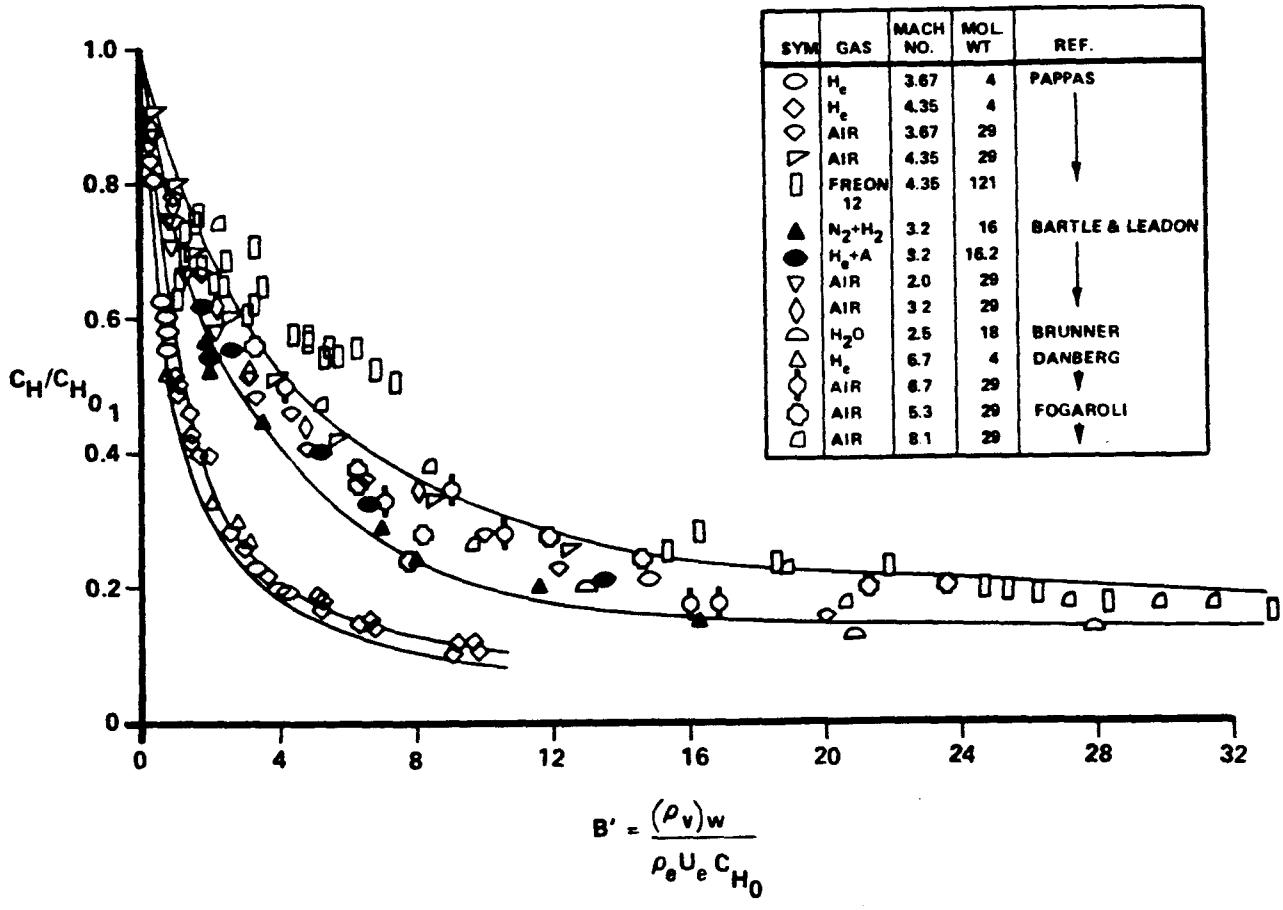


Figure 6 SUMMARY OF BLOCKAGE HEATING FROM EARLIER STUDIES ON FLAT PLATES AND CONES IN TURBULENT FLOW

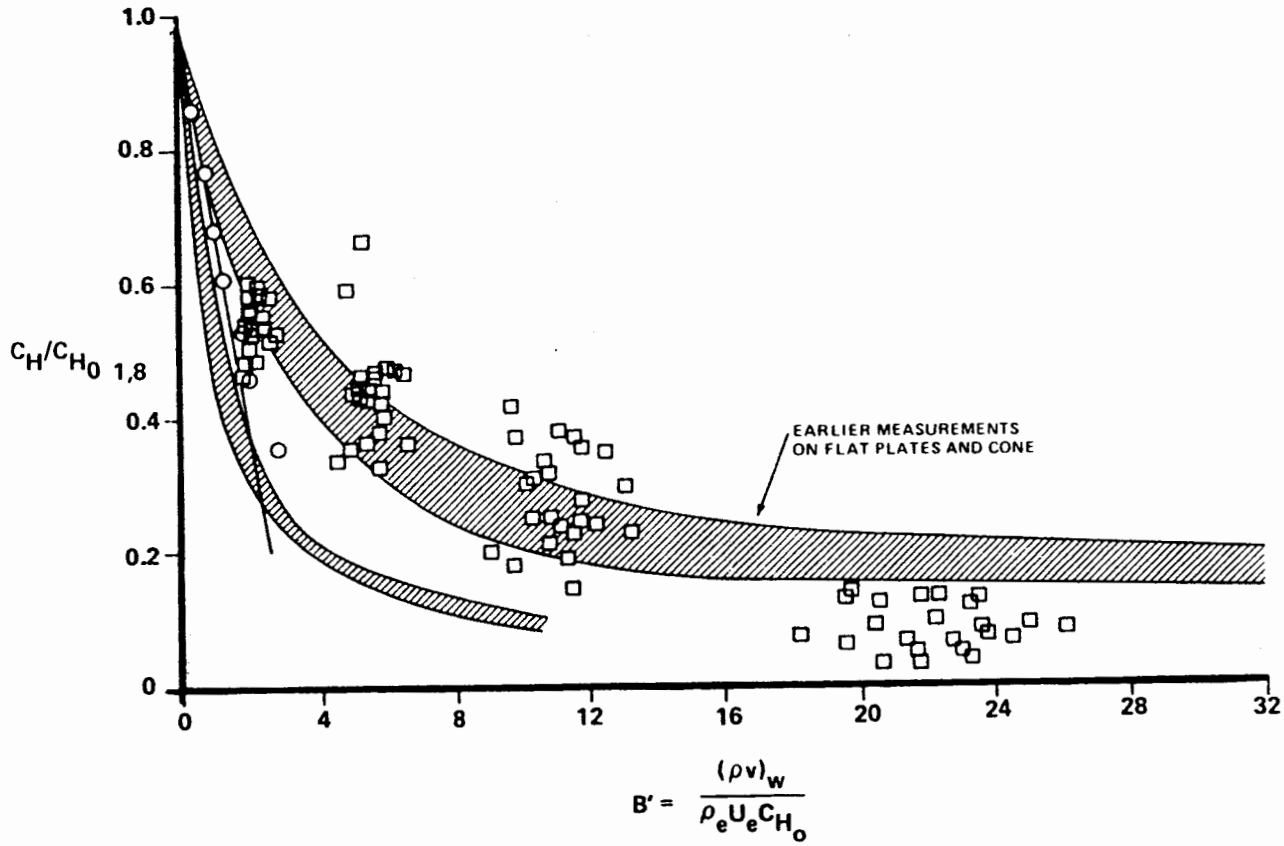


Figure 7 COMPARISON BETWEEN THE MEASUREMENT MADE ON THE TRANSPIRATION-COOLED NOSETIPS WITH NITROGEN INJECTANT AND THE EARLIER BLOCKAGE DATA

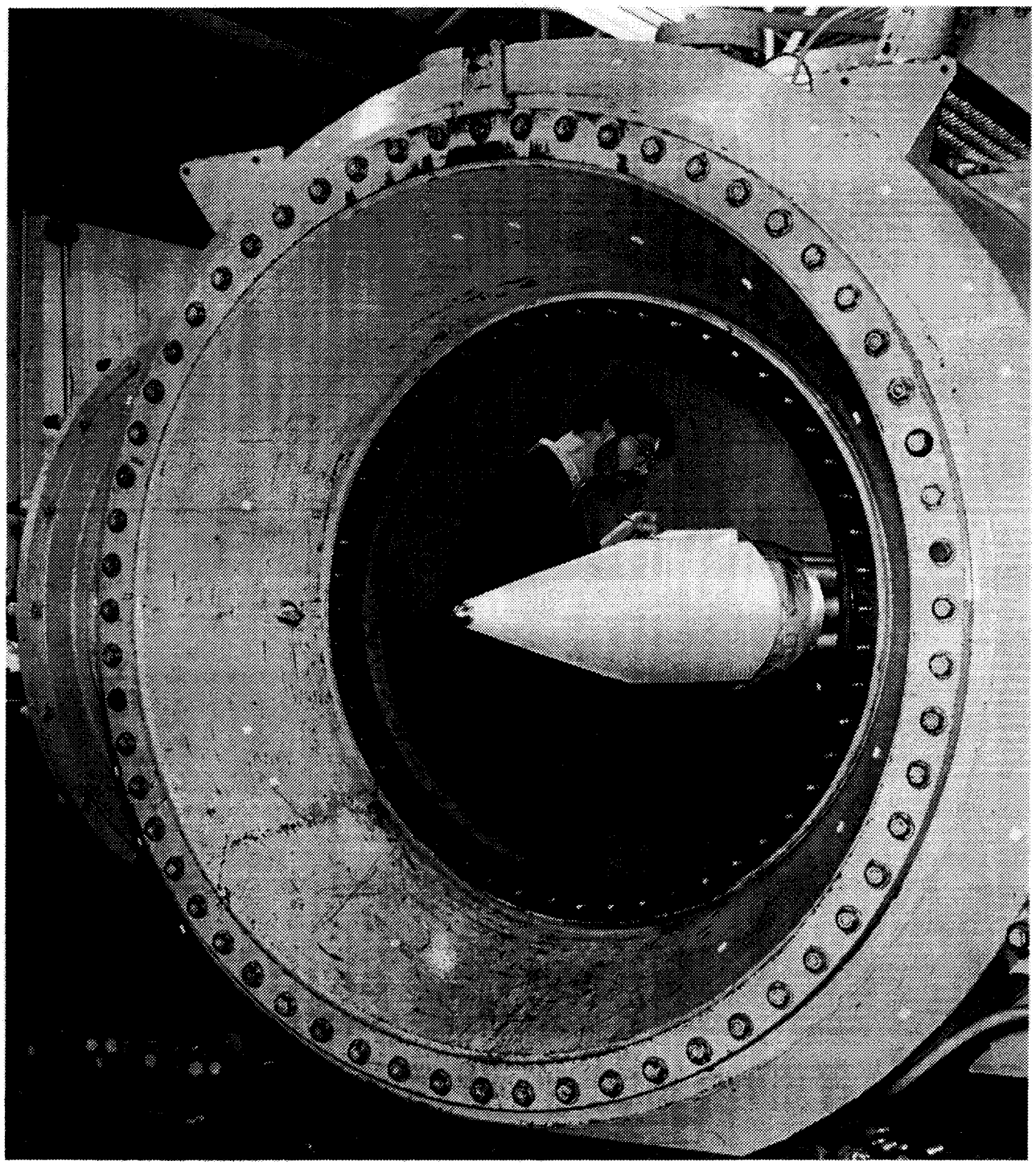


Figure 8 CONE MODEL

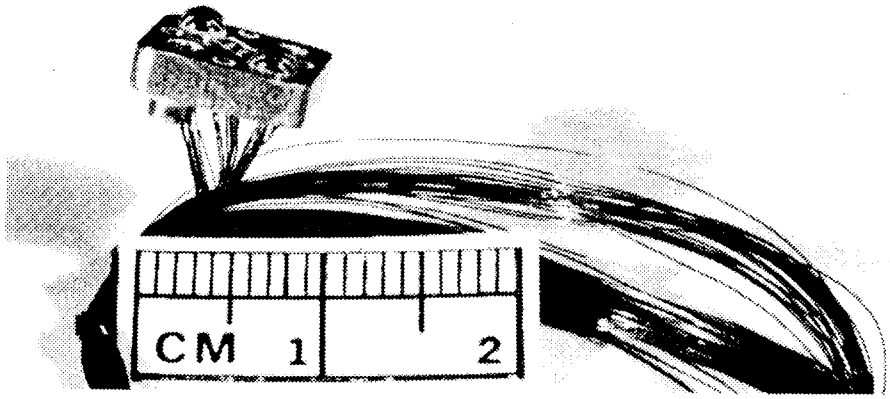


Figure 9 THIN FILM GAGES

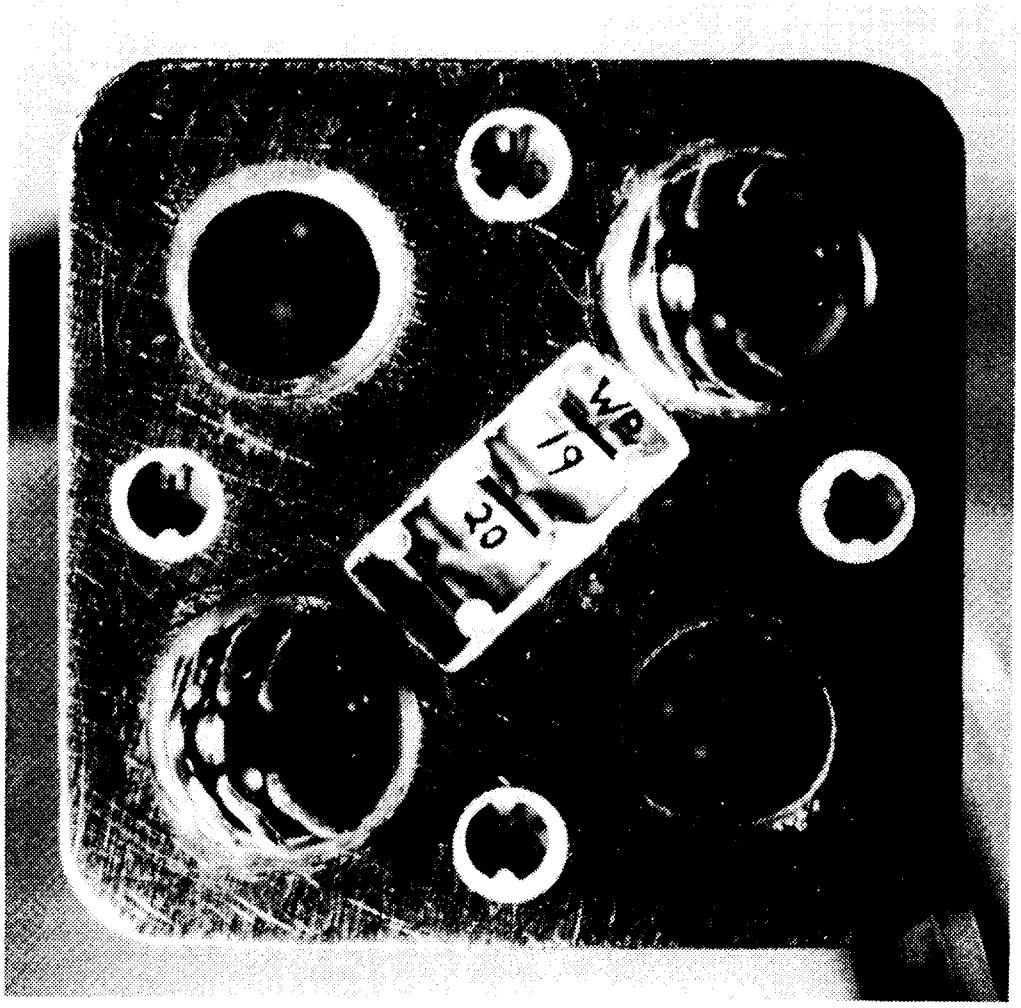


Figure 10 THIN FILM GAGE VIEWED FROM ABOVE

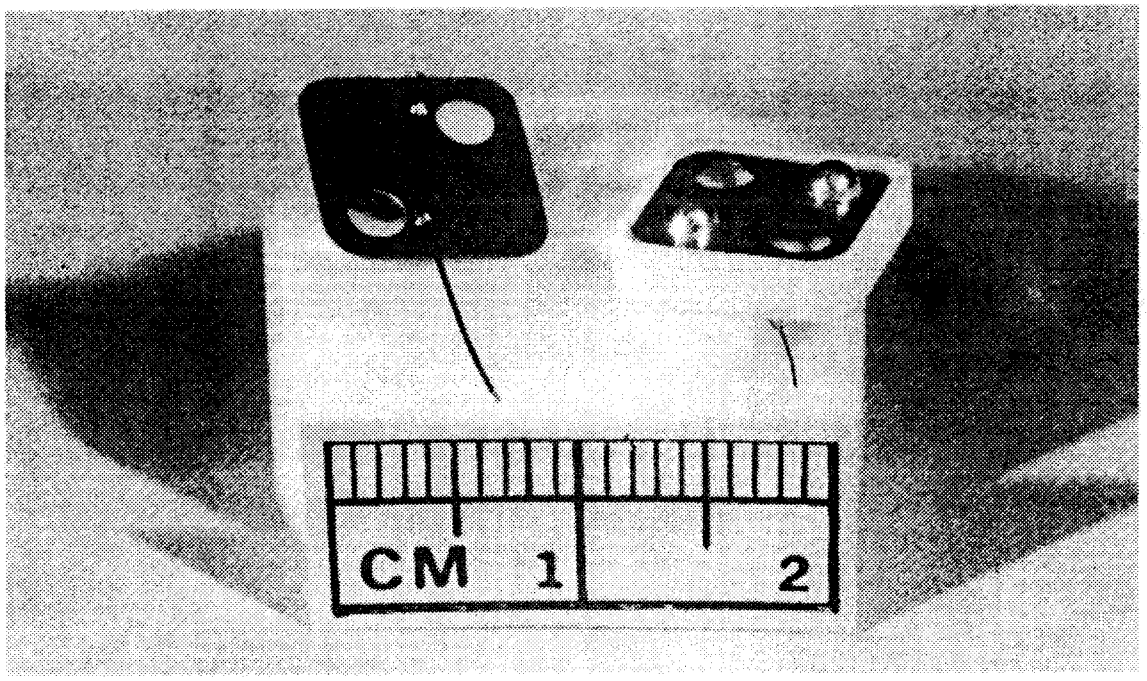


Figure 11 CALORIMETER GAGE

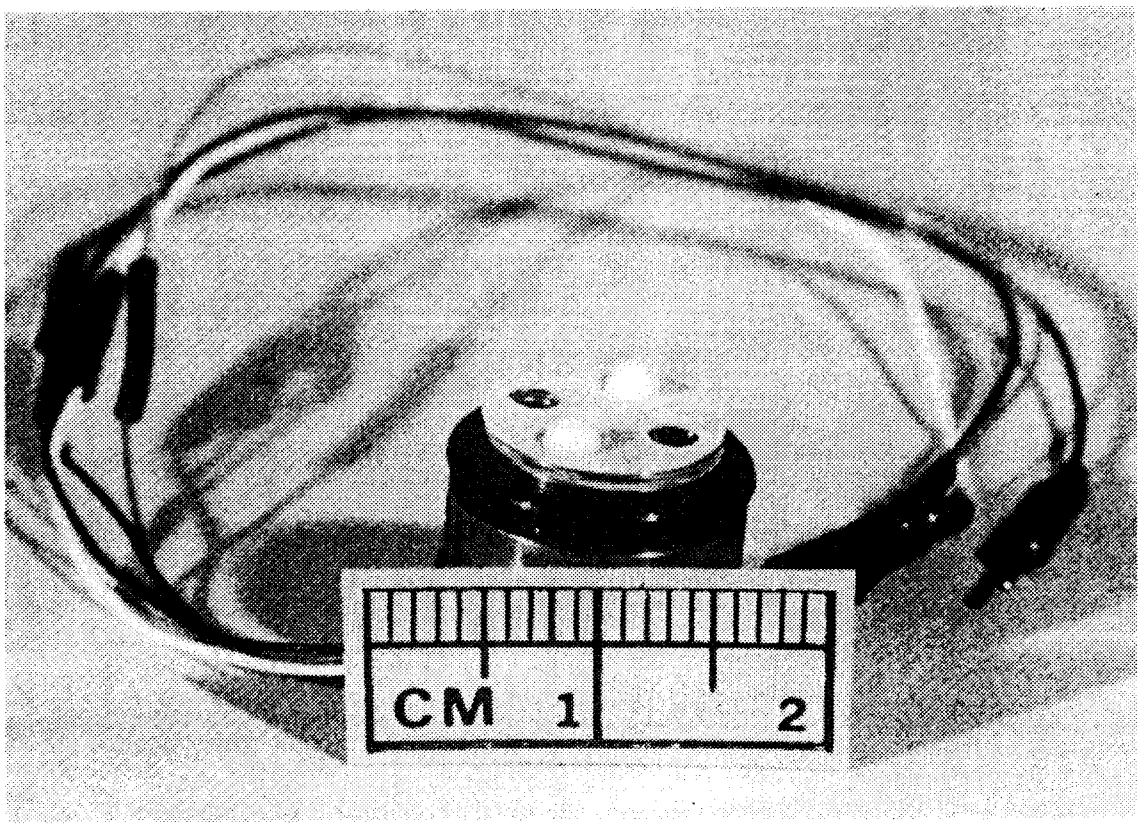
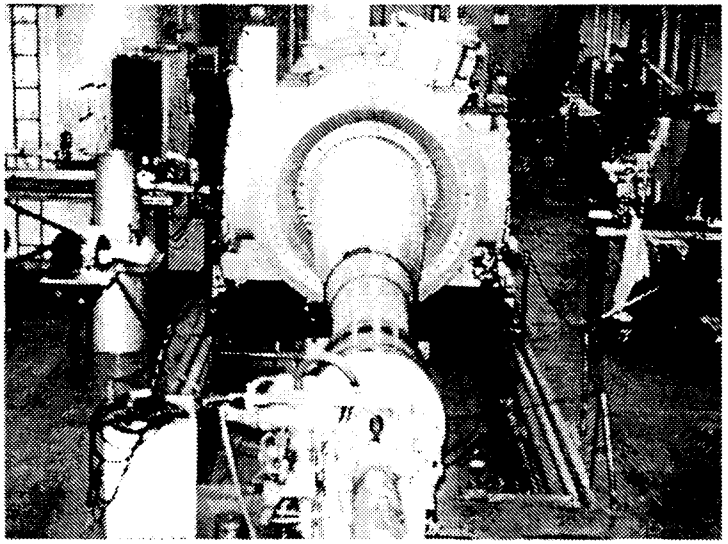
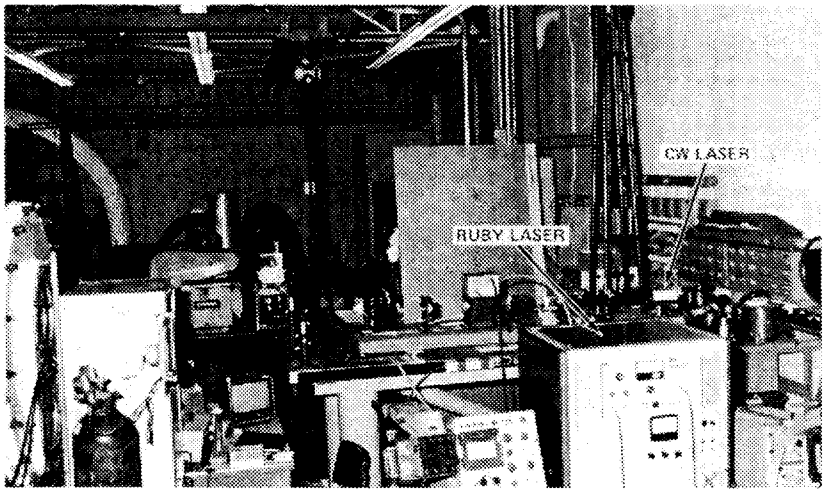


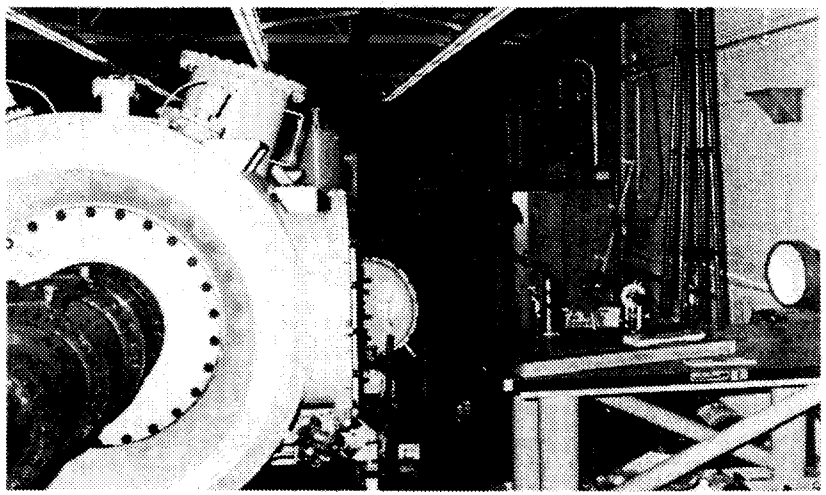
Figure 12 SKIN FRICTION GAGE



96- INCH HYPERSONIC SHOCK TUNNEL WITH LASER HOLOGRAPHIC SYSTEM INSTALLED



SENDING OPTICS



RECEIVING OPTICS

Figure 13 LASER HOLOGRAPHIC SYSTEM

Run#	Mach#	Reynolds# (10^6/Ft)	Blowing Rate	Injectant	Flap Angle (Deg)	A.O.A. (Deg)	Nose Type
2	13	5.0	0	Nitrogen	15	0	Sharp
3	13	5.0	0	Nitrogen	15	0	Sharp
4	13	5.0	.837	Nitrogen	15	0	Sharp
5	13	5.0	.875	Nitrogen	15	0	Sharp
6	13	5.0	1.650	Nitrogen	15	0	Sharp
7	13	5.0	4.043	Nitrogen	15	0	Sharp
9	13	5.0	.479	Nitrogen	25	0	Sharp
10	13	5.0	1.715	Nitrogen	25	0	Sharp
11	13	5.0	4.218	Nitrogen	25	0	Sharp
12	13	5.0	.862	Nitrogen	25	0	Sharp
13	13	5.0	.677	Nitrogen	25	0	Sharp
14	13	5.0	.106	Nitrogen	25	0	Sharp
15	13	5.0	2.765	Nitrogen	25	0	Sharp
16	13	5.0	.678	N2/He	25	0	Sharp
17	13	5.0	.255	N2/He	25	0	Sharp
18	13	5.0	1.586	N2/He	25	0	Sharp
20	13	5.0	.731	Nitrogen	30	0	Sharp
21	13	5.0	.341	Nitrogen	30	0	Sharp
22	13	5.0	5.382	Nitrogen	30	0	Sharp
23	13	5.0	2.015	Nitrogen	30	0	Blunt
24	13	5.0	.721	Nitrogen	30	0	Blunt
25	13	5.0	.352	Nitrogen	35	0	Sharp
26	13	5.0	.711	Nitrogen	35	0	Sharp
27	13	5.0	2.125	Nitrogen	35	0	Sharp
28	13	5.0	.561	N2/He	35	0	Sharp
29	11	4.0	.526	Nitrogen	25	0	Sharp
30	11	4.0	1.126	Nitrogen	25	0	Sharp
31	11	4.0	0	Nitrogen	25	0	Sharp
32	11	10.0	1.686	Nitrogen	25	0	Sharp
33	11	10.0	.678	Nitrogen	25	0	Sharp
34	11	10.0	.222	Nitrogen	25	0	Sharp
35	11	10.0	3.931	Nitrogen	25	0	Sharp
36	11	10.0	.374	N2/He	25	0	Sharp
37	11	10.0	.870	N2/He	25	0	Sharp
38	11	10.0	.202	N2/He	25	0	Sharp
39	11	10.0	1.329	N2/He	25	0	Sharp
40	15	1.5	.398	Nitrogen	25	0	Sharp
41	15	1.5	.560	Nitrogen	25	0	Sharp
42	15	1.5	1.329	Nitrogen	25	0	Sharp
43	13	7.0	.637	Nitrogen	25	3	Sharp
44	13	7.0	.305	Nitrogen	25	3	Sharp
45	13	7.0	2.102	Nitrogen	25	3	Sharp
46	13	6.0	.681	Nitrogen	25	3	Blunt
47	13	6.0	.652	Nitrogen	25	6	Blunt
48	13	6.0	.305	Nitrogen	25	6	Sharp
49	13	6.0	.679	Nitrogen	25	6	Sharp
50	13	6.0	2.216	Nitrogen	25	6	Sharp

Figure 14 TEST MATRIX FOR THE BLOWING AND ROUGHNESS STUDIES

Run	Po/10^4 PSIA	Ho/10^7 (FT/SEC)^2	To °R	M∞	U∞ FT/SEC	T∞ °R	P∞ PSIA	Rho∞ SLUGS/FT3
3	1.7860	2.1181	3063.0	12.968	6417.3	101.820	.076982	6.3447(-5)
4	1.8555	2.0879	3027.2	13.026	6372.1	99.500	.078939	6.6579(-5)
5	1.7671	2.0745	3011.5	13.003	6351.4	99.205	.075545	6.3905(-5)
6	1.8268	2.1142	3061.5	13.019	6412.0	100.870	.077164	6.4199(-5)
7	1.8100	2.0850	3025.8	13.028	6367.8	99.339	.076571	6.4686(-5)
9	1.7010	2.0126	2939.2	13.052	6256.5	95.549	.071542	6.2835(-5)
10	1.7815	2.1206	3073.3	13.031	6421.9	100.990	.074230	6.1681(-5)
11	1.8303	2.2023	3174.1	13.014	6544.2	105.150	.075595	6.0334(-5)
12	1.8320	2.1121	3059.7	13.029	6409.0	100.620	.077089	6.4291(-5)
13	1.8383	2.1364	3091.3	13.034	6445.9	101.710	.076666	6.3259(-5)
14	1.8366	2.1037	3048.2	13.025	6396.1	100.280	.077665	6.4996(-5)
15	1.8235	2.1237	3075.0	13.028	6426.6	101.180	.076408	6.3372(-5)
16	1.7693	2.0641	2997.2	12.997	6335.2	98.793	.076130	6.4669(-5)
17	1.8415	2.1037	3048.2	13.027	6396.1	100.250	.077837	6.5160(-5)
18	1.8528	2.1195	3069.3	13.037	6420.3	100.840	.077633	6.4605(-5)
20	1.7933	2.1111	3059.9	13.025	6407.5	100.630	.075254	6.2760(-5)
21	1.7983	2.1333	3088.5	13.028	6441.0	101.630	.074909	6.1853(-5)
22	1.7878	2.0882	3030.6	13.026	6372.6	99.531	.075458	6.3623(-5)
23	1.8753	2.1386	3092.2	13.032	6449.0	101.830	.078565	6.4745(-5)
24	1.8060	2.0894	3033.7	13.047	6374.8	99.273	.075531	6.3850(-5)
25	1.6813	2.0606	2996.3	12.994	6329.9	98.672	.071683	6.0966(-5)
26	1.8528	2.1216	3073.2	13.048	6423.7	100.790	.077172	6.4254(-5)
27	1.7658	2.0785	3023.0	13.062	6358.3	98.528	.073123	6.2282(-5)
28	1.7993	2.0640	3002.2	13.058	6335.9	97.905	.075321	6.4562(-5)
29	.72940	1.7309	2598.2	11.006	5768.8	114.250	.092331	6.7820(-5)
30	.74835	1.7749	2655.0	10.998	5841.5	117.320	.094527	6.7618(-5)
31	.75930	1.7521	2625.9	11.012	5804.1	115.520	.095770	6.9572(-5)
32	1.7363	1.6998	2575.8	11.347	5723.5	105.800	.207250	1.6439(-4)
33	1.7585	1.7248	2609.1	11.354	5765.6	107.230	.208200	1.6294(-4)
34	1.7423	1.7265	2612.3	11.356	5768.4	107.290	.205350	1.6062(-4)
35	1.7263	1.7357	2624.8	11.353	5783.7	107.930	.202780	1.5767(-4)
36	1.7350	1.7200	2602.3	11.343	5757.4	107.130	.206210	1.6154(-4)
37	1.7435	1.7341	2621.6	11.351	5781.0	107.870	.205710	1.6004(-4)
38	1.6933	1.7223	2607.9	11.348	5761.3	107.180	.199160	1.5593(-4)
39	1.7123	1.6855	2557.2	11.343	5699.2	104.980	.204900	1.6380(-4)
40	1.7995	4.0076	5484.8	15.867	8869.7	129.930	.012272	7.9261(-6)
41	1.8348	2.9248	4028.1	15.414	7572.9	100.370	.019757	1.6519(-5)
42	1.8133	2.9680	4080.7	15.396	7628.5	102.090	.019439	1.5979(-5)
43	2.0123	1.7570	2646.7	13.319	5849.1	80.192	.083197	8.7065(-5)
44	2.0323	1.7925	2693.2	13.326	5908.0	81.733	.082990	8.5211(-5)
45	2.0693	1.8393	2754.4	13.338	5984.8	83.714	.083076	8.3280(-5)
46	1.9748	1.8215	2733.0	13.329	5955.6	83.012	.079161	8.0027(-5)
47	2.0848	1.8666	2788.7	13.326	6028.8	85.111	.083643	8.2473(-5)
48	2.0440	1.8476	2766.1	13.341	5998.3	84.058	.081471	8.1337(-5)
49	2.0260	1.8653	2789.9	13.342	6027.0	84.854	.080084	7.9203(-5)
50	2.0383	1.8941	2827.2	13.343	6073.2	86.142	.079889	7.7829(-5)

Figure 15 EXPERIMENTAL TEST CONDITIONS

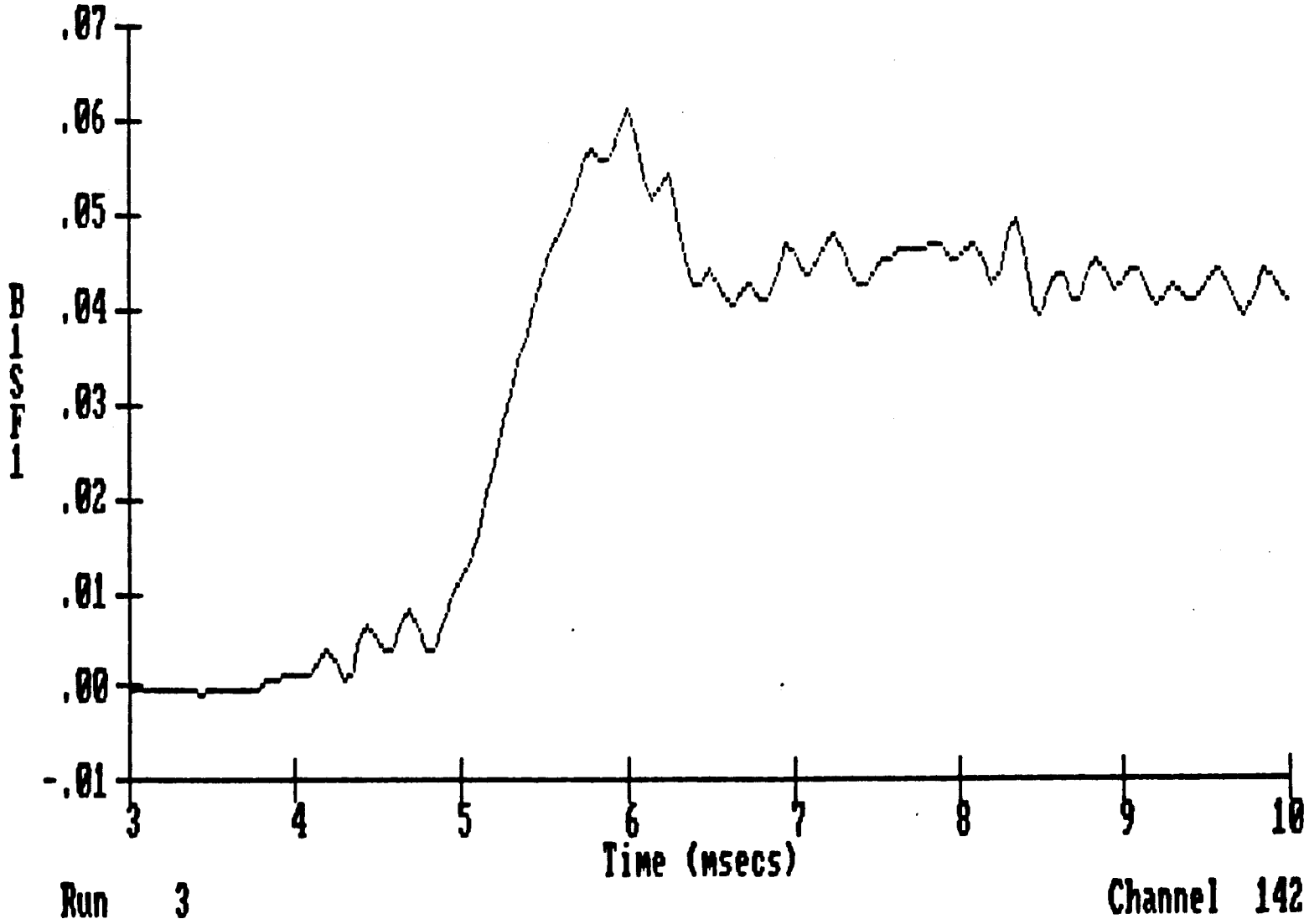


Figure 16 TIME HISTORY TRACE FROM SKIN FRICTION INSTRUMENTATION

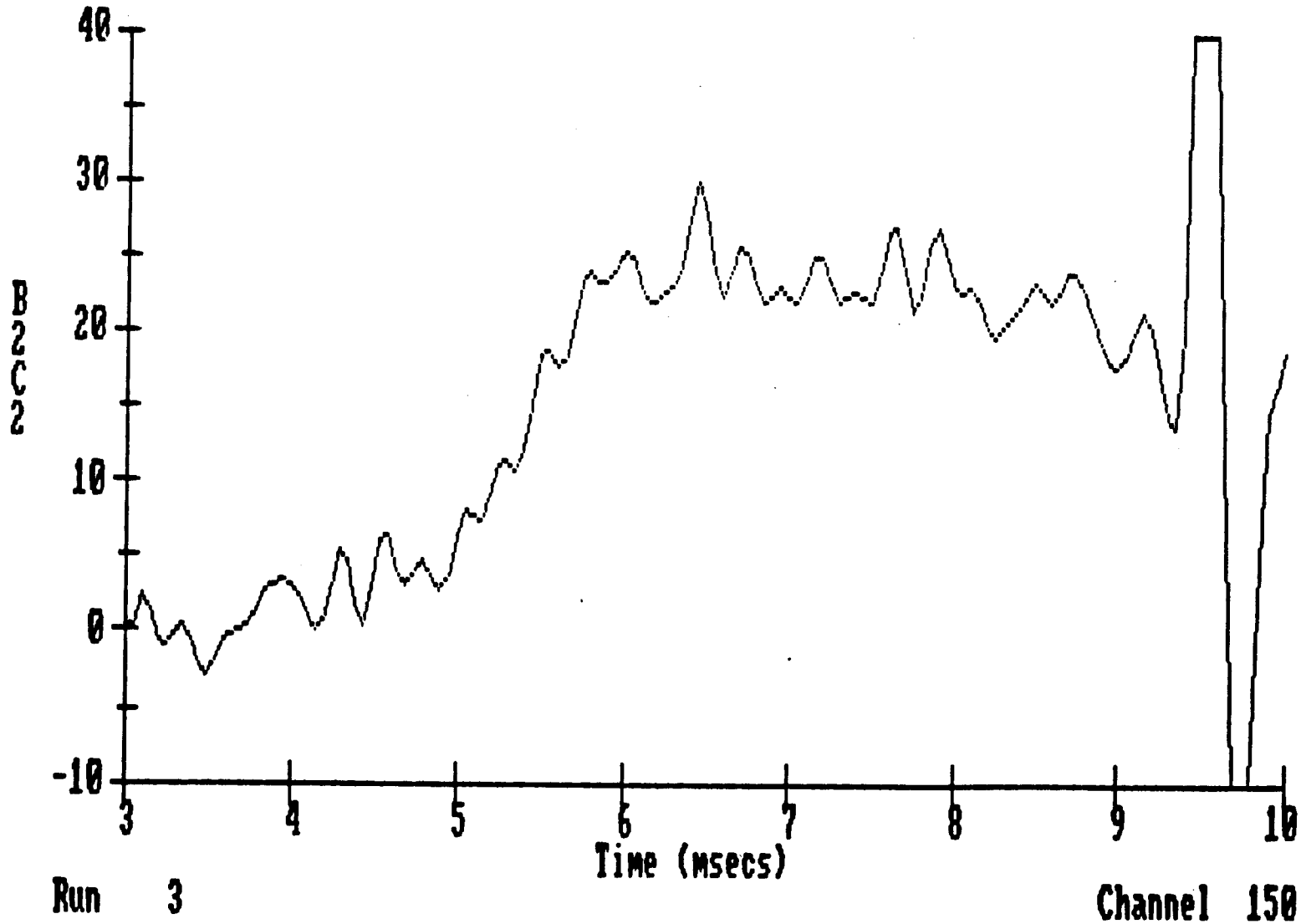


Figure 17 HEAT TRANSFER TIME HISTORY FROM CALORIMETER INSTRUMENTATION

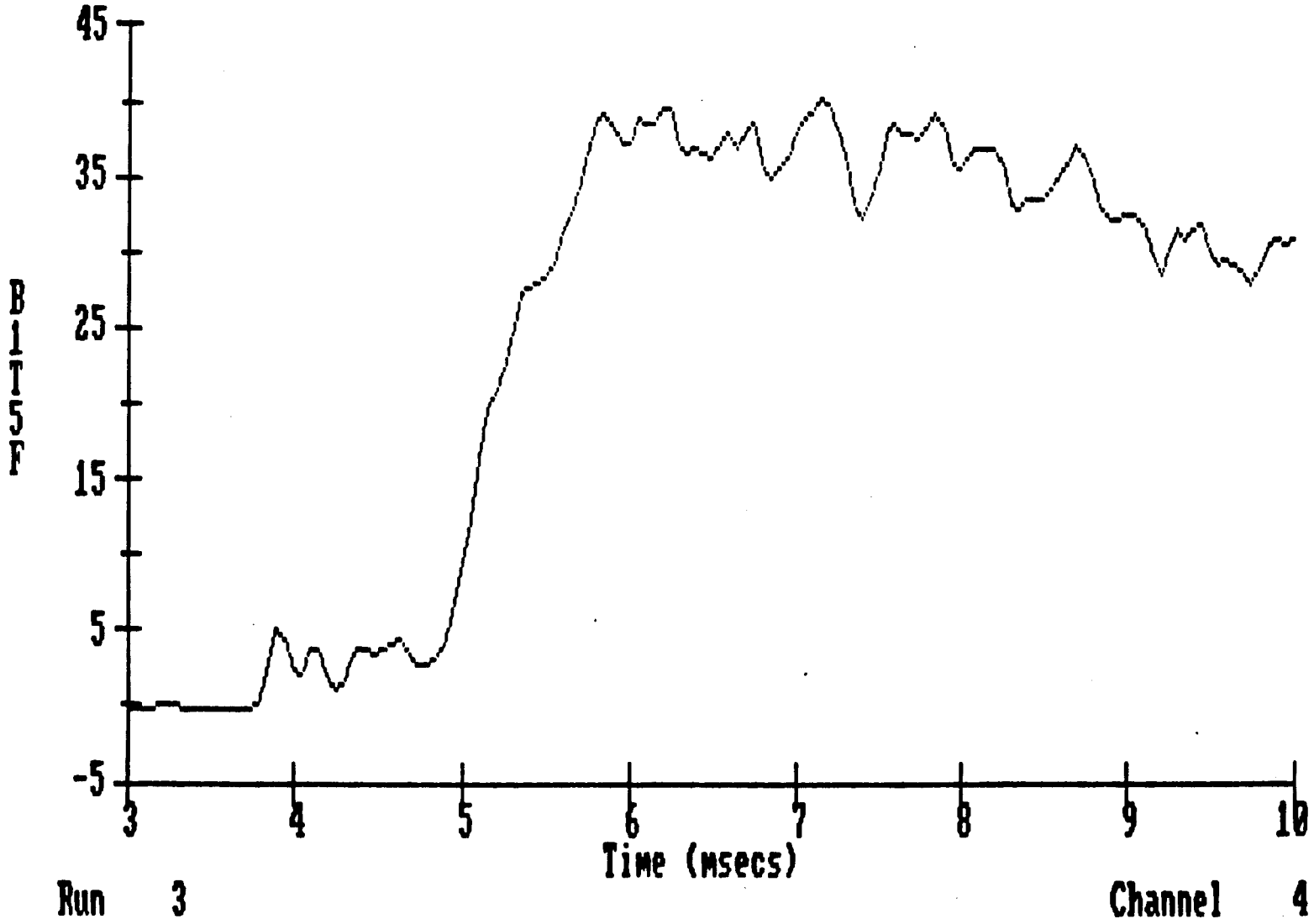


Figure 18 HEAT TRANSFER TIME HISTORY FROM THIN FILM INSTRUMENTATION

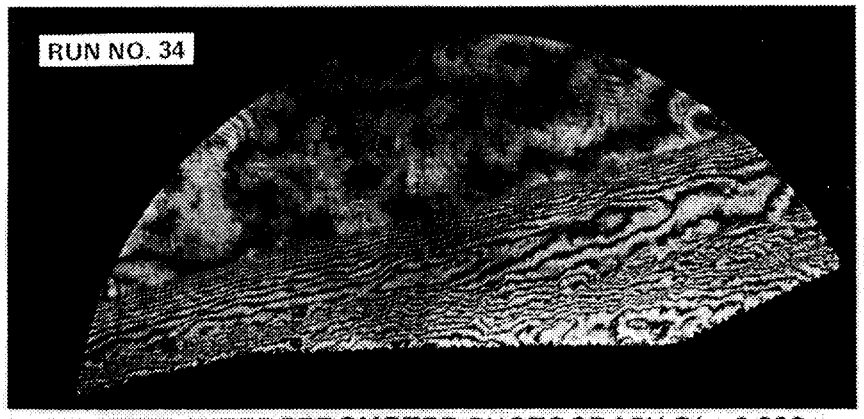


Figure 19 INTERFEROMETER PHOTOGRAPH B′ = 0.222

Figure 20 INTERFEROMETER PHOTOGRAPH B′ = 0.0

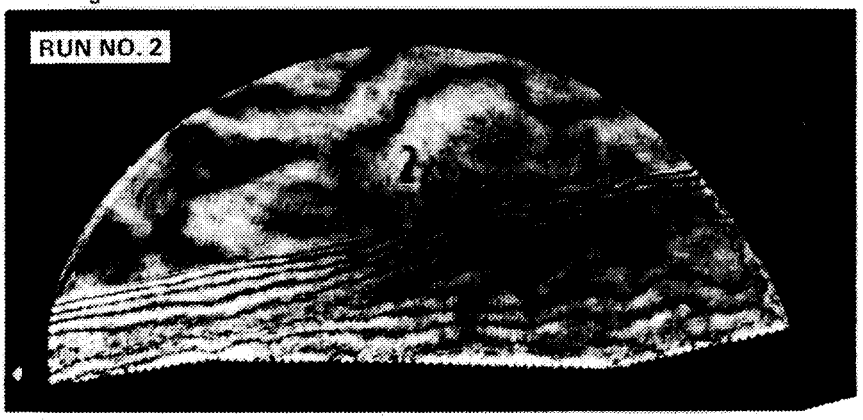


Figure 21 INTERFEROMETER PHOTOGRAPH B′ = 0.0

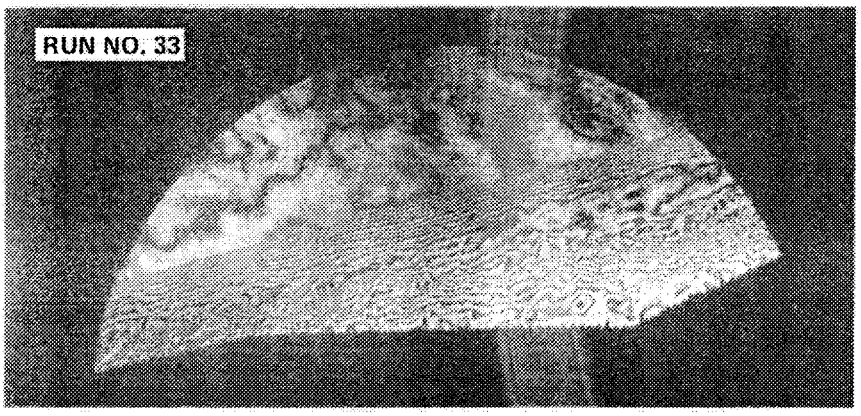


Figure 22 INTERFEROMETER PHOTOGRAPH B′ = 0.678

Figure 23 INTERFEROMETER PHOTOGRAPH B′ = 1.686

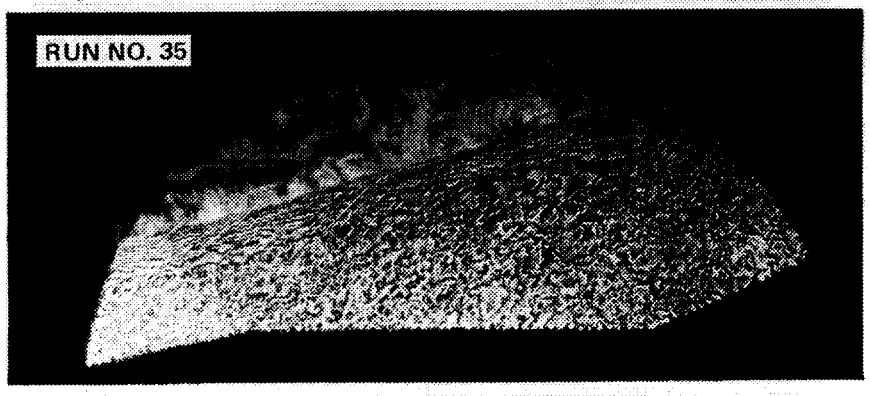


Figure 24 INTERFEROMETER PHOTOGRAPH B′ = 3.931

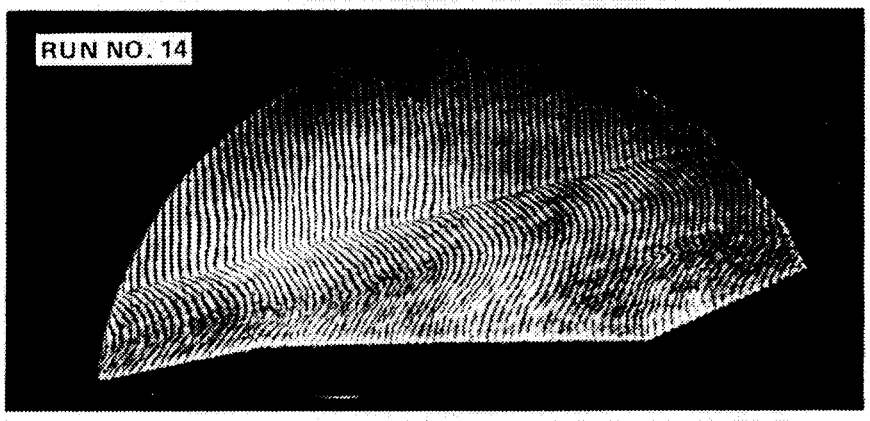


Figure 25 INTERFEROMETER PHOTOGRAPH B′ = 0.106

Figure 26 INTERFEROMETER PHOTOGRAPH B′ = 0.526

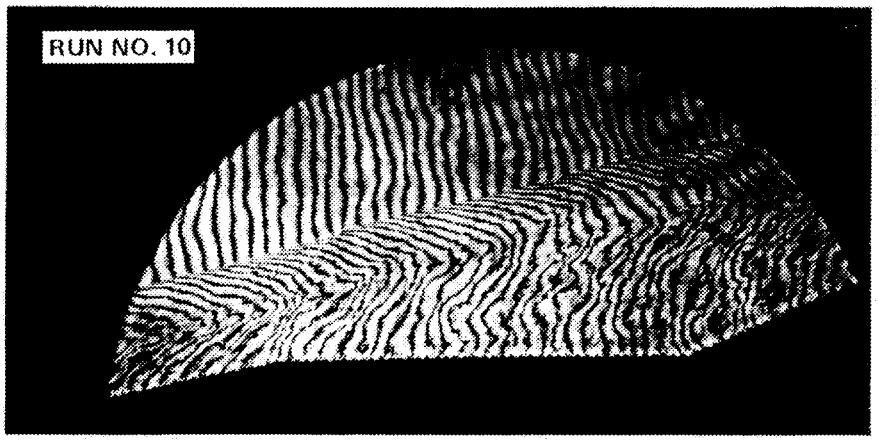


Figure 27 INTERFEROMETER PHOTOGRAPH B′ = 1.715

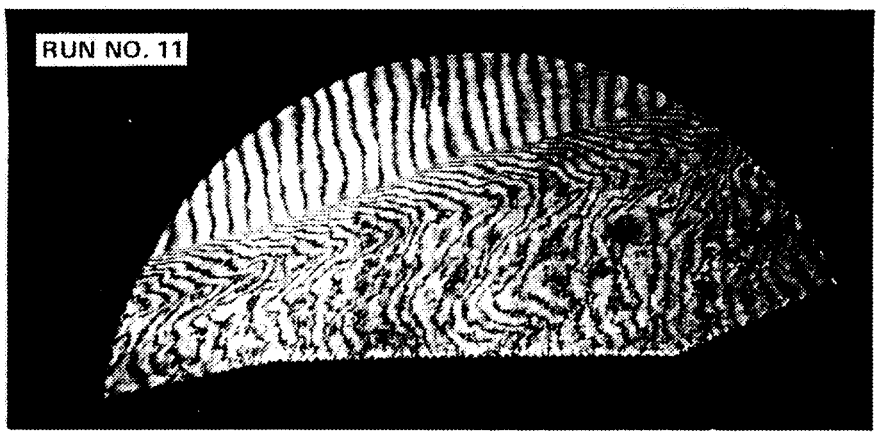


Figure 28 INTERFEROMETER PHOTOGRAPH B′ = 4.218

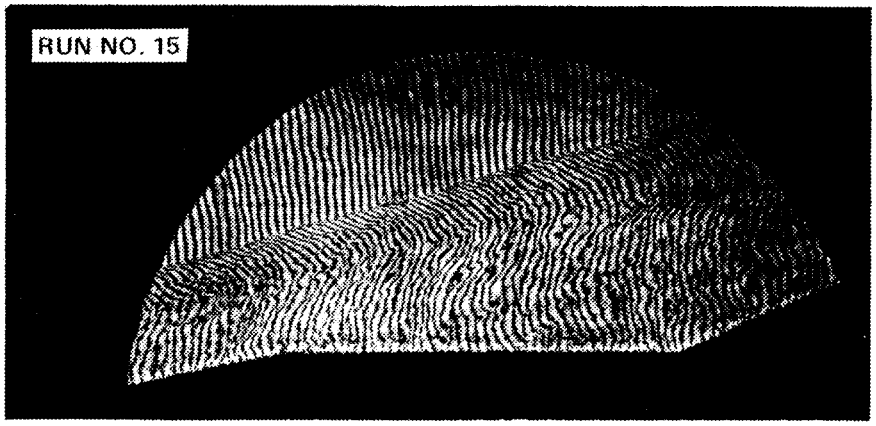


Figure 29 INTERFEROMETER PHOTOGRAPH B′ = 2.765

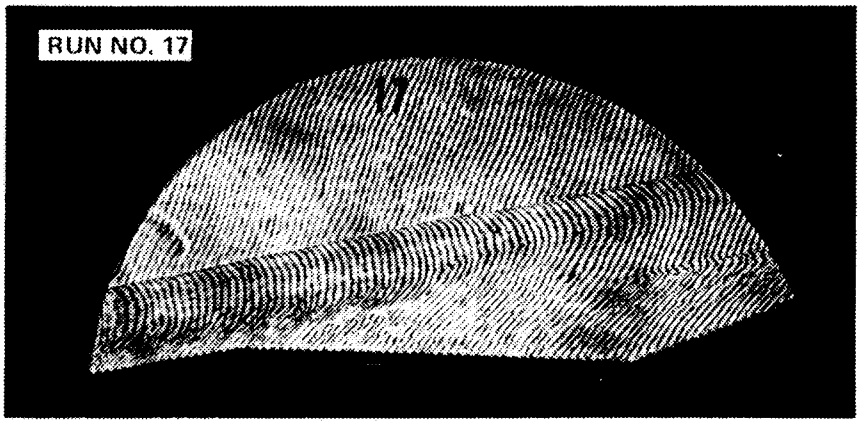


Figure 30 INTERFEROMETER PHOTOGRAPH B′ = 2.549

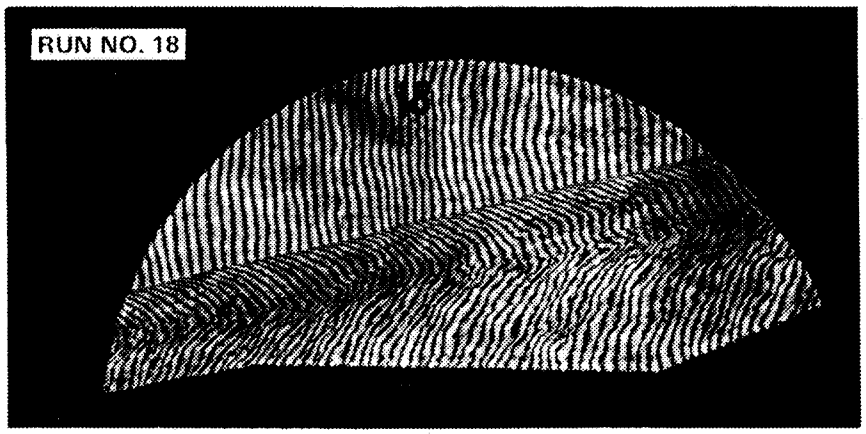


Figure 31 INTERFEROMETER PHOTOGRAPH B′ = 1.586

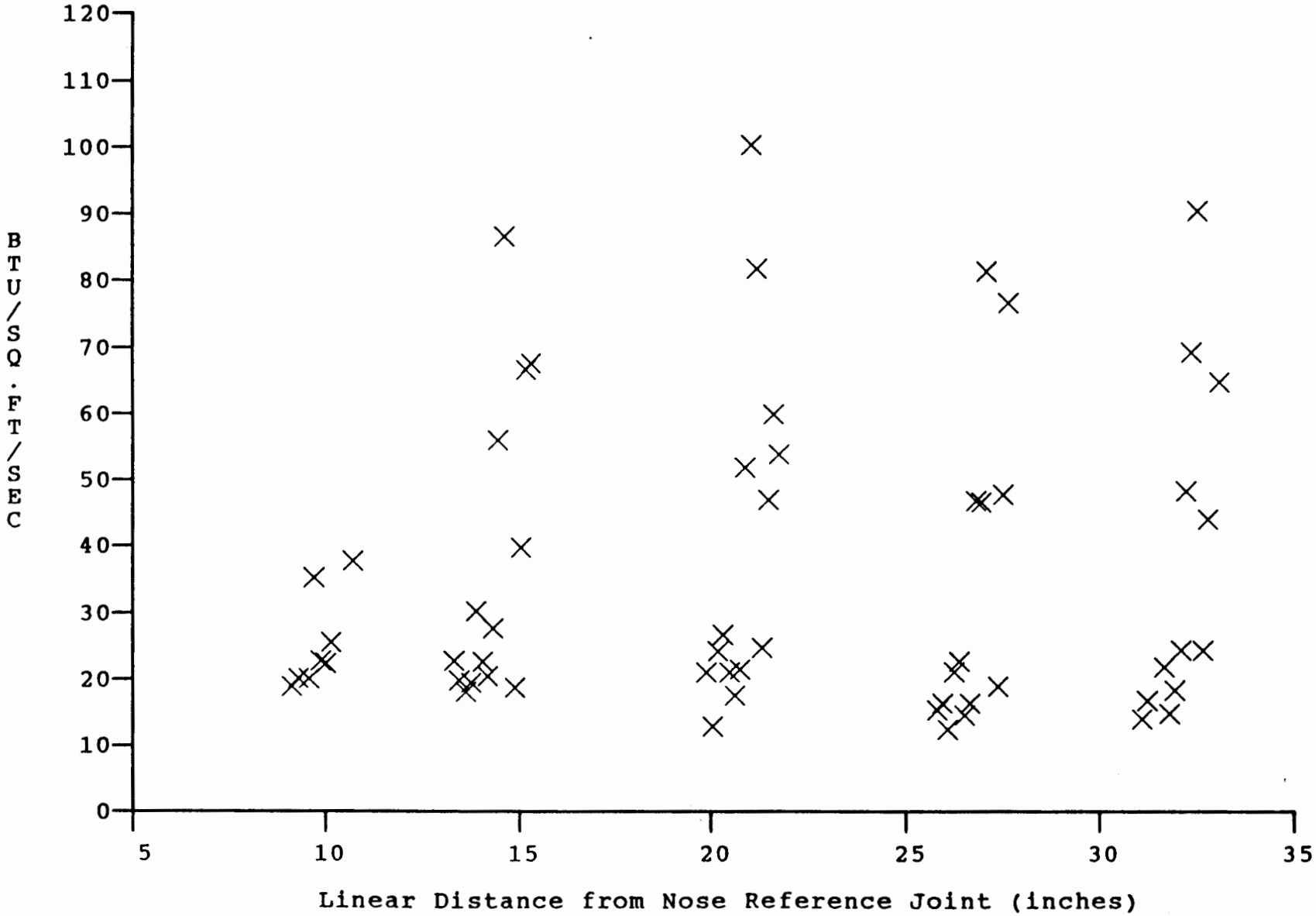


Figure 32a DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 34, B'=0.231, M=11, Re/FT=1.0X10^7)

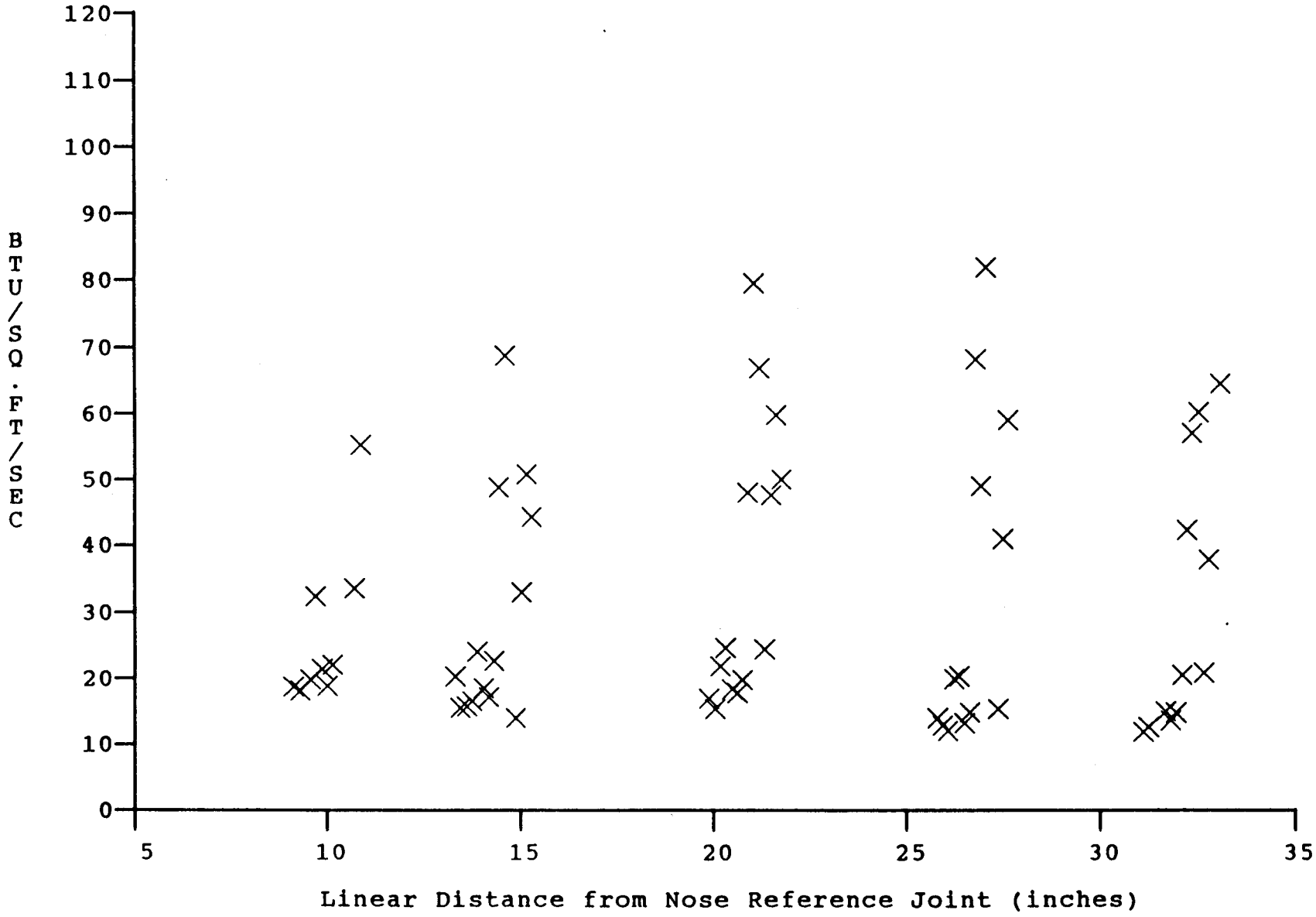


Figure 32b DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 33, B'=0.705, M=11, Re/FT=1.0X10^7)

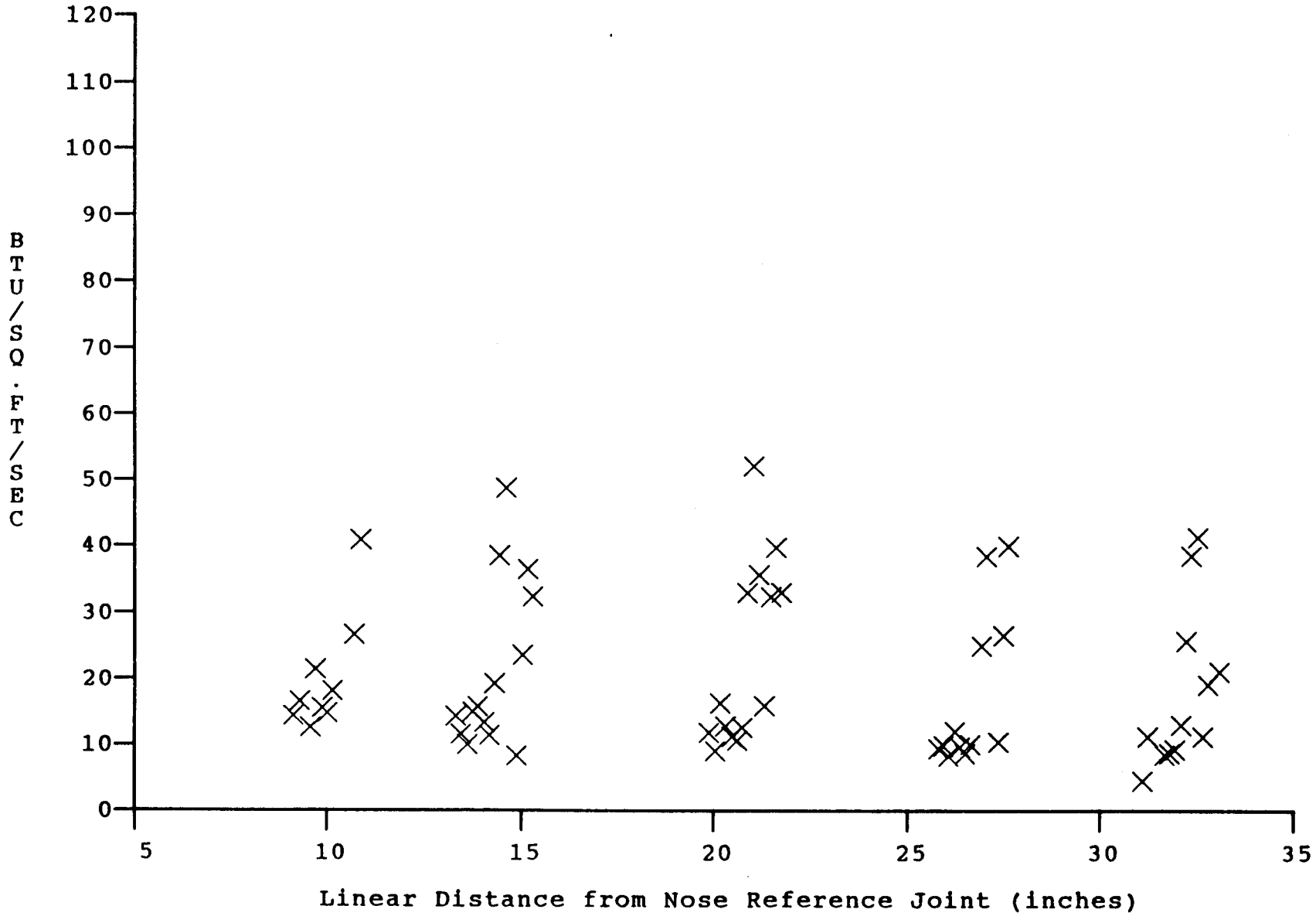


Figure 32c DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 32, B'=1.751, M=11, Re/FT=1.0X10^7)

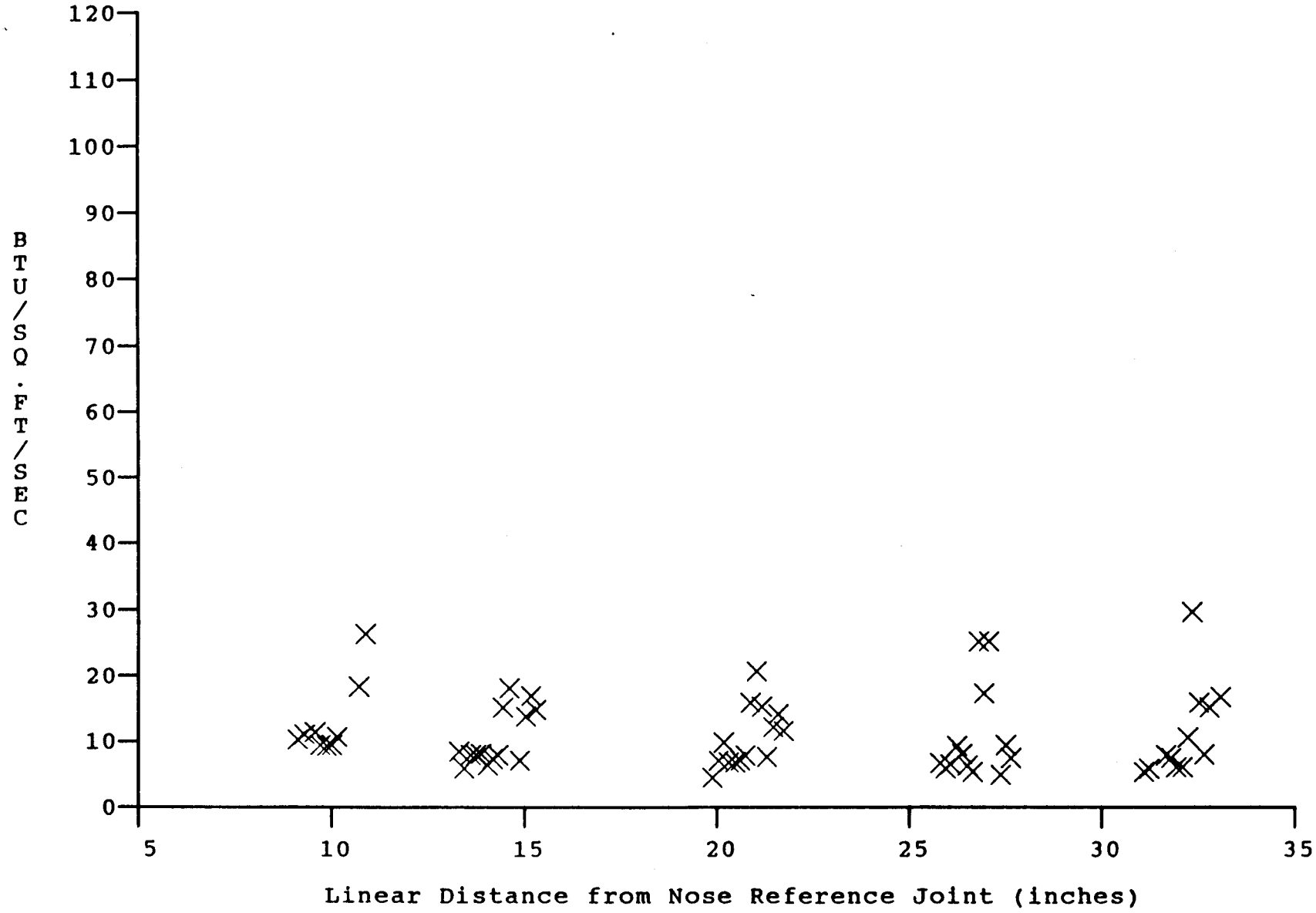


Figure 32d DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 35, B'=4.083, M=11, Re/FT=1.0X10^7)

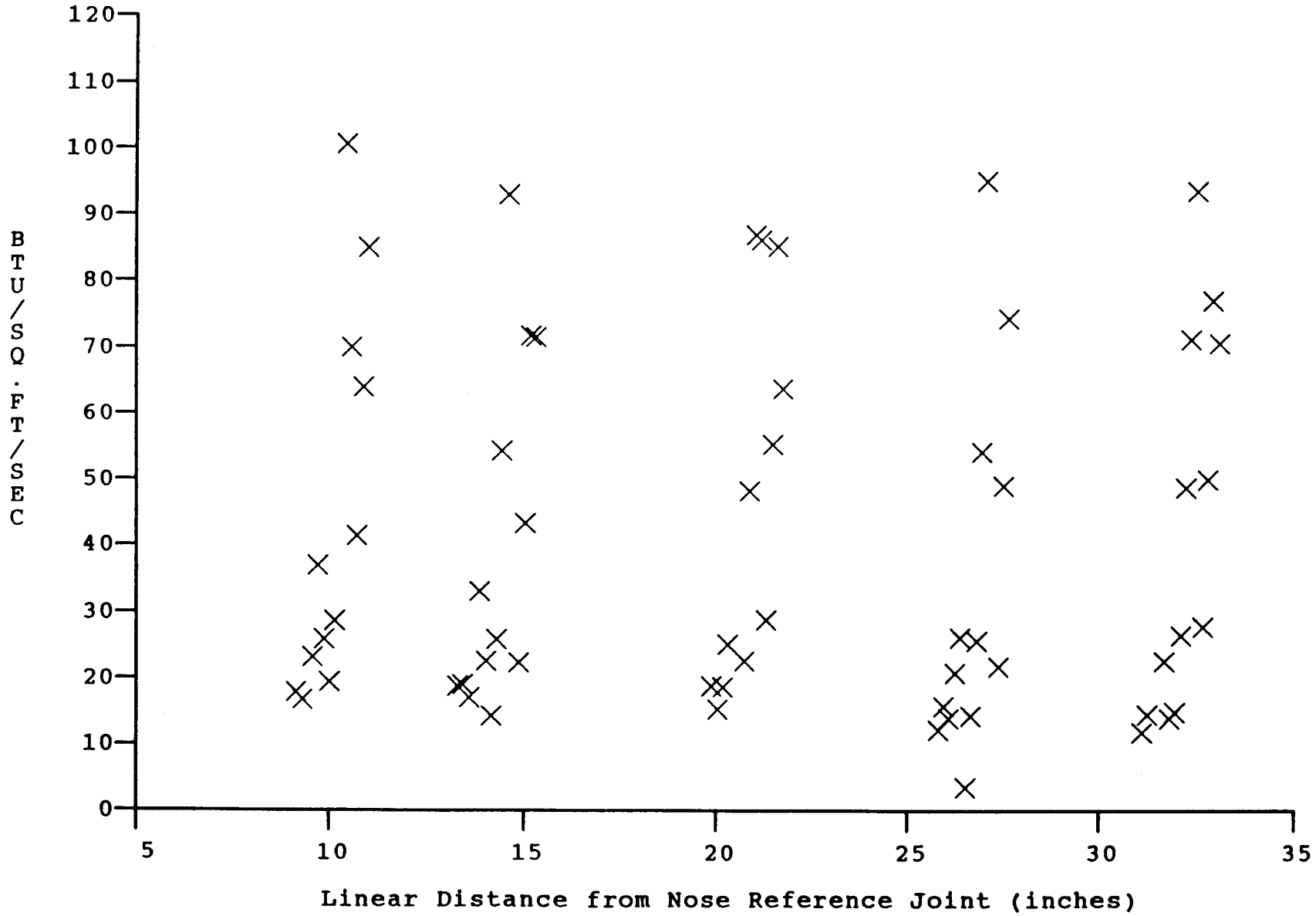


Figure 33a DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 3, B'=0.000, M=13, Re/FT=5.0X10^6)

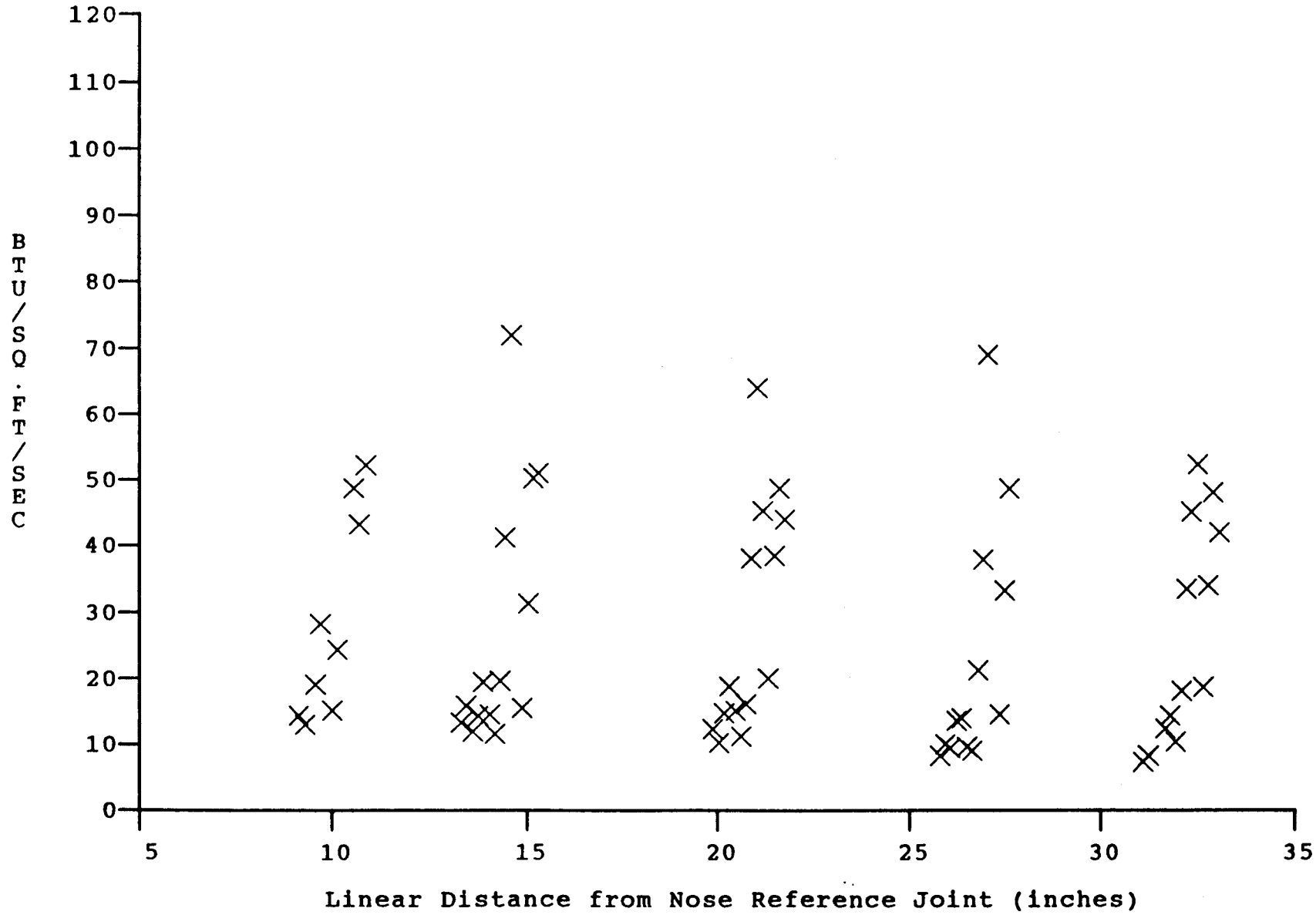


Figure 33b DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 9, B'=0.366, M=13, Re/FT=5.0X10^6)

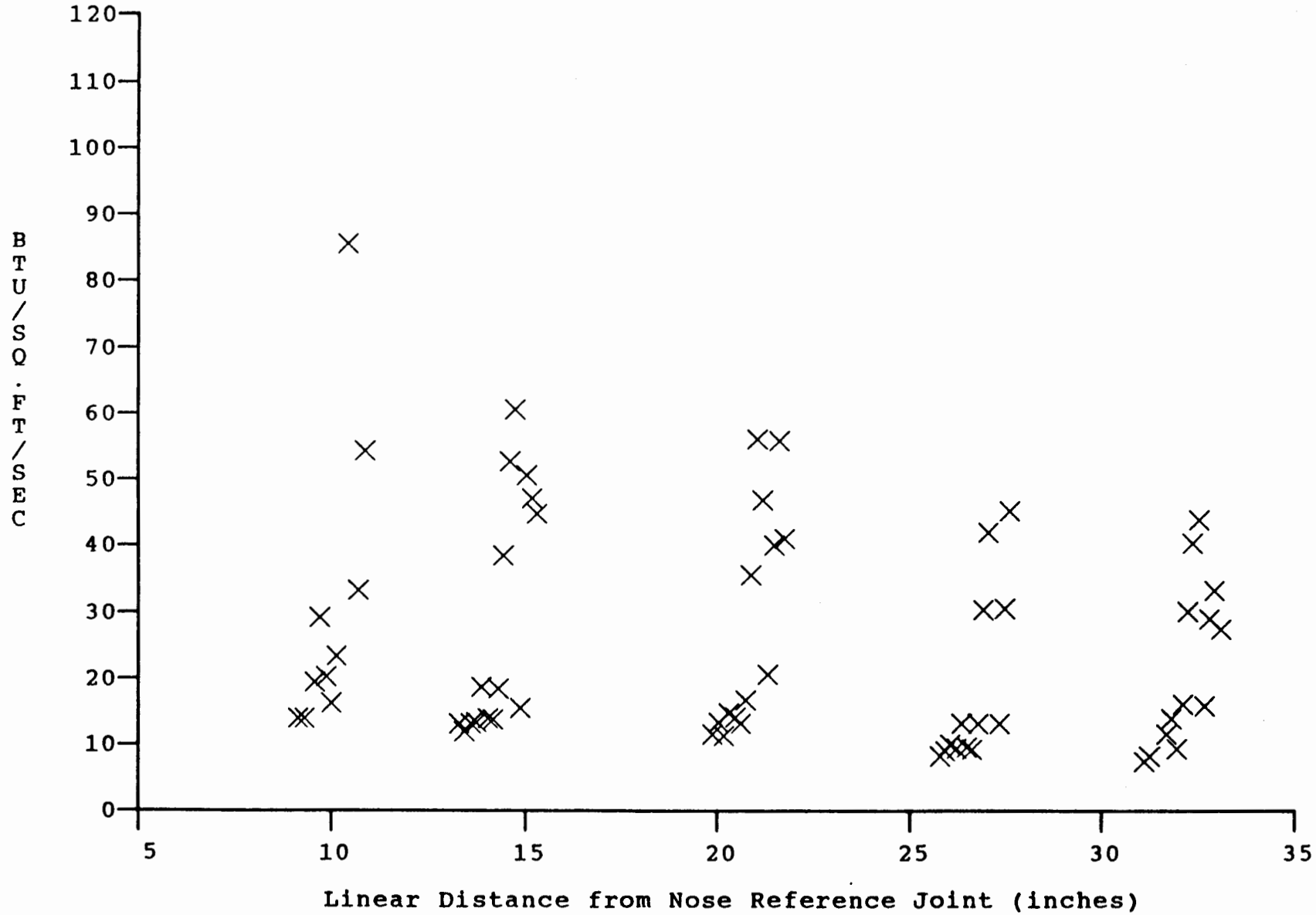


Figure 33c DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 4, B'=0.639, M=13, Re/FT=5.0X10^6)

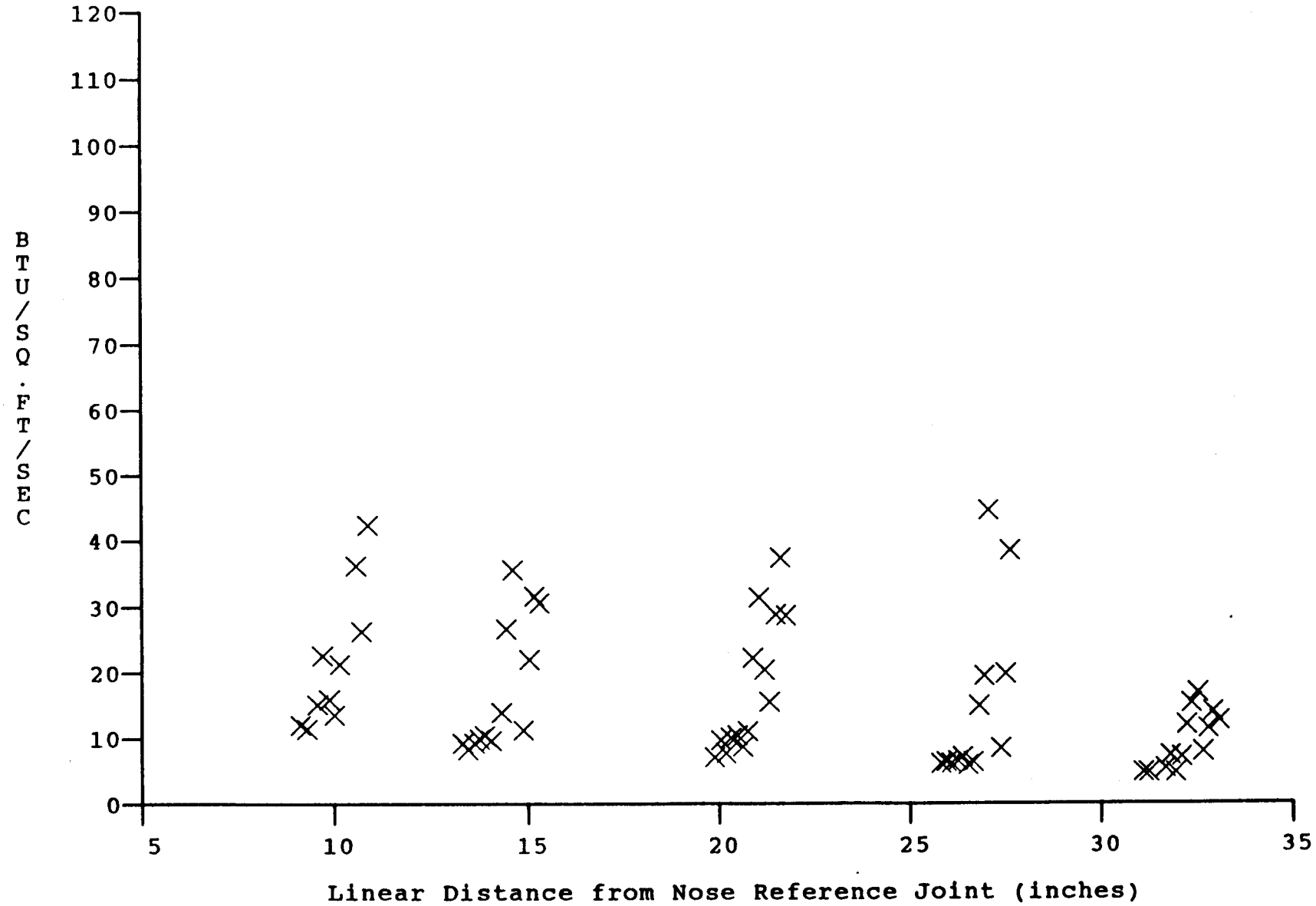


Figure 33d DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 6, B'=1.259, M=13, Re/FT=5.0X10^6)

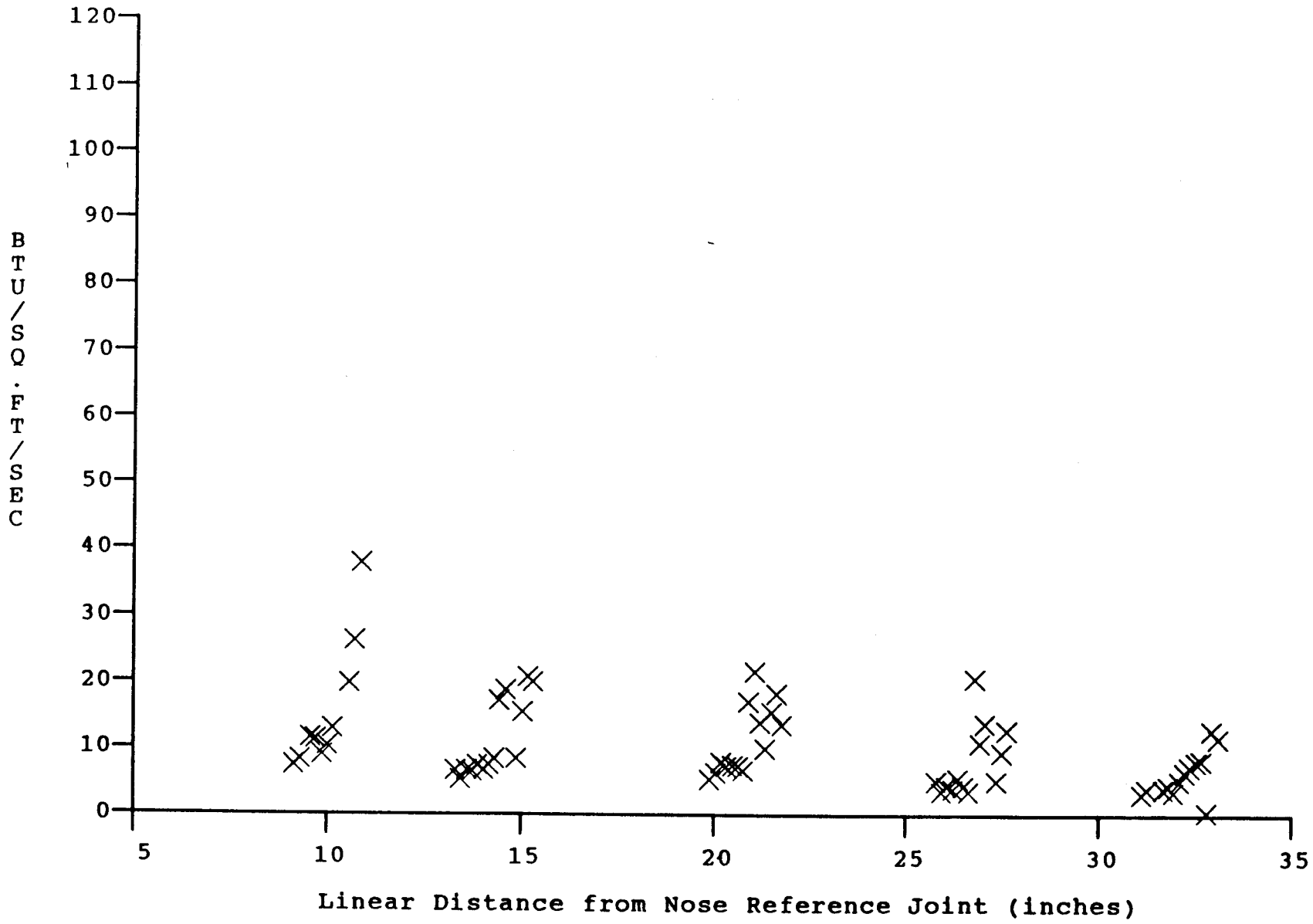


Figure 33e DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 15, B'=2.111, M=13, Re/FT=5.0X10^6)

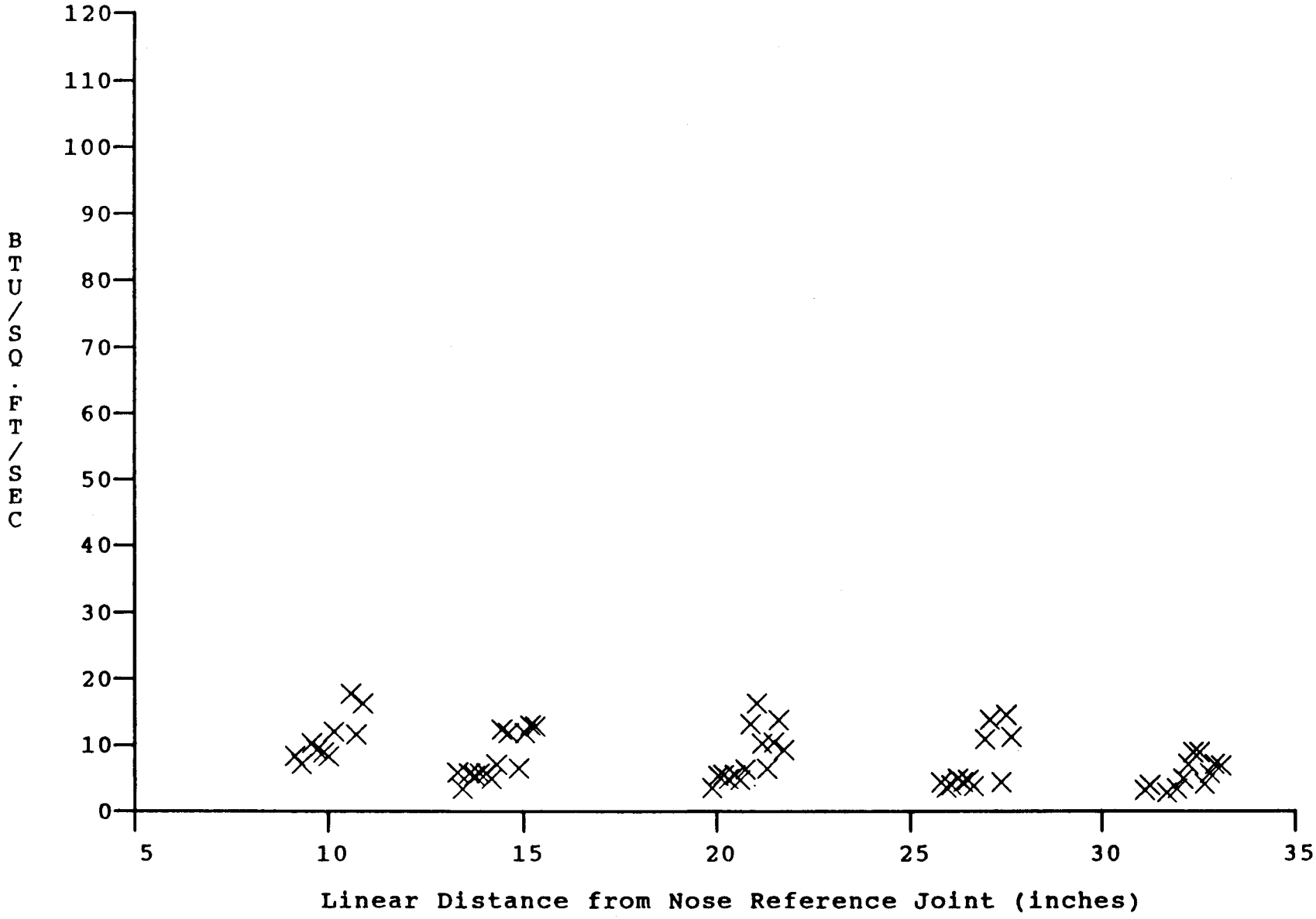


Figure 33f DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 7, B'=3.086, M=13, Re/FT=5.0X10^6)

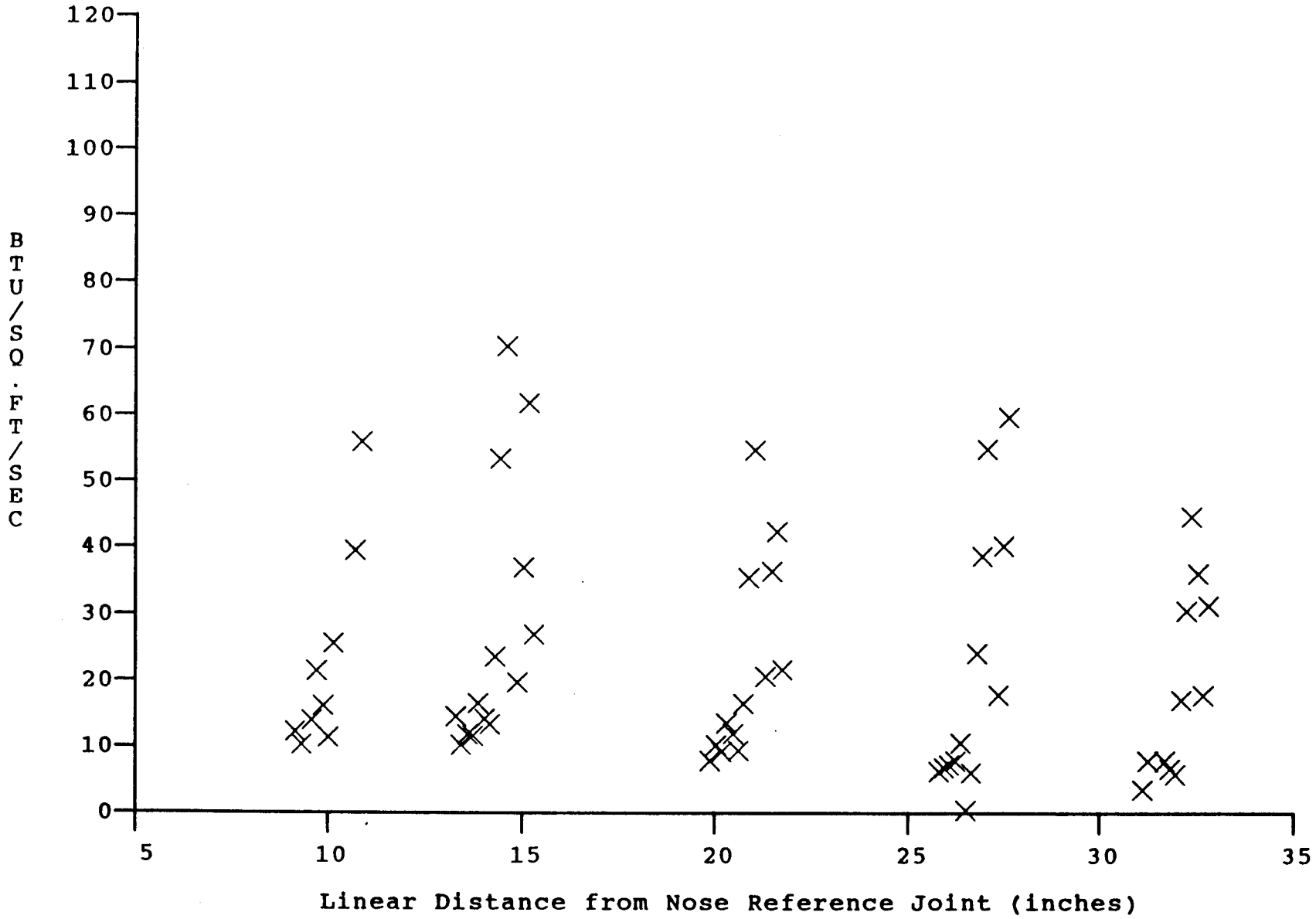


Figure 34a DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 40, B'=0.398, M=15, Re/FT=1.5X10^6)

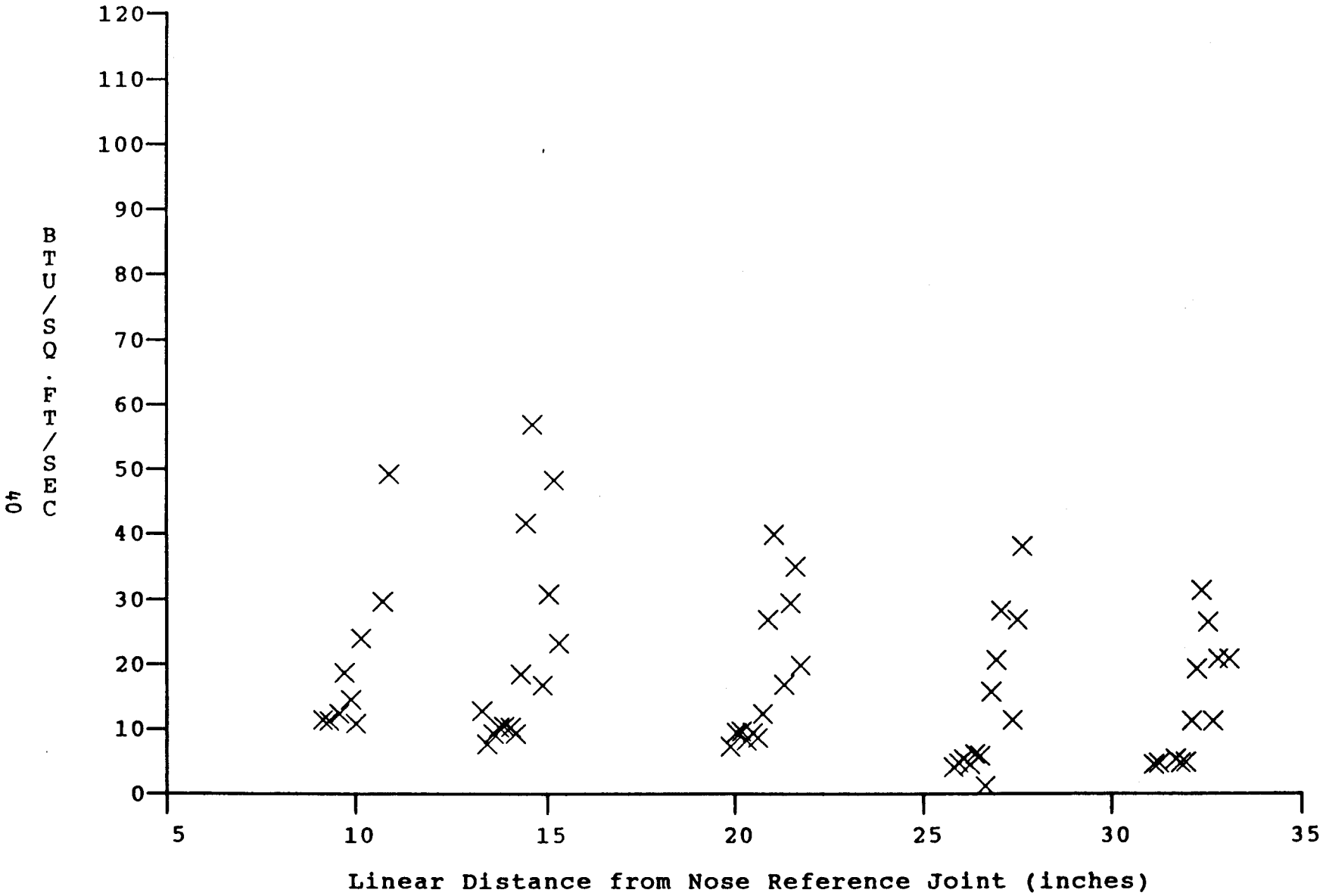


Figure 34b DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 41, B'=0.560, M=15, Re/FT=1.5X10^6)

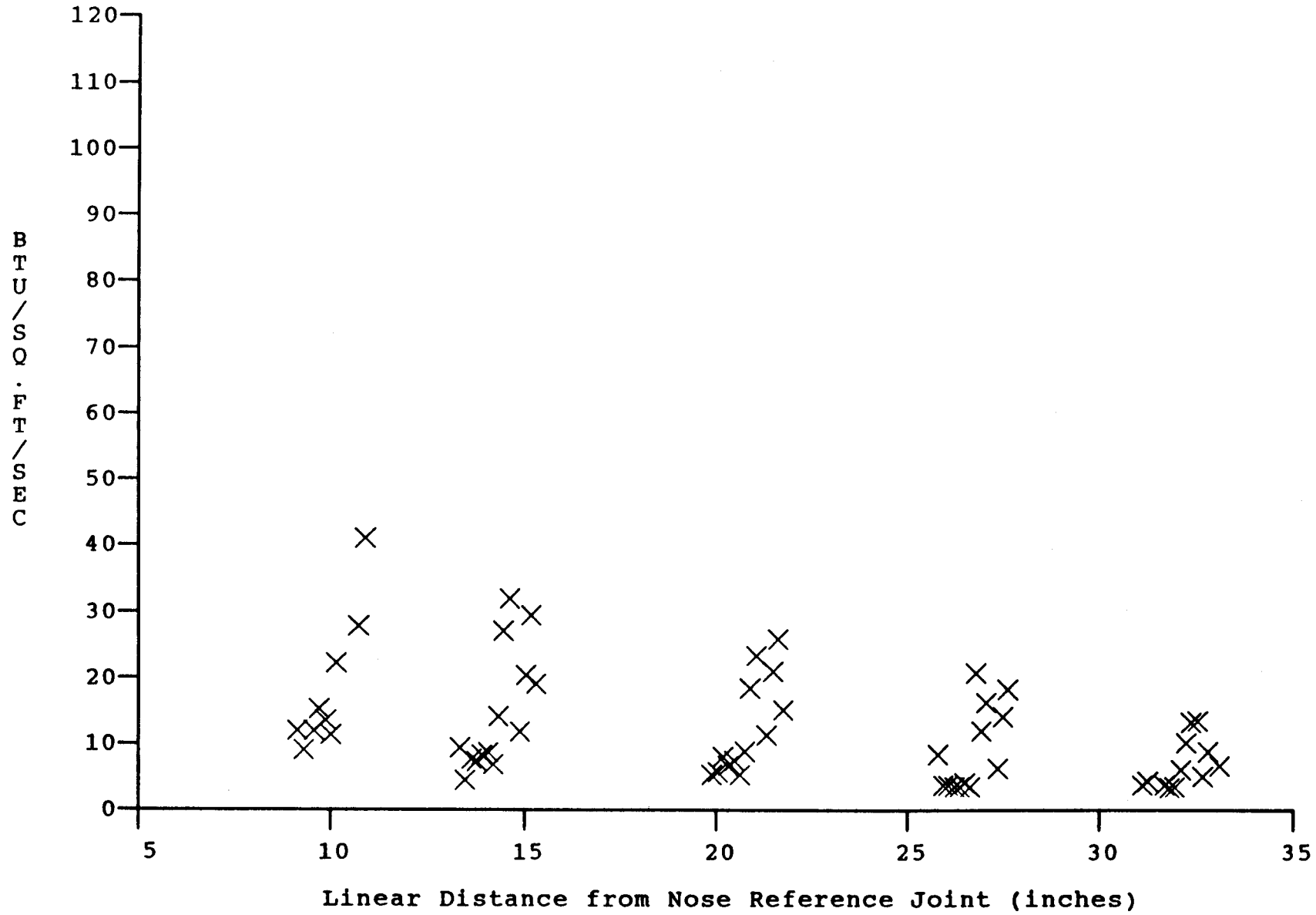


Figure 34c DISTRIBUTION OF HEATING ALONG MODEL SURFACE
(RUN # 42, B'=1.329, M=15, Re/FT=1.5X1J^6)

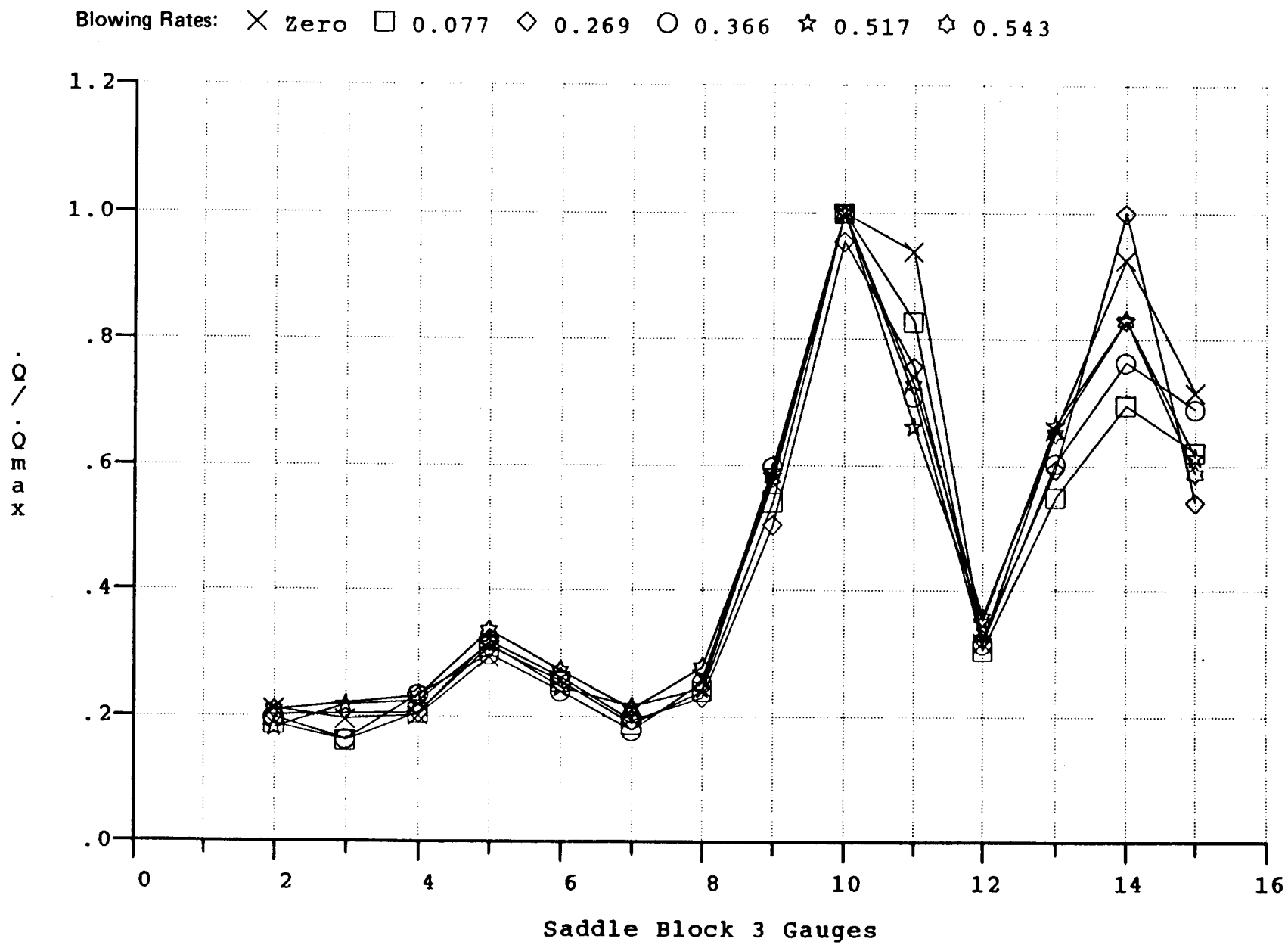


Figure 35a NORMALIZED HEAT RATE DISTRIBUTIONS FOR VARYING BLOWING RATE
M=13, Re/FT = 5X10 6, N2 INJECTANT, SHARP NOSE

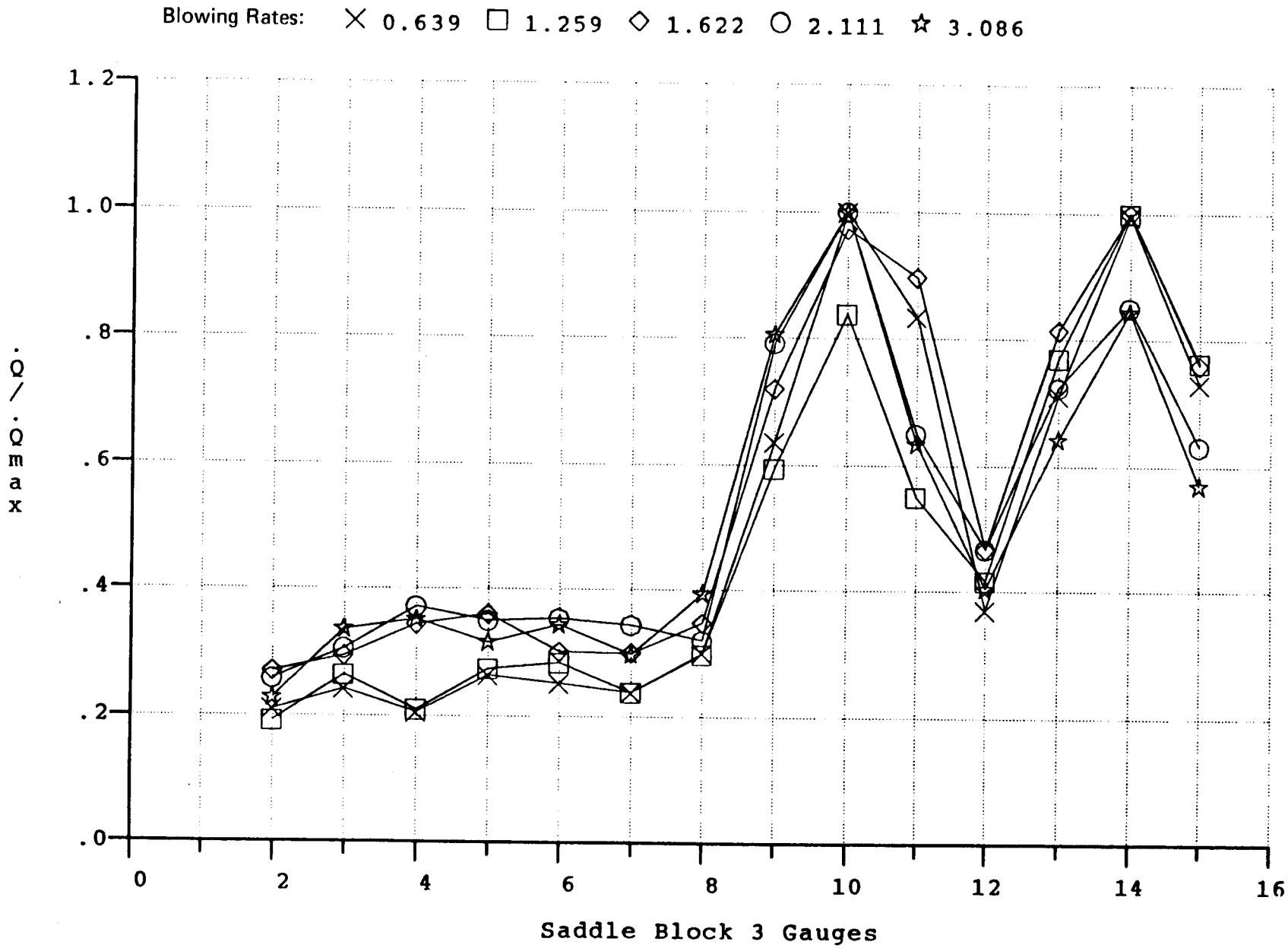


Figure 35b NORMALIZED HEAT RATE DISTRIBUTIONS FOR VARYING BLOWING RATE
M=13, Re/FT = 5X10 6, N2 INJECTANT, SHARP NOSE

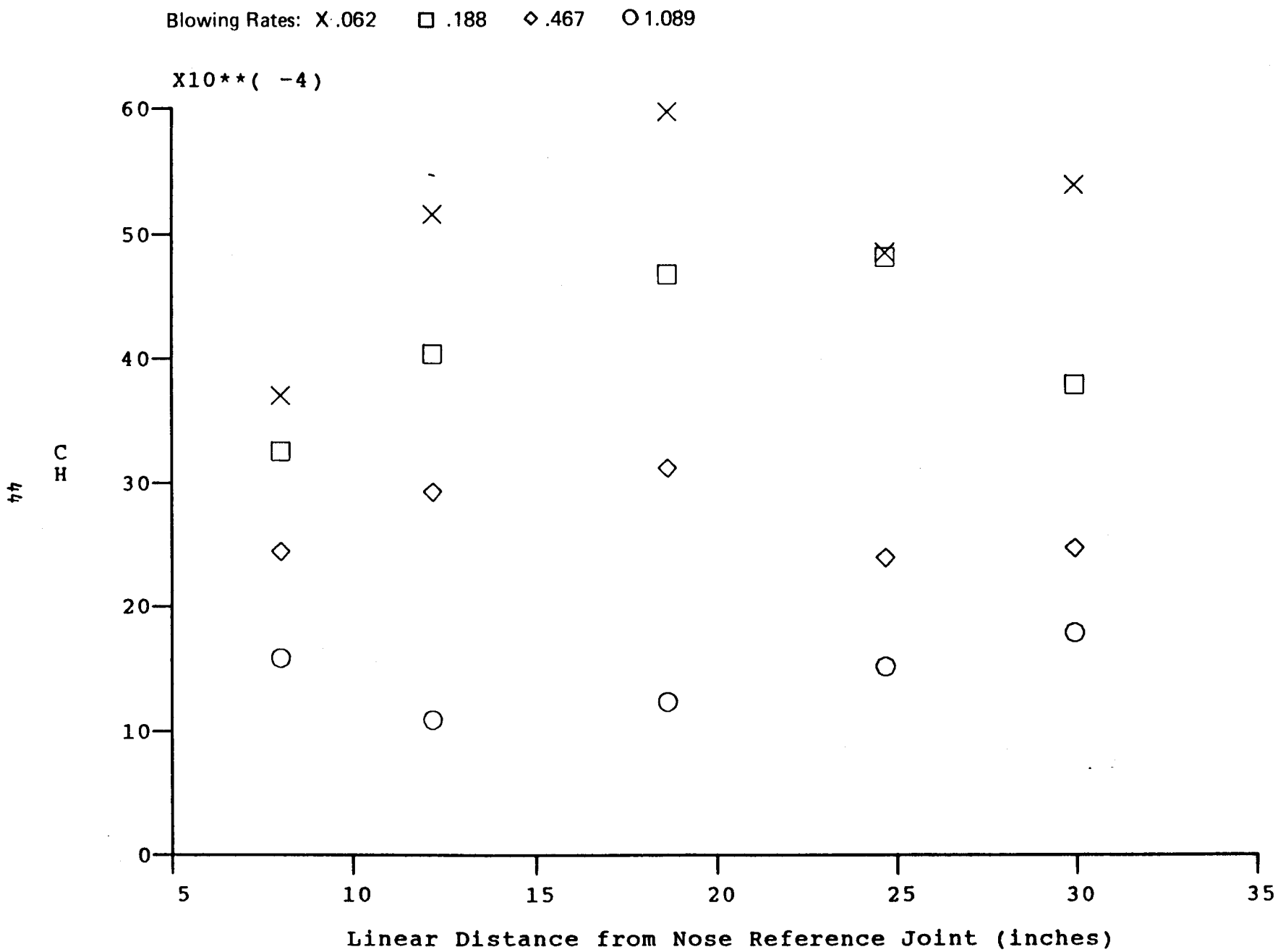


Figure 36a PEAK HEATING RATE ON ROUGHNESS ELEMENTS ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 11)

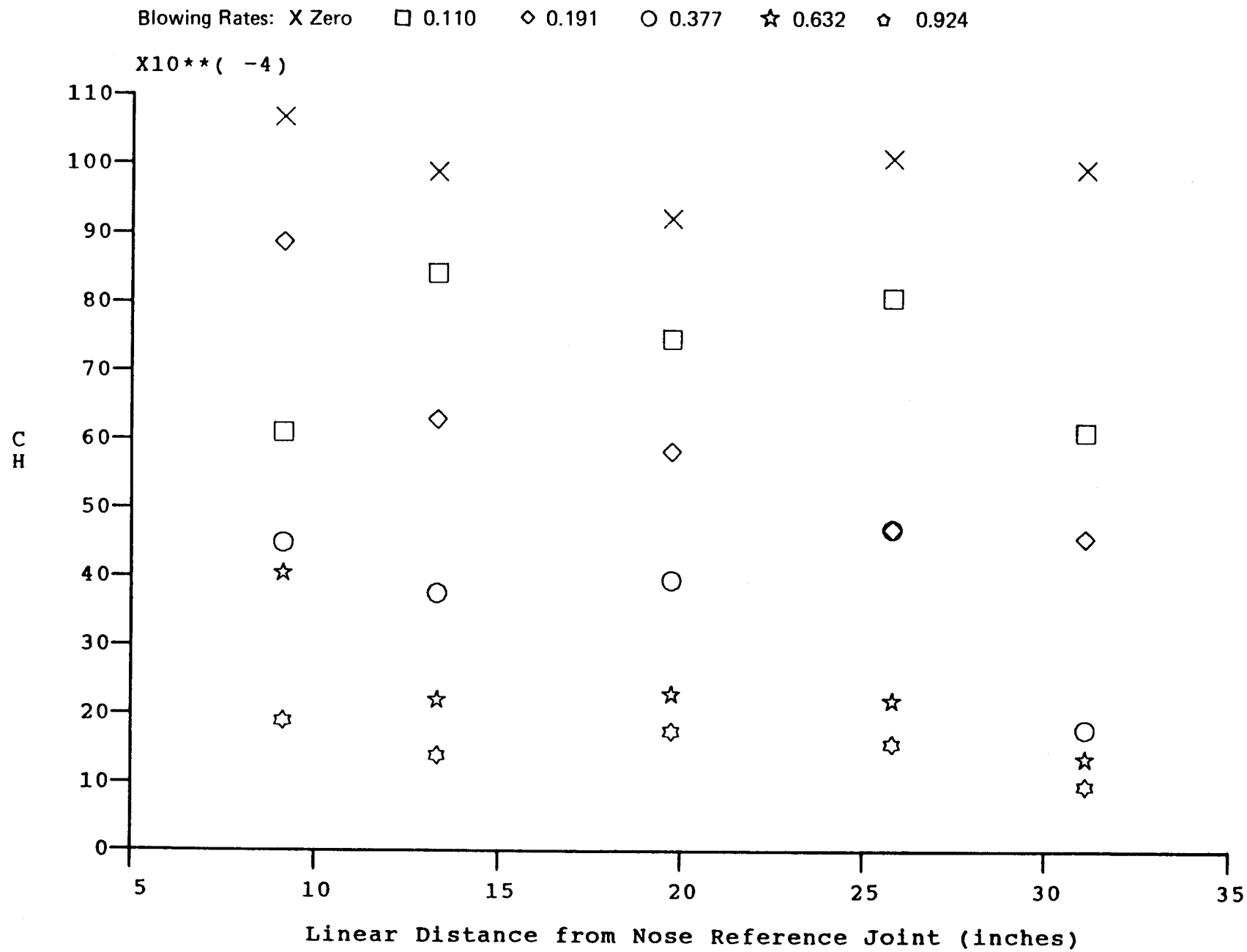


Figure 36b PEAK HEATING RATE ON ROUGHNESS ELEMENTS ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 13)

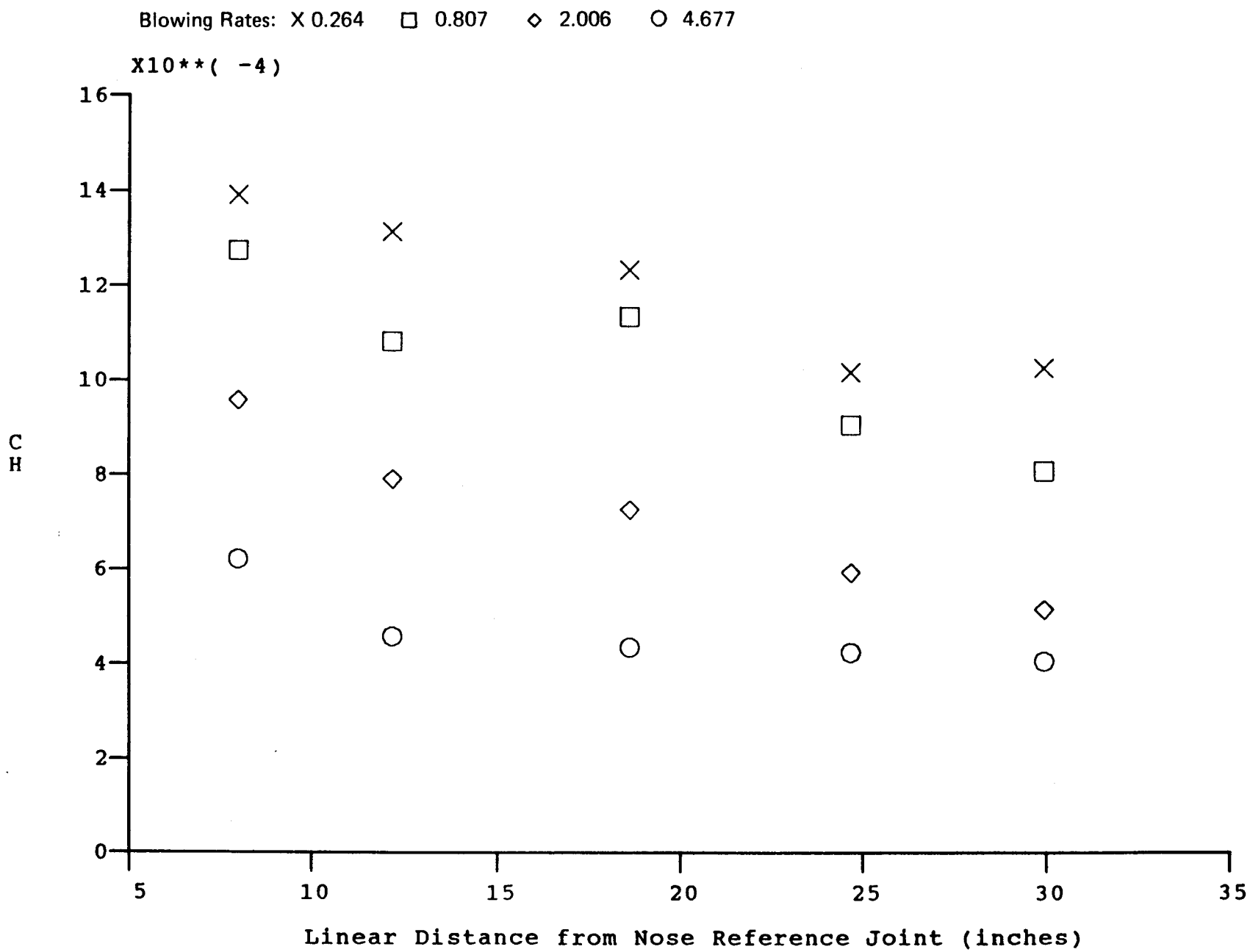


Figure 37a HEATING RATE ON BASE ELEMENTS ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 11)

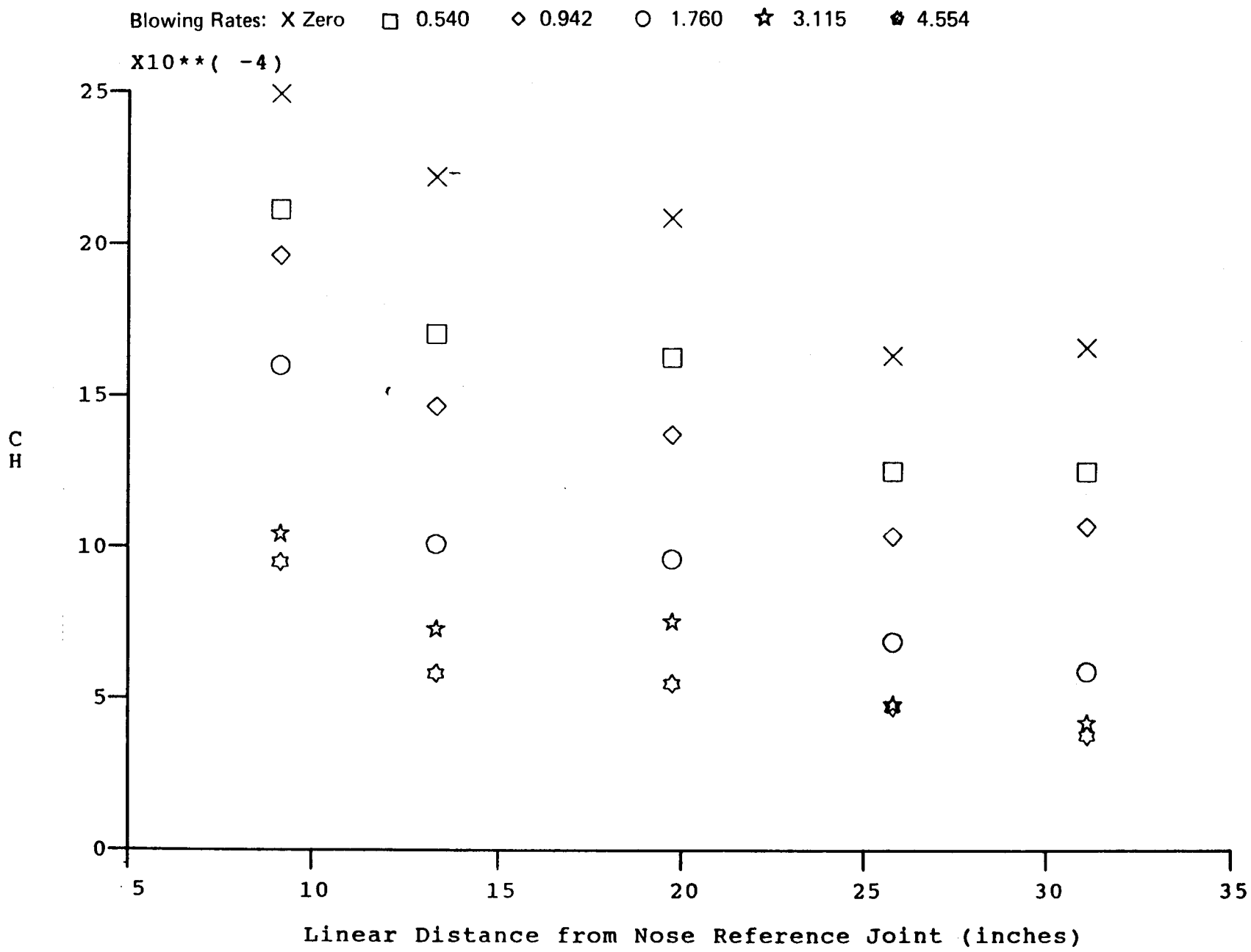


Figure 37b HEATING RATE ON BASE ELEMENTS ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 13)

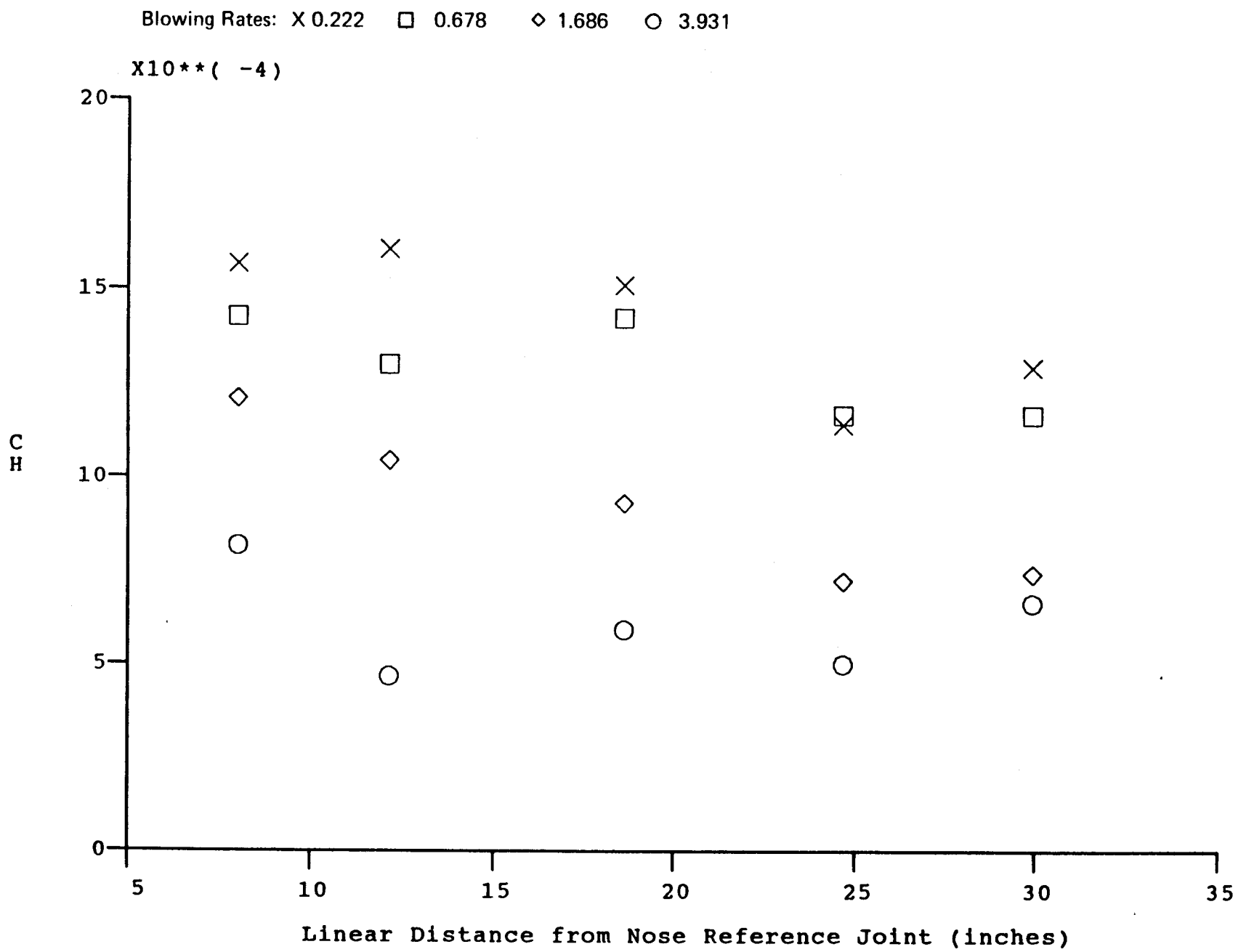


Figure 38a CALORIMETER HEATING RATE ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 11)

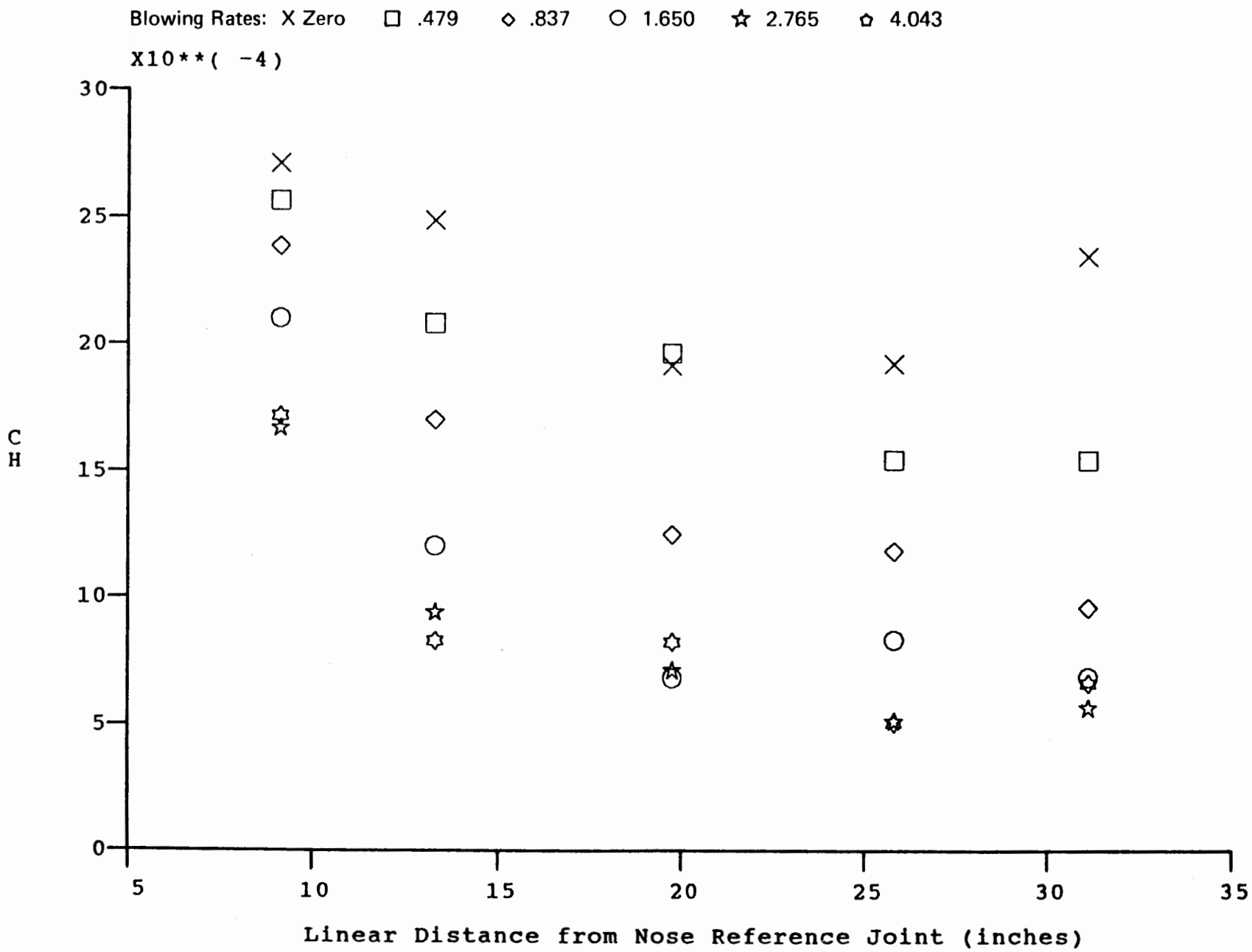


Figure 38b CALORIMETER HEATING RATE ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 13)

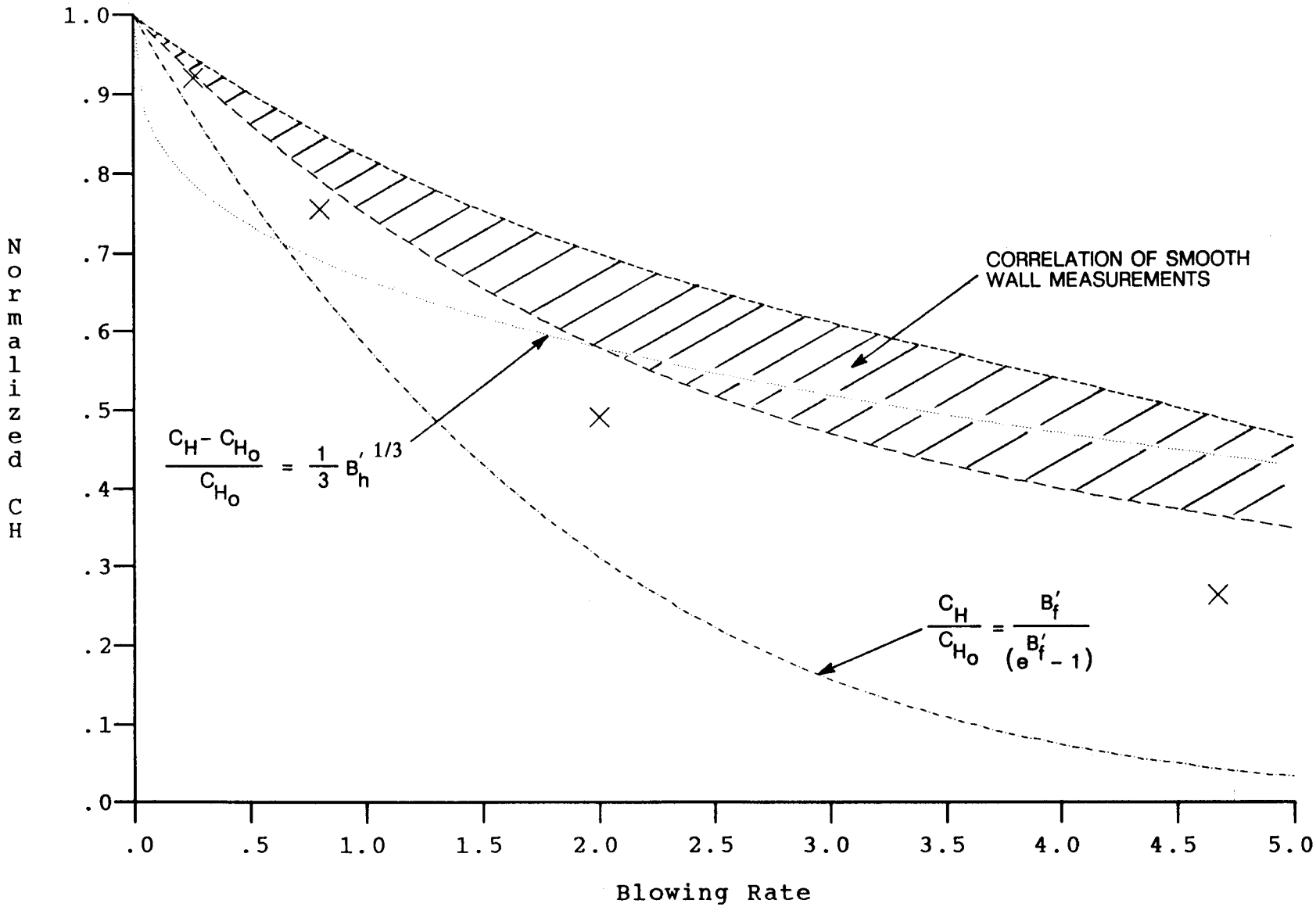


Figure 39a VARIATION OF NORMALIZED PEAK HEATING WITH BLOWING RATE MACH # = 11 (NORMALIZED CH = CH/CHo)

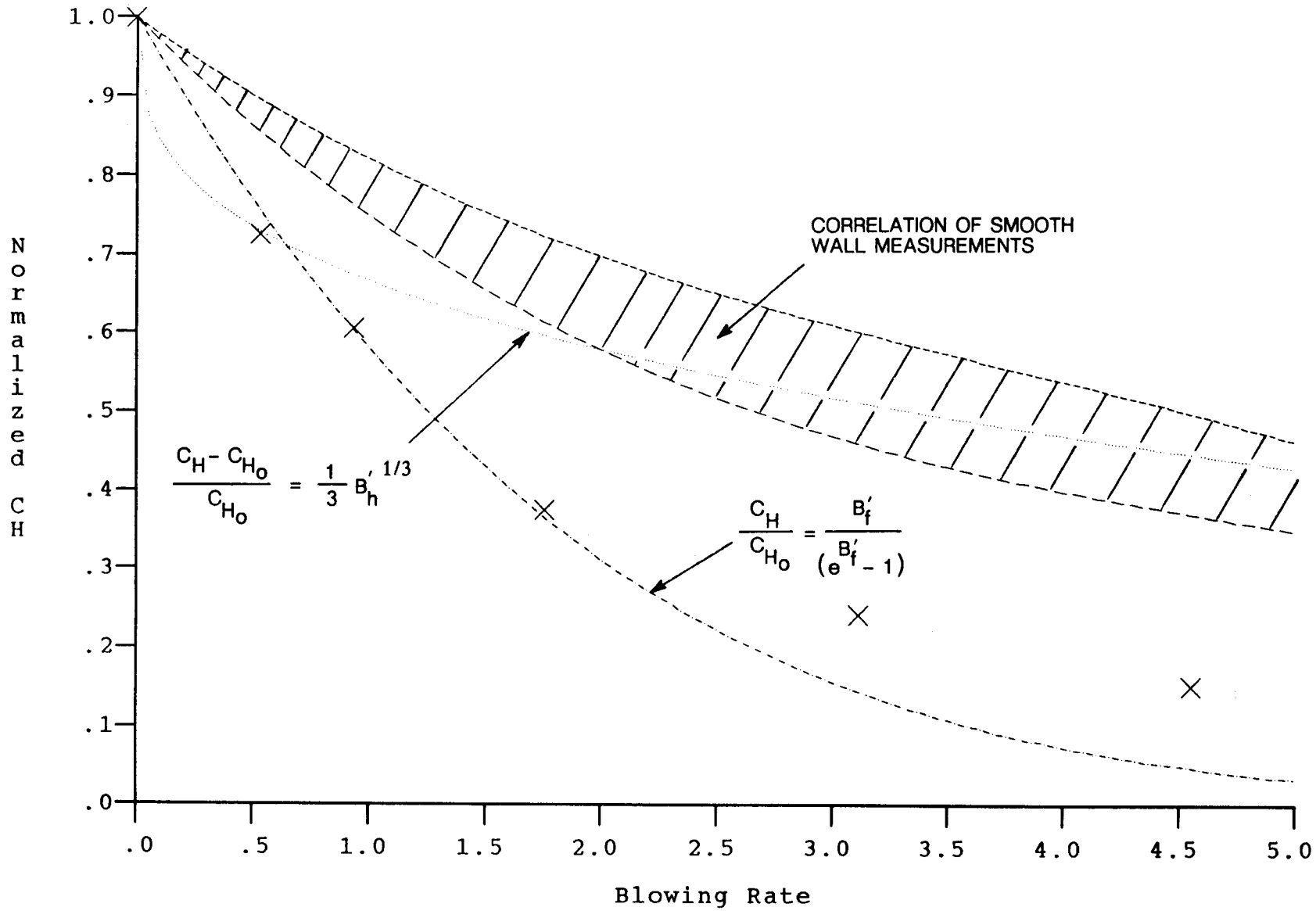


Figure 39b VARIATION OF NORMALIZED PEAK HEATING WITH BLOWING RATE
MACH # = 13 (NORMALIZED CH = CH/CHo)

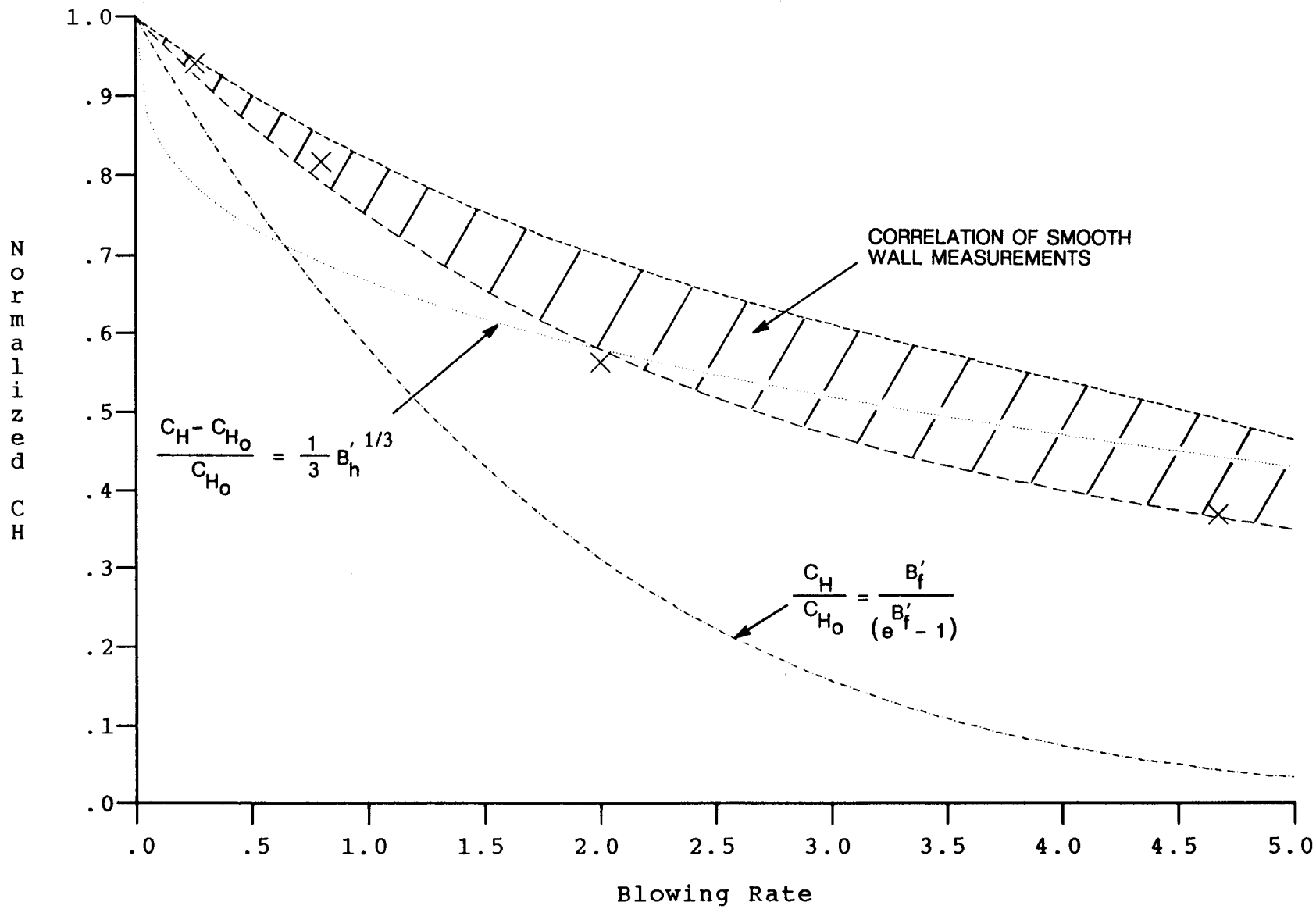


Figure 40a VARIATION OF NORMALIZED BASE HEATING WITH BLOWING RATE MACH # = 11 (NORMALIZED CH = CH/CHo)

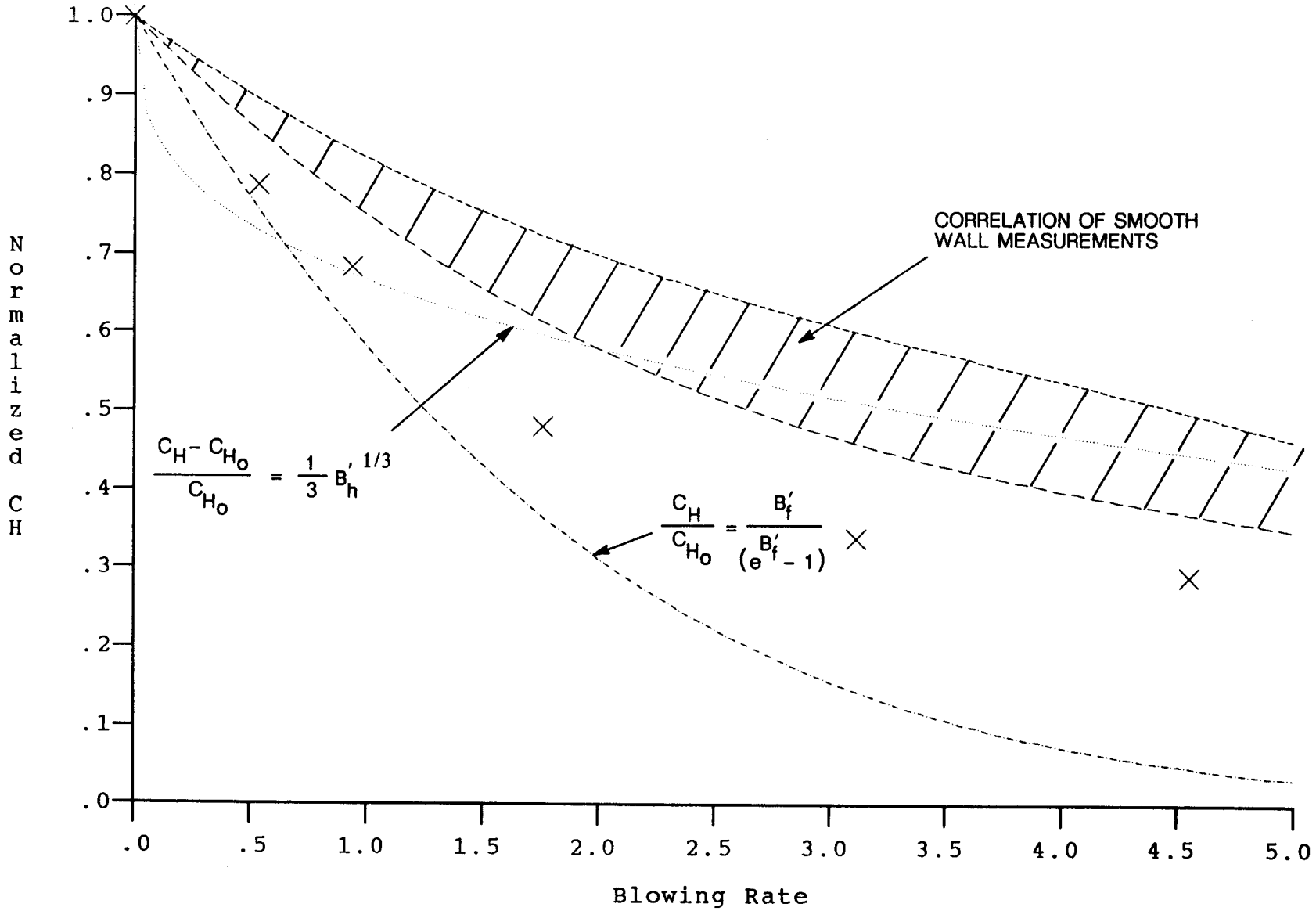


Figure 40b VARIATION OF NORMALIZED BASE HEATING WITH BLOWING RATE
MACH # = 13 (NORMALIZED CH = CH/CHo)

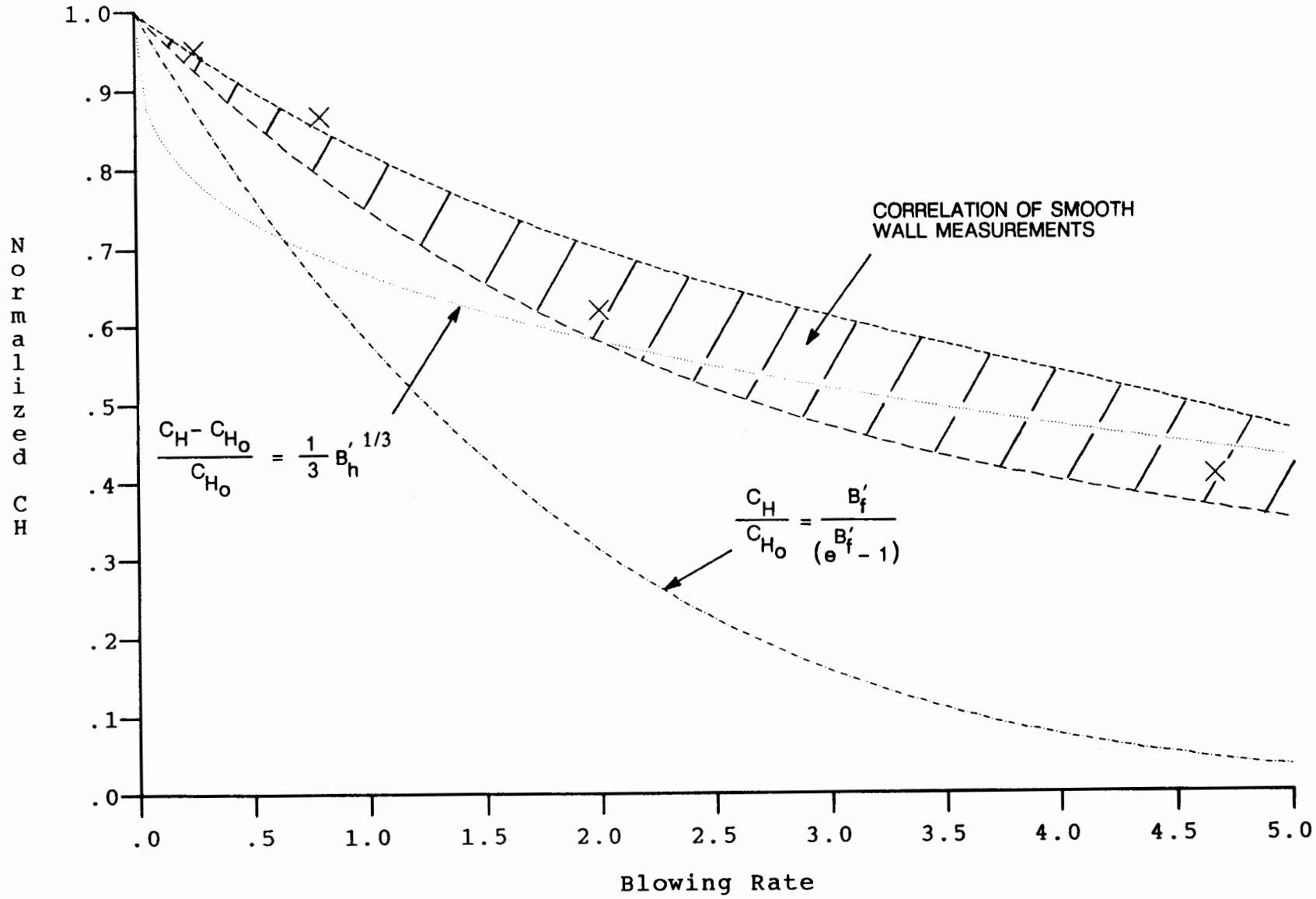


Figure 41a VARIATION OF NORMALIZED CALORIMETER HEATING WITH BLOWING RATE MACH # = 11 (NORMALIZED CH = CH/CHo)

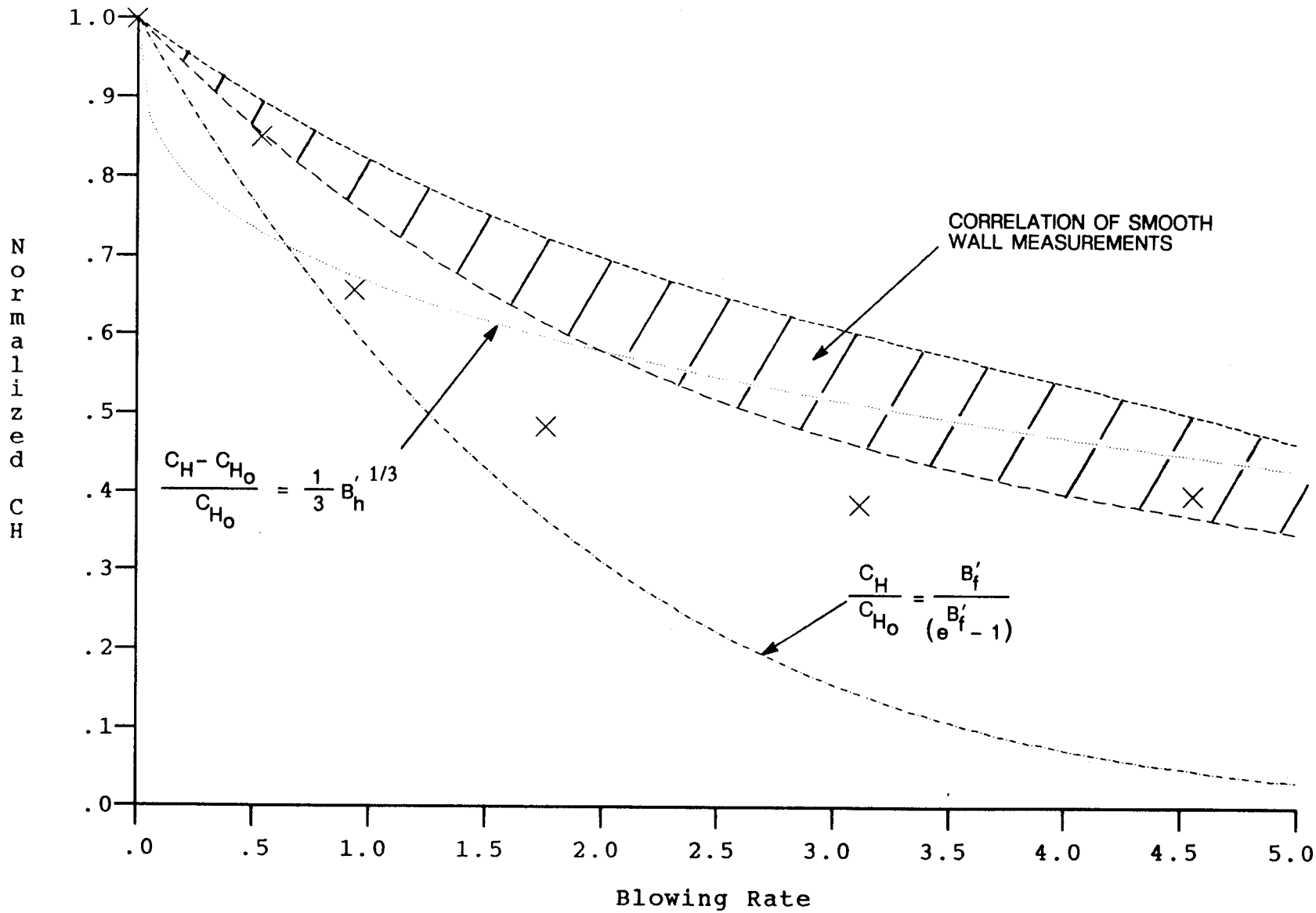


Figure 41b VARIATION OF NORMALIZED CALORIMETER HEATING WITH BLOWING RATE
MACH # = 13 (NORMALIZED CH = CH/CHo)

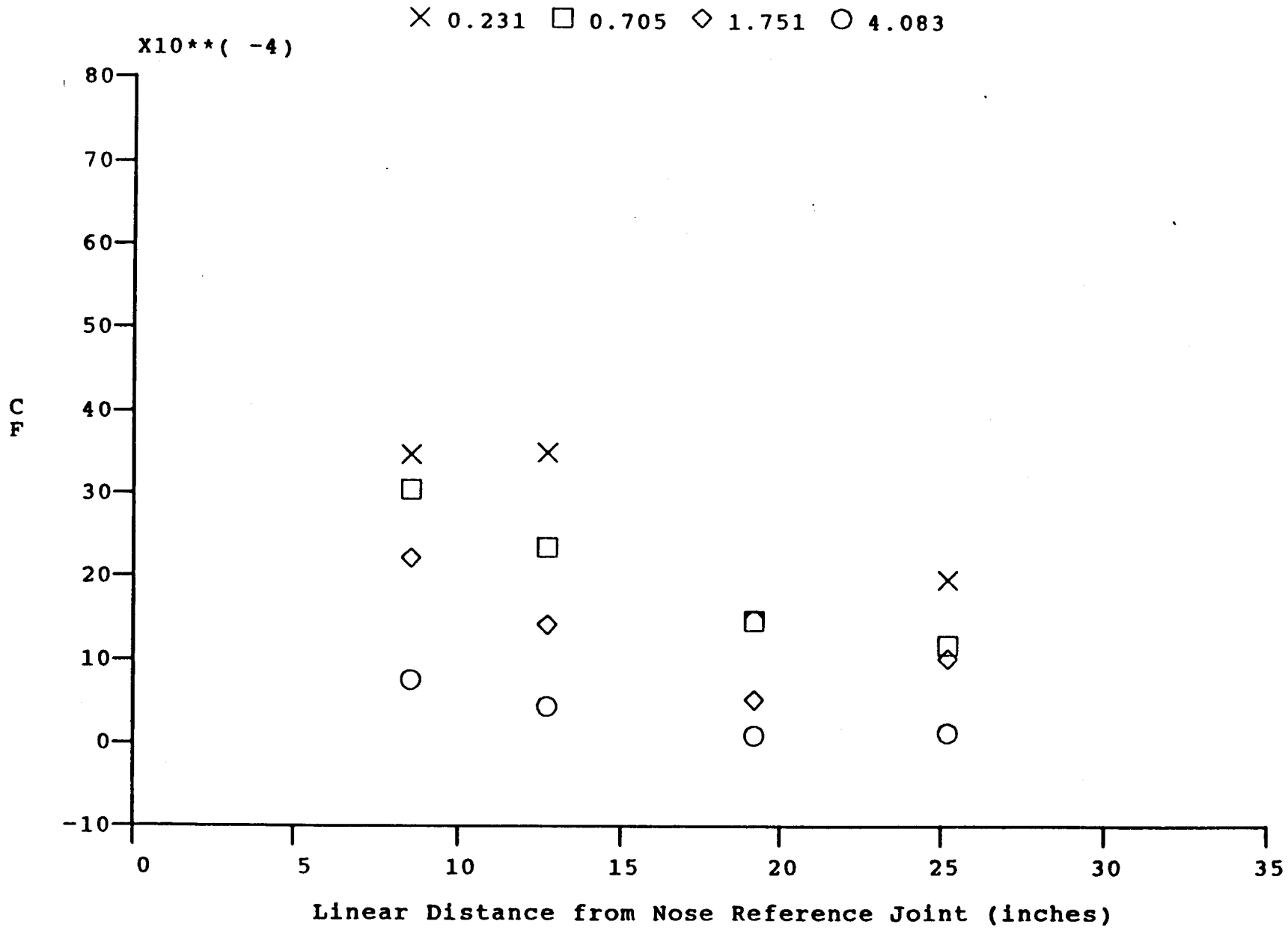


Figure 42a SKIN FRICTION ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 11)

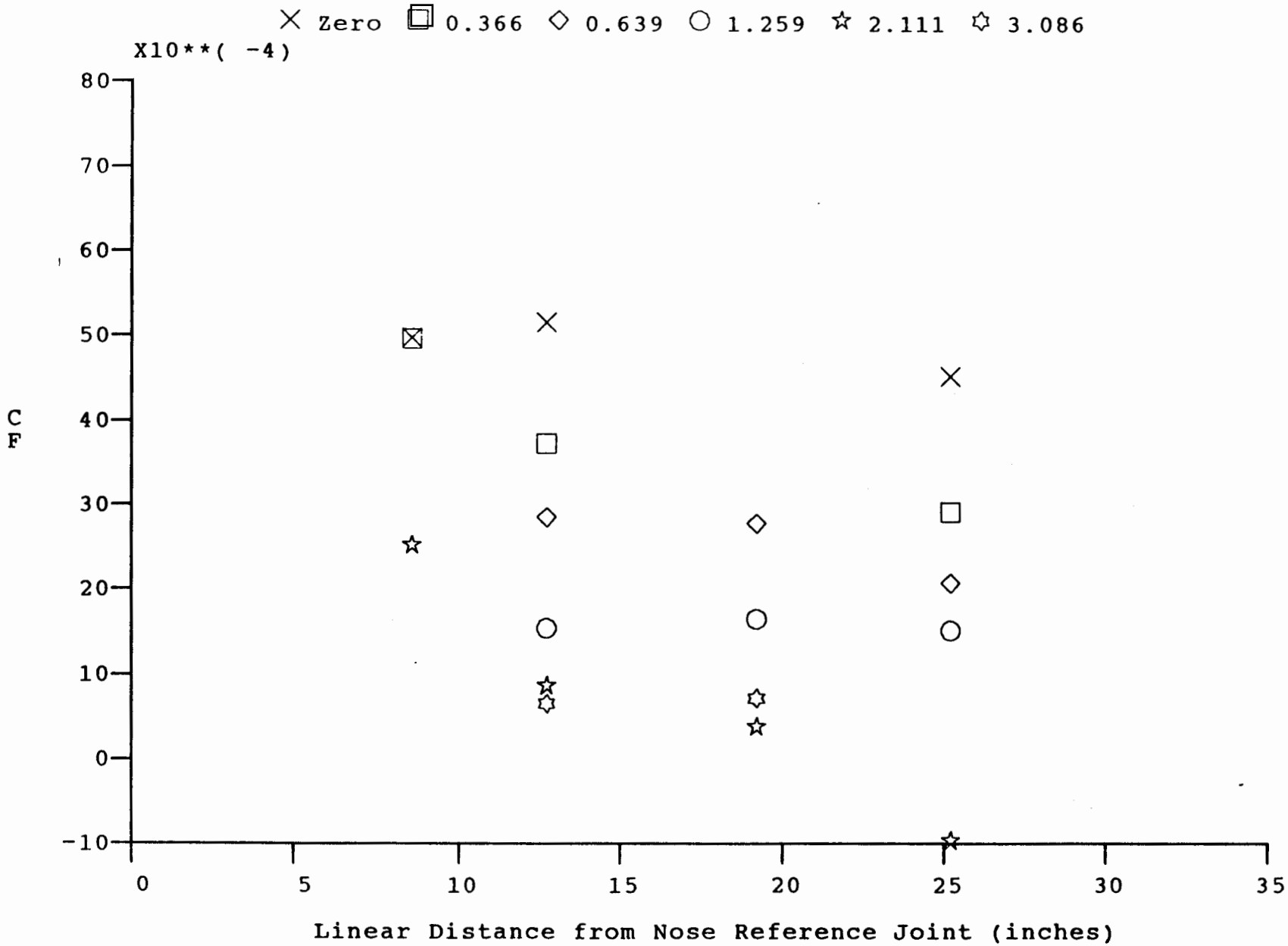


Figure 42b SKIN FRICTION ALONG MODEL FOR VARIOUS BLOWING RATES (MACH # = 13)

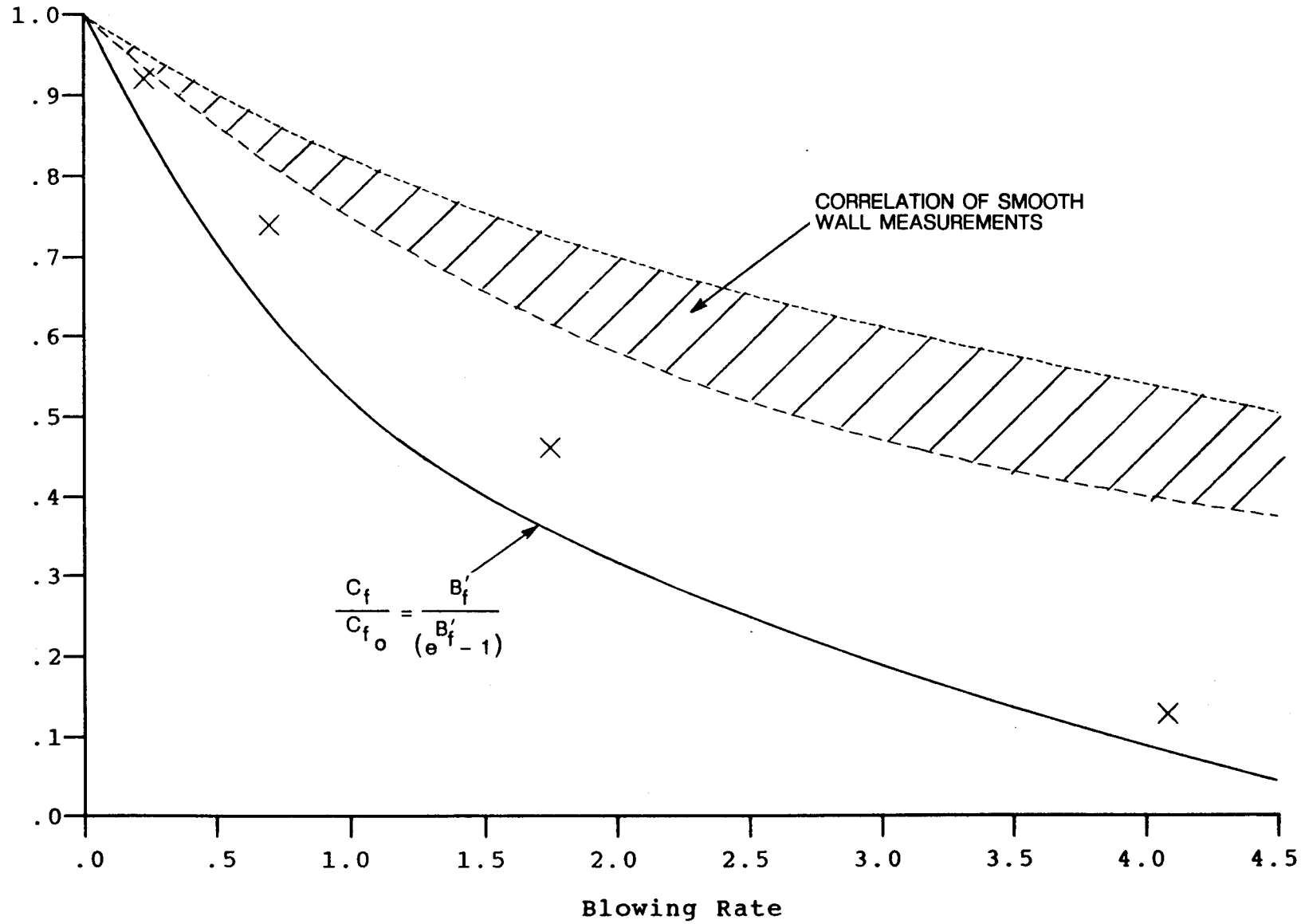


Figure 43a VARIATION OF NORMALIZED SKIN FRICTION WITH BLOWING RATE MACH # = 11 (NORMALIZED CF = CF/CFo)

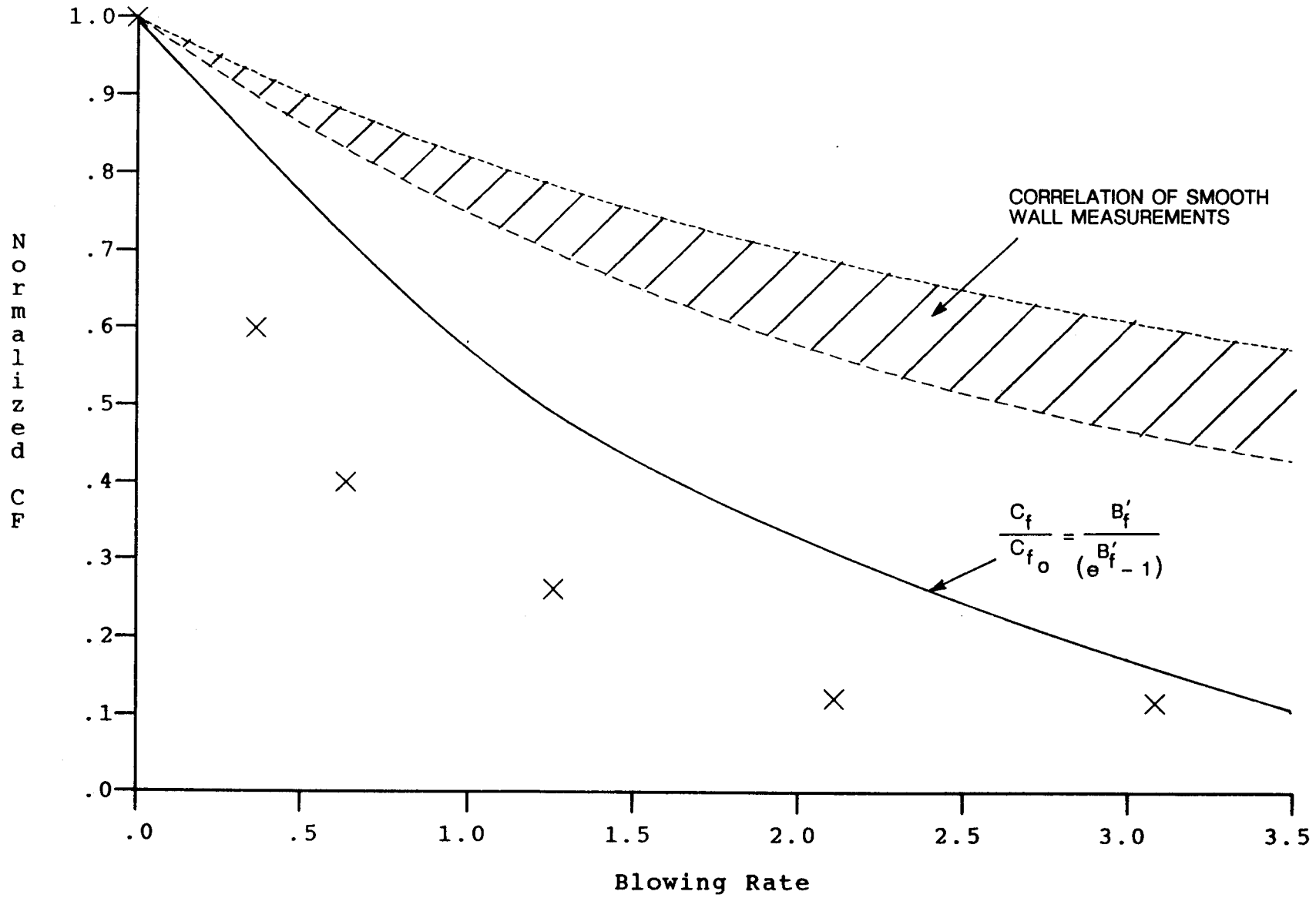


Figure 43b VARIATION OF NORMALIZED SKIN FRICTION WITH BLOWING RATE
MACH # = 13 (NORMALIZED CF = CF/CFo)

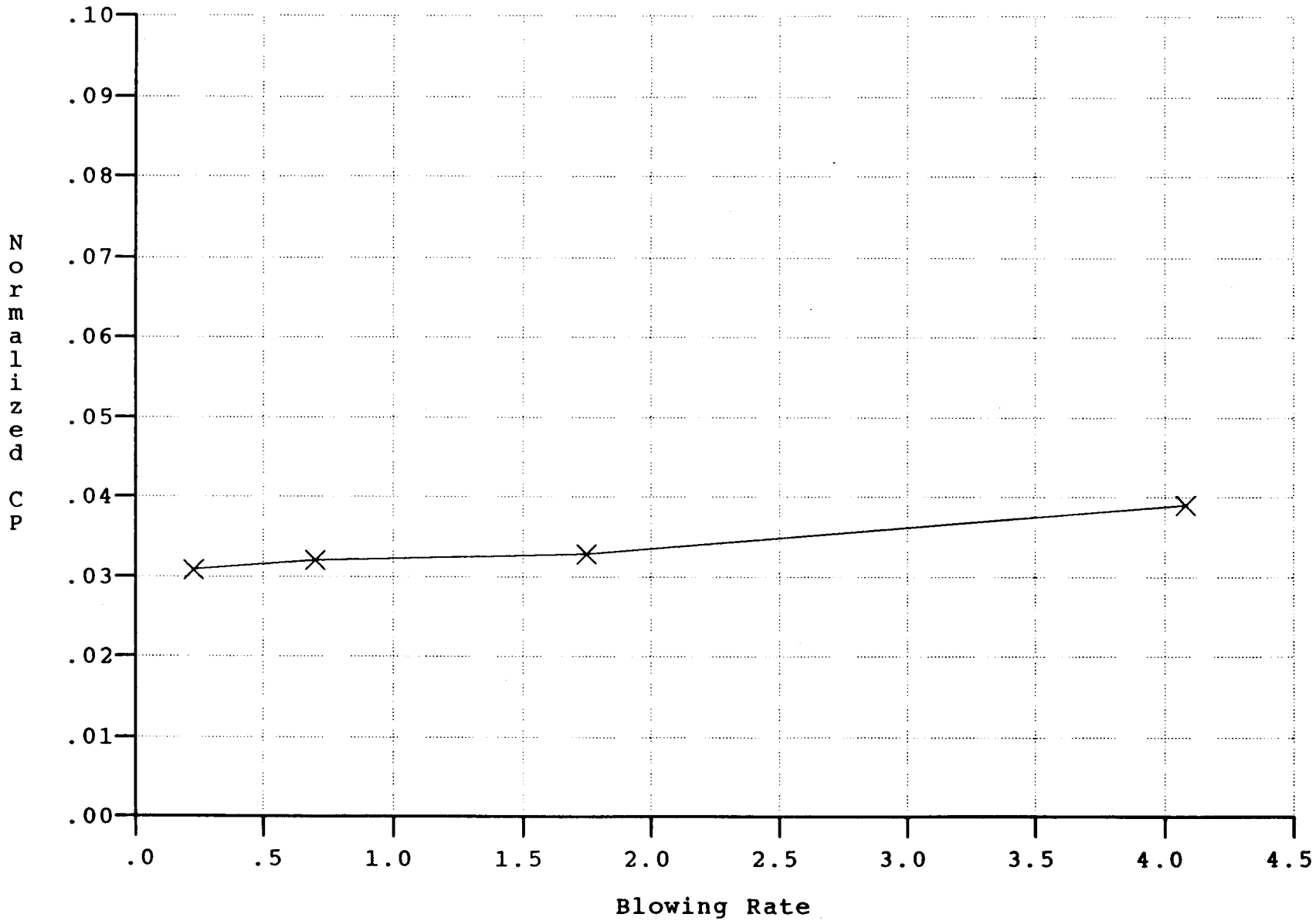


Figure 44a VARIATION OF CP WITH BLOWING RATE (MACH # = 11)

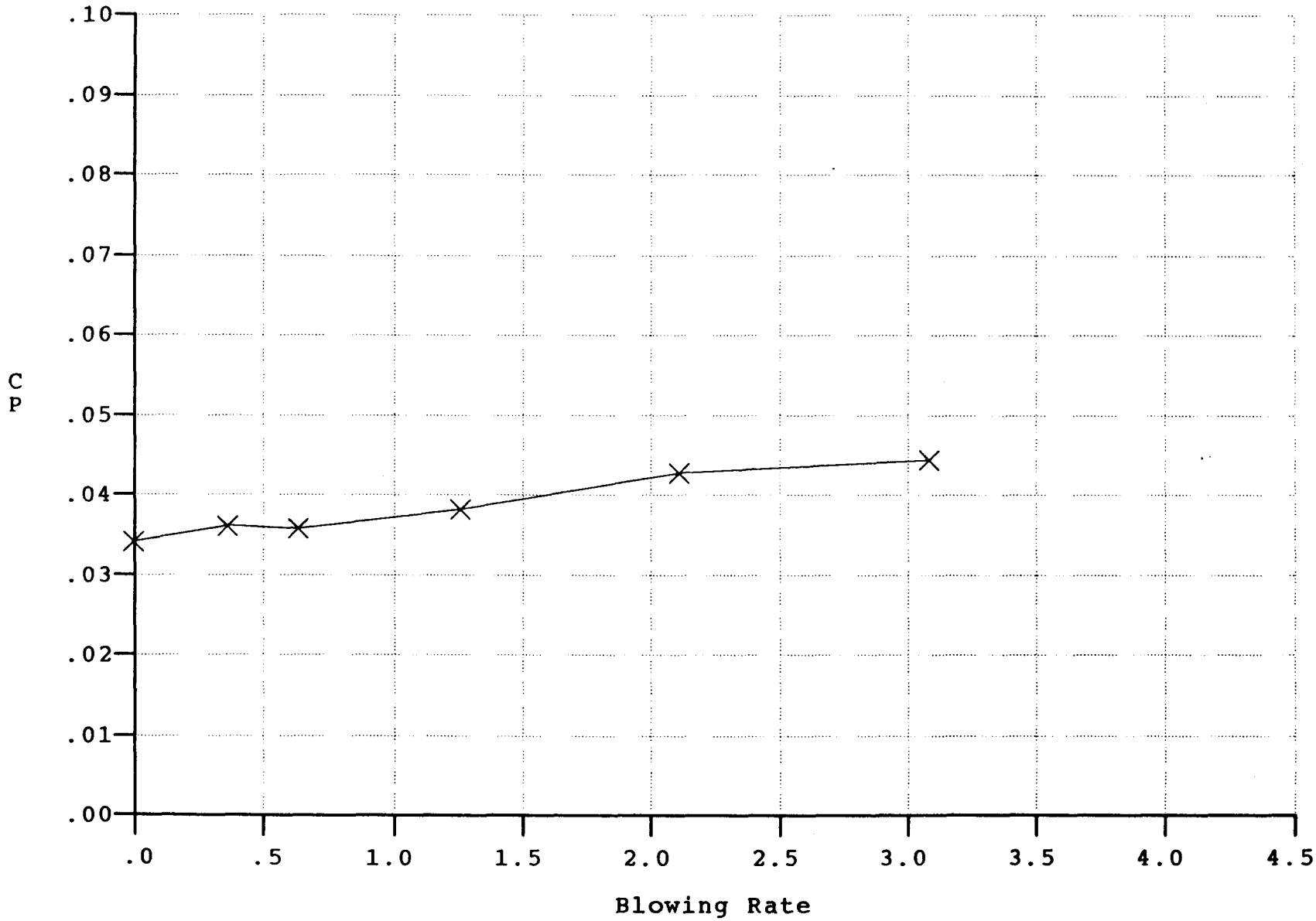


Figure 44b VARIATION OF CP WITH BLOWING RATE (MACH # = 13)